

Chapter-1

INTRODUCTION OF MICROSOFT EXCEL

Structure

- 1.0 Learning Objectives
 - 1.1 Introduction
 - 1.2 Basic of Microsoft Excel
 - 1.2.1 Microsoft Excel
 - 1.2.2 Features of Microsoft Excel
 - 1.3 How to Open Microsoft Excel?
 - 1.3.1 Open Microsoft Excel in Windows 7
 - 1.3.2 Open Microsoft Excel in Windows 8,8.1,10
 - 1.3.3 Open Microsoft Excel in Windows 11
 - 1.4 Understanding Microsoft Excel Basic Function
 - 1.5 Understanding the Ribbon
 - 1.6 Summary
 - 1.7 Answer to Self-Check exercises
 - 1.8 Suggested Readings

1.0 LEARNING OBJECTIVES

After going through this lesson, you will be able to:

- Understand the basic of Microsoft Excel
- Understand how to open Microsoft Excel
- Understand the basic function of Microsoft Excel
- Understand the concept of Ribbon

1.1 Introduction

Microsoft Excel is an application for creating spreadsheets that employs tables to store and analyse numerical and statistical data using formulae and functions.

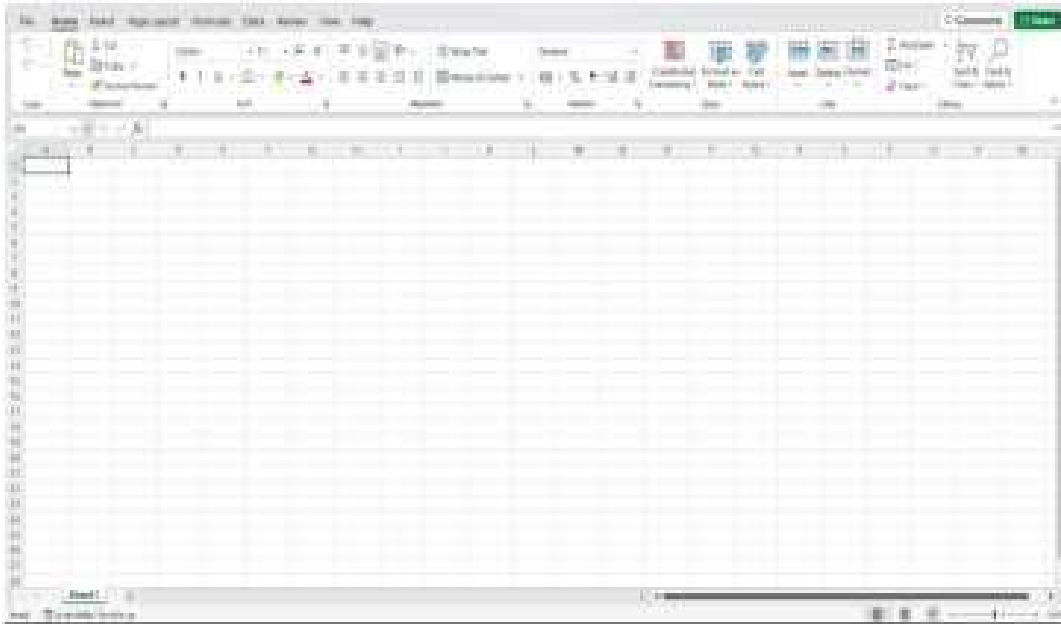
Simply said, MS Excel is a robust spreadsheet application that comes with Microsoft Office and is primarily used to enter data into tables. It has a number of rows and columns where data can be entered. The programme is regarded as useful for a variety of activities, including bookkeeping, creating payslips, sorting and filtering data interactively, and calculating weekly expenses.

1.2 Basic of Microsoft Excel

1.2.1 Microsoft Excel

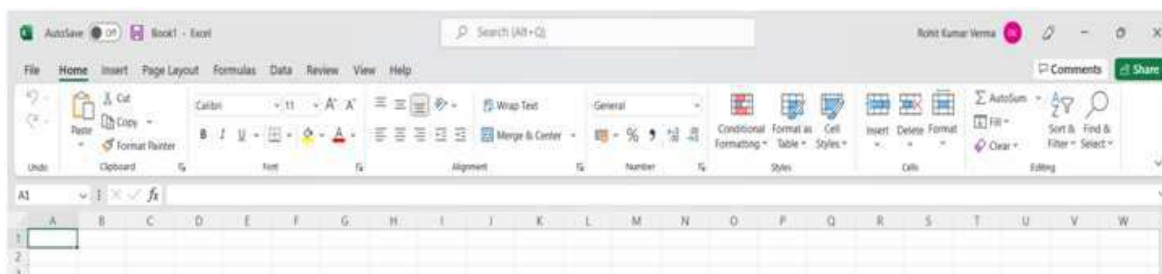
Microsoft's Excel spreadsheet programme is a part of the Office family of business software programmes. Users of Microsoft Excel may format, arrange, and compute data in a spreadsheet.

One can enter data in the form of tables using the spreadsheet tool MS Excel. An Excel spreadsheet makes it simple to analyse data. An Excel spreadsheet looks like this, as shown in the image below:



1.2.2 Features of Microsoft Excel

The following are some of Microsoft Excel's key attributes:



Home

Font size, font styles, font colour, background colour, alignment, formatting options and styles, cell insertion and deletion, and editing options all are available under the home option.

Insert

Insert includes features like table format and style, image and figure insertion, graph, chart, and sparkline addition, header and footer choice, equation and symbol options, and more.

Page Layout

Page Layout comprises options like themes, orientation and page setup.

Formulas

In order to generate a value, a cell can contain formulas, cell references, or functions. Formulas must be introduced with an equal's sign ("=").

Data

This category offers tools for adding external data (from the web), filtering choices, and data.

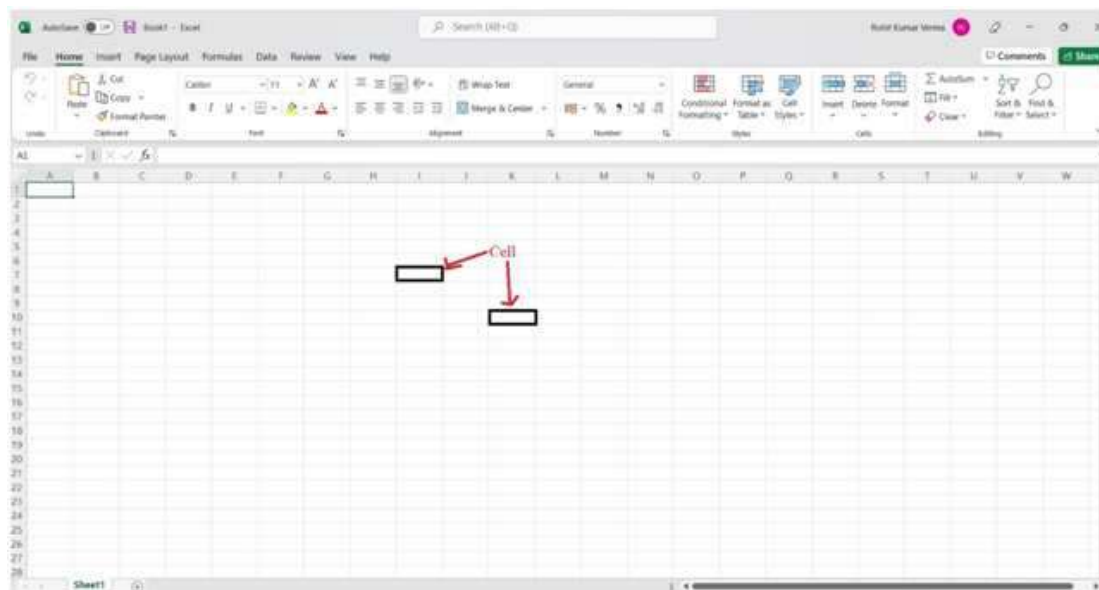
Review

A reader may write comments in this section after proofreading an excel sheet (using a feature similar to spell check).

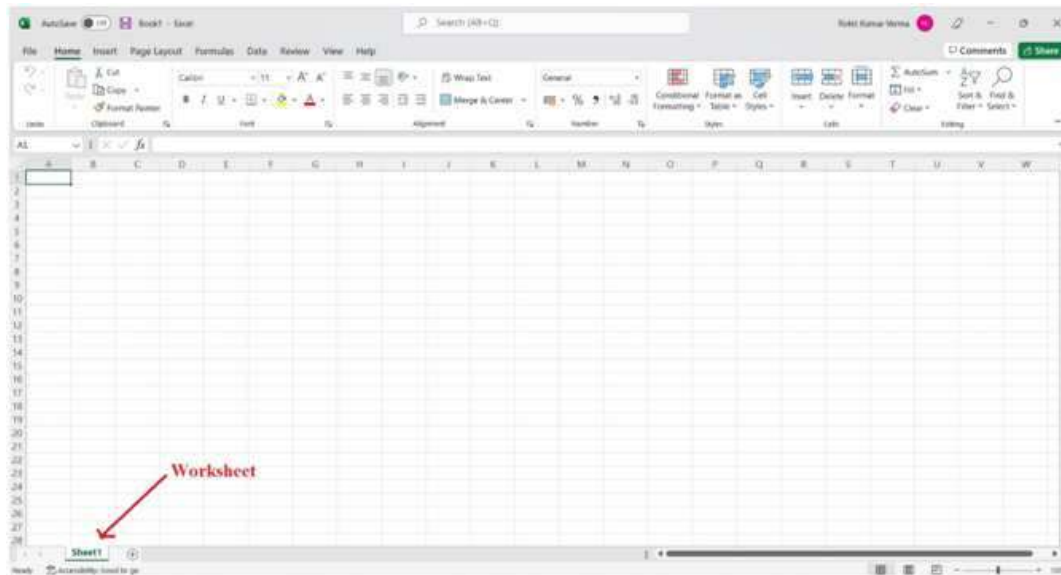
View

Here, we can adjust the various views in which we want the spreadsheet to be shown. Under this area, options for pane organisation and zooming in and out are provided. You must first comprehend the three most crucial Excel features:

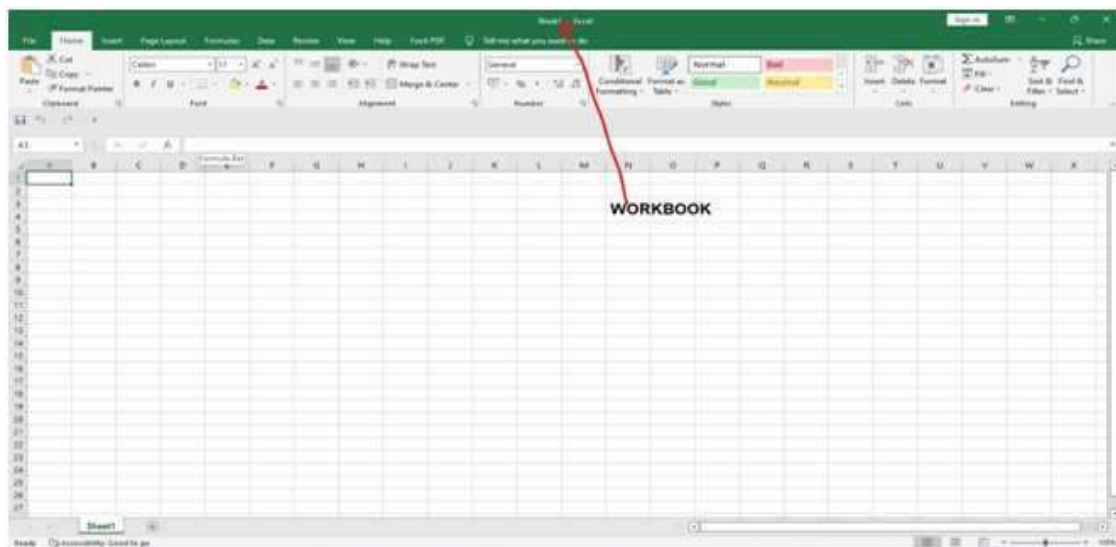
1. **Cell:** The smallest but most useful component of a spreadsheet is a cell. You have the option of typing or copying and pasting your data into a cell. Text, numbers, and dates all qualify as data. Additionally, you can alter its size, font colour, background colour, borders, and other features. A cell's address, which includes its column number and row number, serves as a unique identifier for each individual cell (If a cell is on 11th row and on column AB, then its address will be AB11).



2. **Worksheet:** A worksheet is made up of individual cells may contain text, a formula, or a value. Additionally, it contains an unseen sketch layer that stores diagrams, charts, and photos. Clicking the tab at the bottom of the workbook window will take you to each worksheet in the workbook. A chart sheet, which displays a single chart and is accessed by clicking a tab, can also be stored in a workbook.



3. **Workbook:** Every application has a separate file, just like a workbook does. There are one or more worksheets in every workbook. A workbook is often referred to as a collection of worksheets or as a single worksheet. Worksheets can be added or removed, hidden without being deleted from the workbook, and the arrangement of your worksheets can be modified.



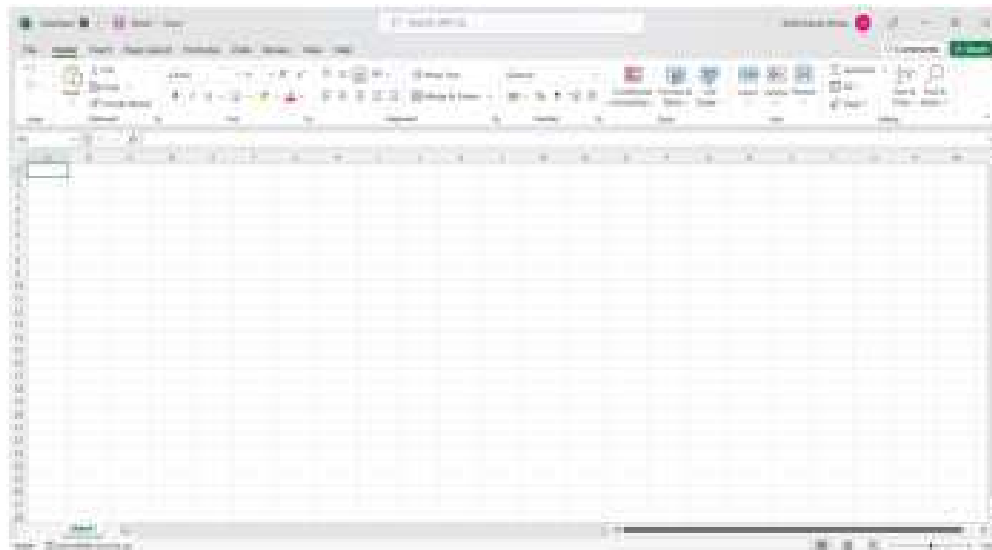
1.3 How to open MS Excel?

1.3.1 Open Microsoft Excel in Windows 7

The procedures listed below should help you open Microsoft Excel on your computer:

- Click Start, followed by All Programs.
- Next step is to click on MS Office
- Finally, select MS-Excel as your output type.

Your Microsoft Excel programme will start, and you'll see the Microsoft Excel window shown below.

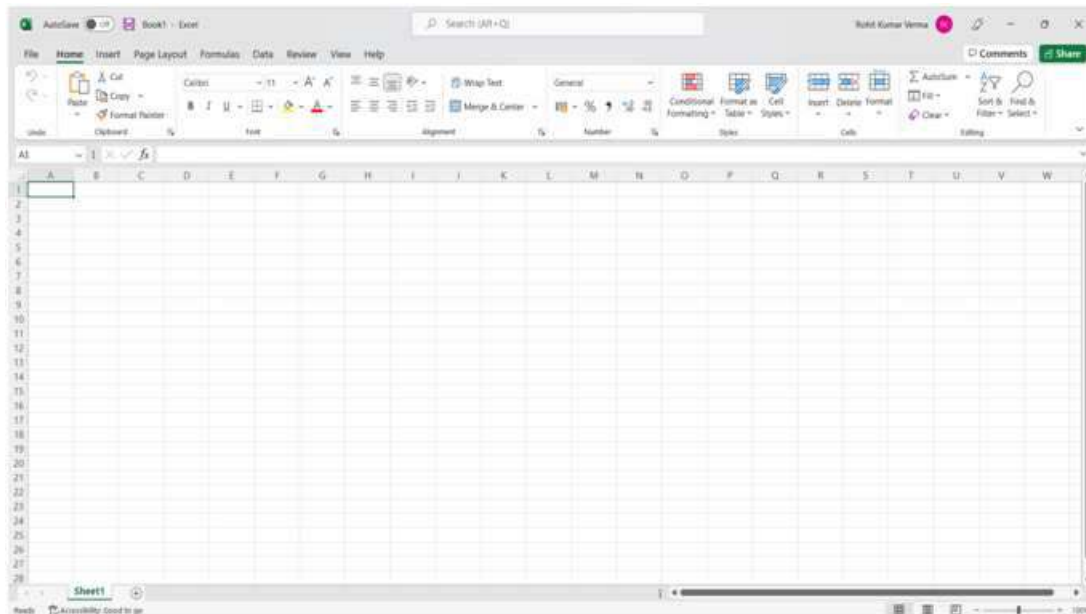


1.3.2 Open Microsoft Excel in Windows 8,8.1,10

- Press the Windows key on the keyboard or click the Windows button in the taskbar's lower-left corner.
- In the lower-left corner of the screen, select "Search Box."
- Type "Excel".
- Click the Excel icon as shown in the screenshot below.



Your Microsoft Excel programme will start, and you'll see the Microsoft Excel window shown below.

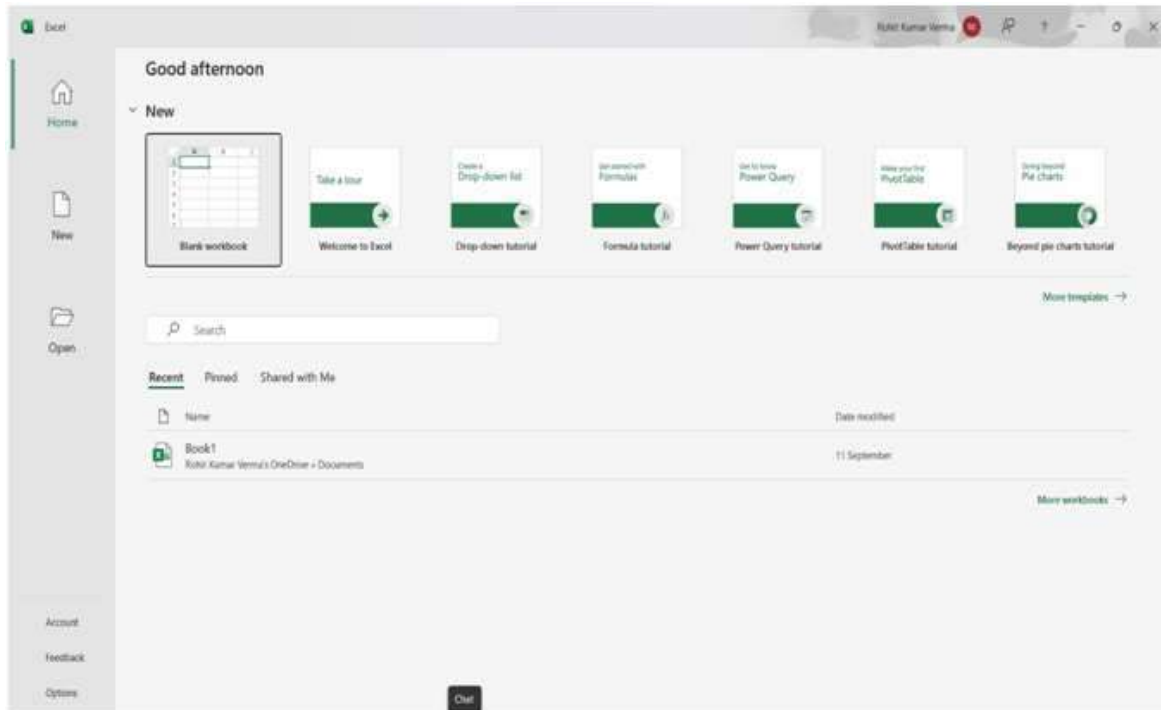


1.3.3 Open Microsoft Excel in Windows 11

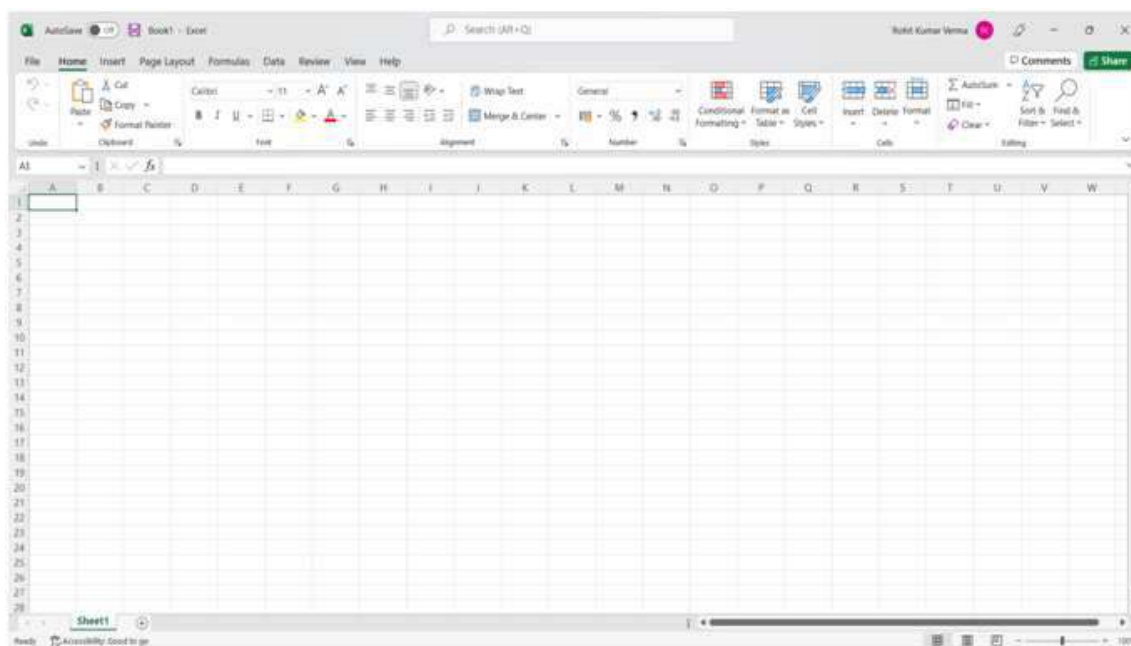
- Press the Windows key on the keyboard or click the Windows button in the taskbar's lower-left corner.
- Click the Excel icon as shown in the screenshot below.



- Next step is to click on Blank Workbook as shown in the screenshot below:



- Your Microsoft Excel programme will start, and you'll see the Microsoft Excel window shown below.



1.4 Understanding Microsoft Excel Basic Function

Excel has two primary methods for performing calculations: formulas and functions.

FORMULA:

A formula in Excel is an expression that manipulates values in a cell or a range of cells. Consider the formula =A1+A2+A3, which calculates the sum of the values in cells A1 through A3.

FUNCTIONS:

In Excel, functions are predefined formulas. They provide formulas human-friendly labels and do away with tedious manual entry. For instance, =SUM (A1:A3). The formula adds up each value from A1 to A3, or array A.

One of Excel's most crucial features is the functions. You can use it to conduct both simple and complicated computations.

SUM: It gives back the total of the cell's numerical values. You can insert the values directly into the function or refer to the cells where you have data. Using the Excel AutoSum button is the quickest and most straightforward approach to add a group of cells. The selected cell receives an automatic Excel SUM function entry. With the SUM function, you may add up any number in a set of cells.

COUNT: It gives back the number of numerical values found in a cell. You can insert the values directly into the function or refer to the cells where you have data. The COUNT function counts cells with numbers in a range.

AVERAGE: It gives back the cell's mean of all the numerical values. The AVERAGEIF function calculates the average of a range based on a true or false condition.

TIME: According to Excel's time format, it returns a legitimate time serial number.

Hours, minutes, and seconds must be specified [...]

DATE: It gives back a legitimate date serial number that follows Excel's time format.

You must select the day, month, and year.

LEFT: With the leftmost character coming first, this method extracts particular characters from a cell or string (start). The text and the quantity of characters to be extracted must be specified.

RIGHT: Starting from the right, this method removes particular characters from a cell or string (last). The text and the quantity of characters to be extracted must be specified.

VLOOKUP: When a value is searched for in a column, it can either return that value directly or a value from the adjacent columns using the same row number. A one-column or one-row range is searched for a close match using the Excel LOOKUP function, which then retrieves the comparable value from another one-column or one-row range.

IF: When a value is searched for in a column, it can either return that value directly or a value from the adjacent columns using the same row number. The IF function in Microsoft Excel returns one value if the condition is TRUE and a different value if it is FALSE.

The IF function, an Excel built-in function, falls under the category of a logical function. It can be used in Excel as a worksheet function (WS). The IF function can be used as a worksheet function to input a formula as part of a formula in a worksheet cell.

Syntax:

The syntax for the IF function in Microsoft Excel is:

IF (condition, [value_if_true], [value_if_false])

Parameters or Arguments

condition

The value that you want to test.

value_if_true

It is the value that is returned if condition evaluates to TRUE.

value_if_false

Optional. It is the value that is returned if condition evaluates to FALSE.

Nested IF: Nested IF functions, meaning one IF function inside of another, allows you to test multiple criteria and increases the number of possible outcomes.

NOW: The cell you insert is updated with the current date and time. It using the settings on your computer [...]

LARGE: When a list is sorted by value in descending order, the Excel LARGE function produces a numeric value based on its position inside the list. In other words, LARGE can return the “nth largest” value, which includes the first, second, third, etc. largest value.

SORT: The SORT function orders the elements of an array or range either in ascending or descending order.

MAX: The MAX function finds the highest number in a range.

SQRT: A number’s square root is a quantity that, when multiplied by itself, yields the original number. Excel’s SQRT function gives the square root of an integer.

Exponential: The EXP function in Excel, which is classified as a Math/Trig function and returns a numerical value equal to e raised to the power of a specified number, is an exponential function. The value of the exponential growth and decay mathematical constant e is roughly 2.71828.

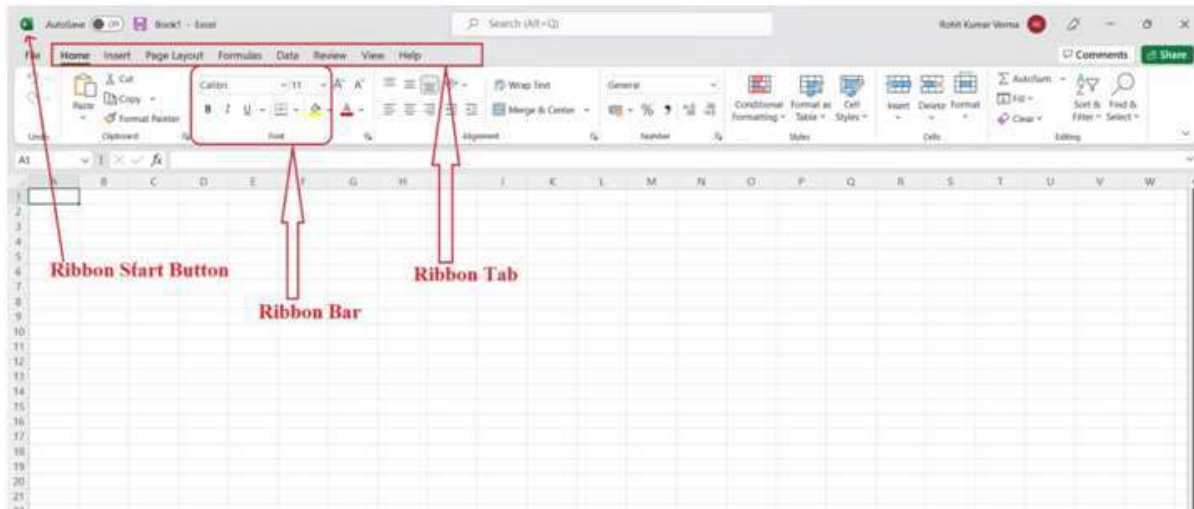
Square: The power function, denoted by the carat sign, in Excel allows you to square a number. Use the equation =N^2, where N might be a number or the value of the cell you want to square. A worksheet can make numerous uses of this formula.

1.5 Understanding the Ribbon

Shortcuts to Excel commands are available via the ribbon. A command is a task that the user carries out. Making a new document, printing a document, etc. are examples of commands. The Excel 2021 ribbon is displayed in the picture below.

Ribbon start button

It is used to access commands, such as those for generating new documents, saving completed work, printing, and accessing Excel customization options, among others.



Ribbon tabs

Similar instructions are grouped together using tabs. The home tab is used for basic operations like sorting and finding specific data inside the spreadsheet as well as formatting the data to make it more attractive.

Ribbon bar

The bars are used to group together similar commands. The instructions used to align data, for instance, are all gathered together in the Alignment ribbon bar.

1.6 Summary

In this chapter, we have studied:

- Microsoft Excel is an effective spreadsheet tool used to collect, process, and store numerical data.
- In Excel, the ribbon is utilised to access multiple commands.
- We've reviewed many procedures for opening Microsoft Excel.

1.7 Answers to self-check exercises

- Q.1 Explain various features of Microsoft Excel?
- Q.2 What are the various function of Microsoft Excel?
- Q.3 Explain the various ways to open Microsoft Excel?
- Q.4 How Ribbon tab is different from Ribbon bar?

1.8 Suggested Reading

- Ron Mansfield, The Compact Guide to Microsoft office; BPB Publication.
- Dienes, Sheila s., Microsoft office, Professional for windows 95.
- Joan Lambert, Curtis Frye, Microsoft Office Step by Step (Office 2021 and Microsoft 365),Microsoft Press Publication.
- Kenneth N. Berk,Data Analysis with Microsoft Excel.
- Wayne L. Winston,Microsoft Excel 2013: Data Analysis and Business Modeling.
- Danielle Stein, Using Excel for business analysis: a guide to financial modelling fundamentals.
- Olafusi M., Microsoft Excel and Business Data Analysis for The Busy Professional
- Rajiv Mattus: Learning Excel 97 for windows step by step BPB publication.
- Lonnie. E. Moseley & David M. Boodey: Mastering office 97.
- Micro Soft Office 97: Unleashed:Techmedia.
- John Walken Bach: Excel 97.

Chapter-2

MICROSOFT EXCEL WORKBOOK AND WORKSHEET

STRUCTURE

2.0 Learning Objectives

2.1 Introduction of Microsoft Excel Workbook

- 2.1.1 How to create a new workbook?
- 2.1.2 How to create a workbook from a template?
- 2.1.3 How to rename a workbook?

2.2 Introduction of Microsoft Excel Worksheet

- 2.2.1 How to select a Worksheet?
- 2.2.2 How to insert a Worksheet?
- 2.2.3 How to rename a Worksheet?
- 2.2.4 How to move a Worksheet?
- 2.2.5 How to Delete a Worksheet?
- 2.2.6 How to Copy a Worksheet?

2.3 Introduction to Row

- 2.3.1 How to insert a row in Excel?
- 2.3.2 How to delete a row in Excel?

2.4 Introduction to Column

- 2.4.1 How to insert column in Excel?
- 2.4.2 How to delete column in Excel?

2.5 Summary

2.6 Answer to Self-Check exercises

2.7 Suggested Readings

2.0 LEARNING OBJECTIVES

After going through this lesson, you will be able to:

- Understand the basic of Microsoft Excel Workbook
- Understand the basic of Microsoft Excel Worksheet
- Understand the basic of Microsoft Excel Row
- Understand the basic of Microsoft Excel Column

2.1 Introduction of Microsoft Excel Workbook

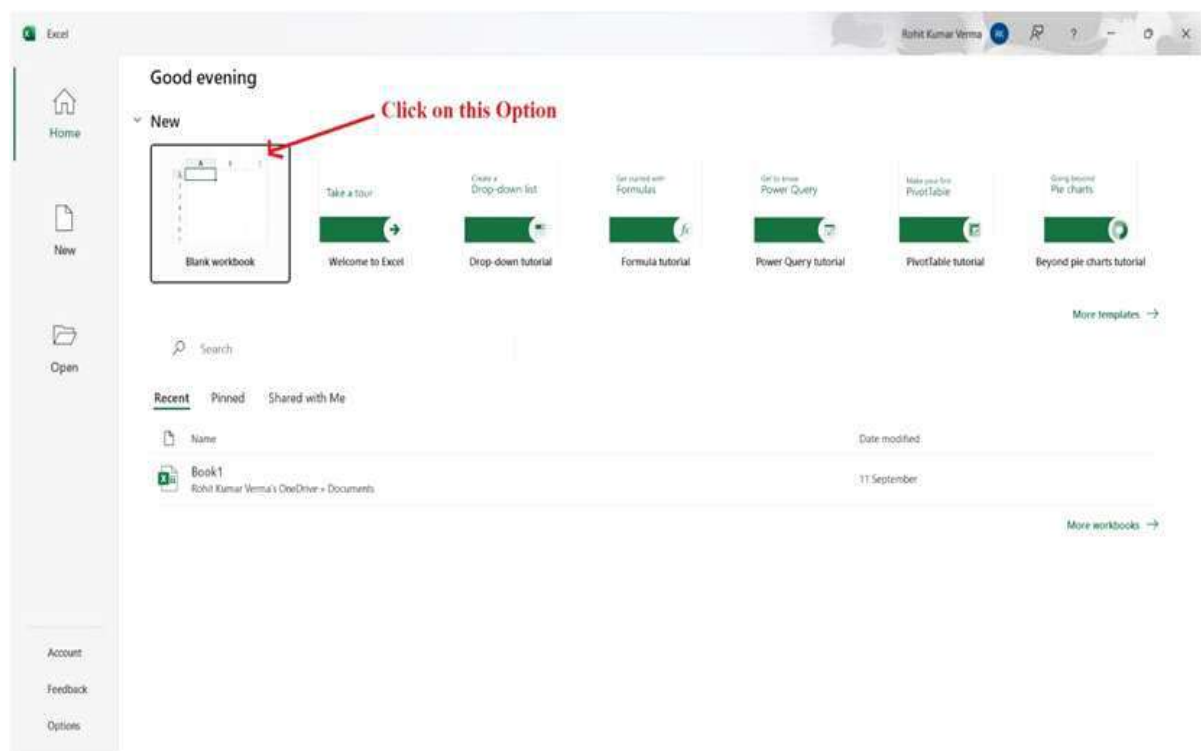
A set of worksheets called an Excel workbook has rows and columns of data storage. By default, a new Excel workbook is titled Book1 (see at the top of the Excel). Depending on the data included inside, you can give and save the workbook. A single Excel worksheet typically has 16,384 columns and 1,048,576 rows.

The worksheets in a workbook are numbered Sheet1, Sheet2, Sheet3,...SheetN. By selecting the + sign next to the worksheets, you can add these worksheets to your workbook. A single worksheet may be in use at once. It indicates that only one worksheet may be active at a time in an Excel spreadsheet. However, a workbook can contain multiple worksheets.

2.1.1 How to create a new workbook?

The procedures listed below should help you to create a new workbook on your computer:

- Open Excel.
- Press Ctrl+N or choose Blank workbook.



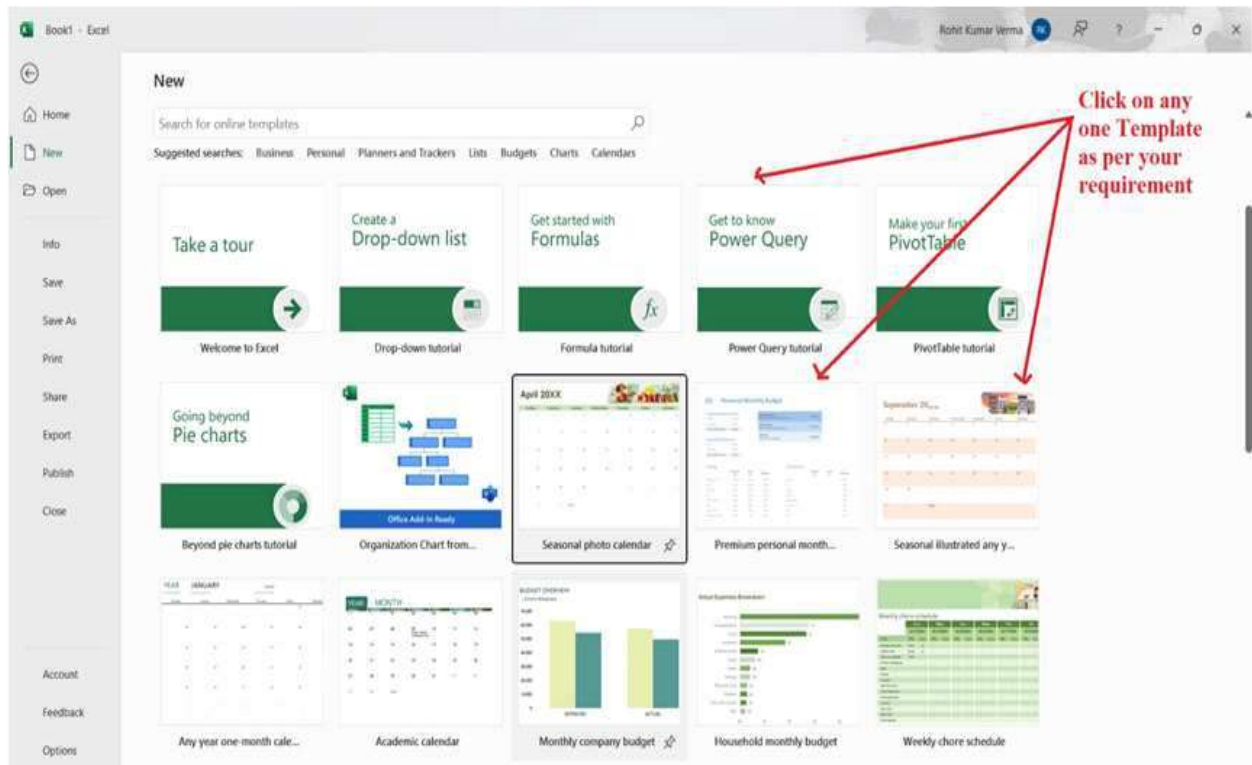
- Start typing

2.1.2 How to create a workbook from a Template?

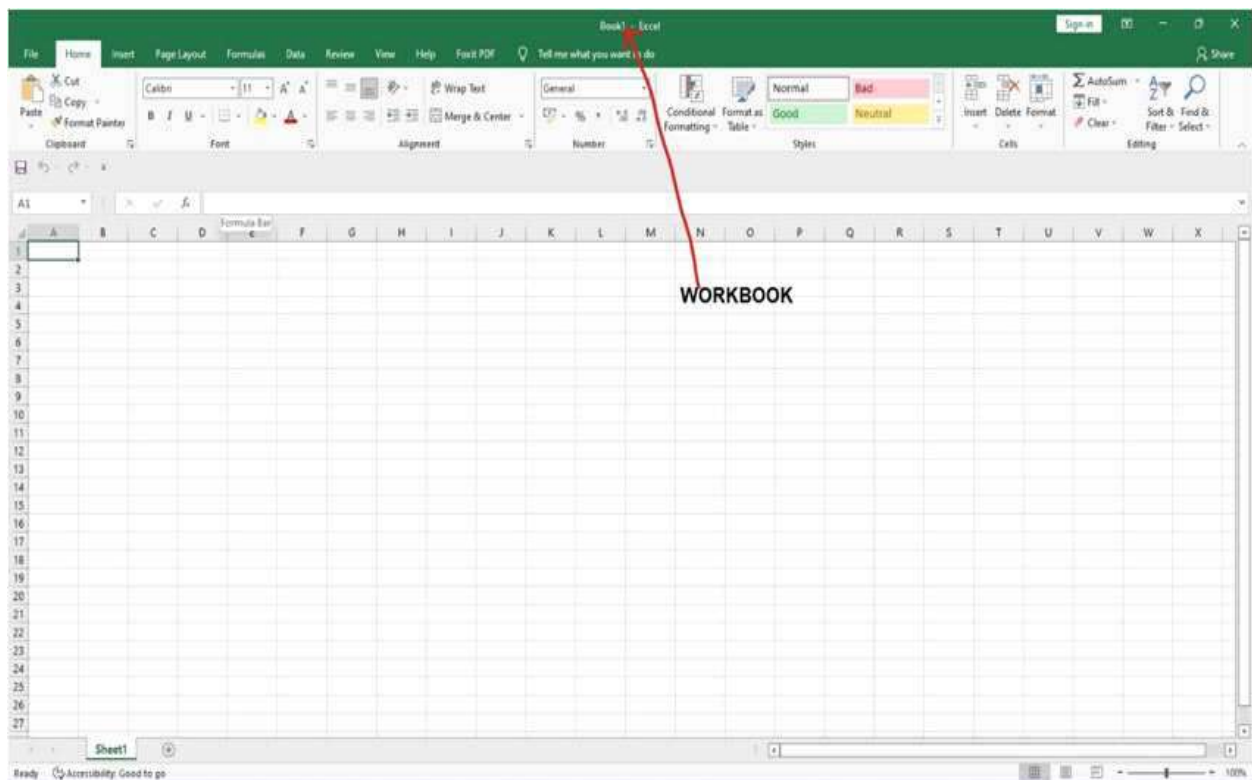
The procedures listed below should help you to create a new workbook from a template on your computer:

- Choosing File > New.

- Double-click a template.



- Click and start typing.

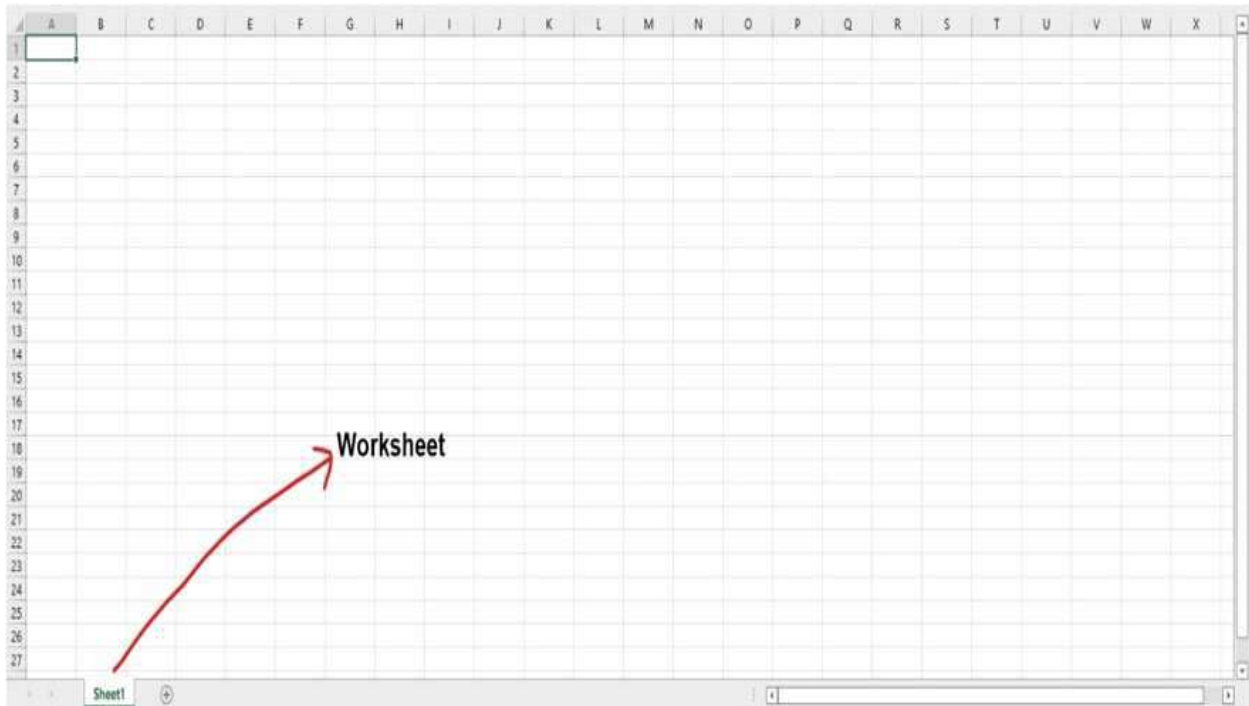


2.1.3 How to rename a workbook?

- To rename a workbook, locate it in Windows Explore, then you can press F2, or right-click and select Rename, then type the new name.
- If your workbook is currently open, you can save it with an alternative name by selecting File > Save As. This will make a duplicate of the current worksheet.

2.2 Introduction of Microsoft Excel Worksheet

A worksheet is a single page with a number of cells where the user can input, modify, and store data. Spreadsheets are also referred to as worksheets. The worksheet has rows, columns, and cells in it.



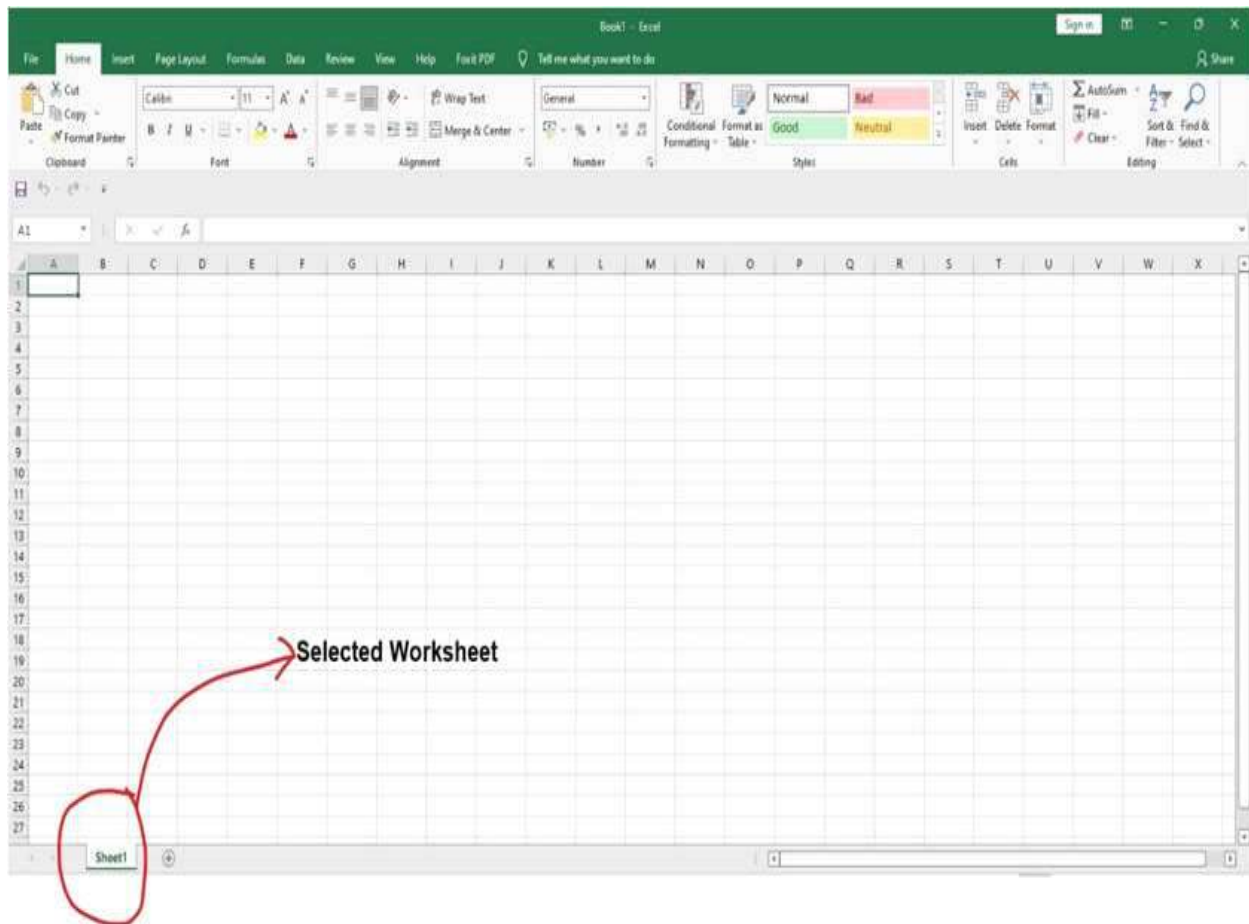
Each individual data cell in an Excel worksheet is defined and arranged with the aid of the vertical column and horizontal row that are combined to produce a cell reference, unlike B1 or A25.

The current versions of Excel Worksheet include the following specifications:

- Each worksheet has 1,048,576 rows.
- There are 16,384 rows per worksheet.
- There are 17,179,869,184 total cells on each worksheet.

2.2.1 How to select a worksheet?

By default, Excel opens the Sheet 1 worksheet whenever you start an Excel workbook. The worksheet's name, "Sheet1," is shown on its sheet tab at the bottom of the windowpane, as seen in the image below.



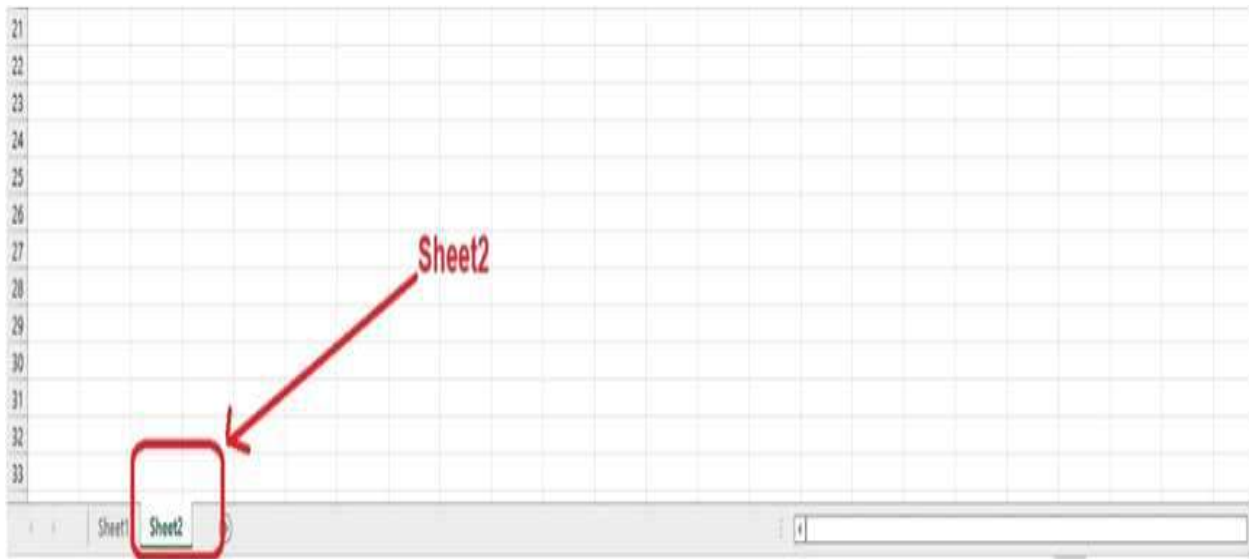
2.2.2 How to insert a worksheet?

Excel gives the user the option to insert as many spreadsheets as they need. Click on the addition (+) sign at the bottom of the window to add a new worksheet to Excel. The default name of the new worksheet is “SheetN,” where N is one of the integers 1, 2, 3, 4, 5, 6, 7,..., n.

In the example below, Sheet2 is the name of a new worksheet that has been added.

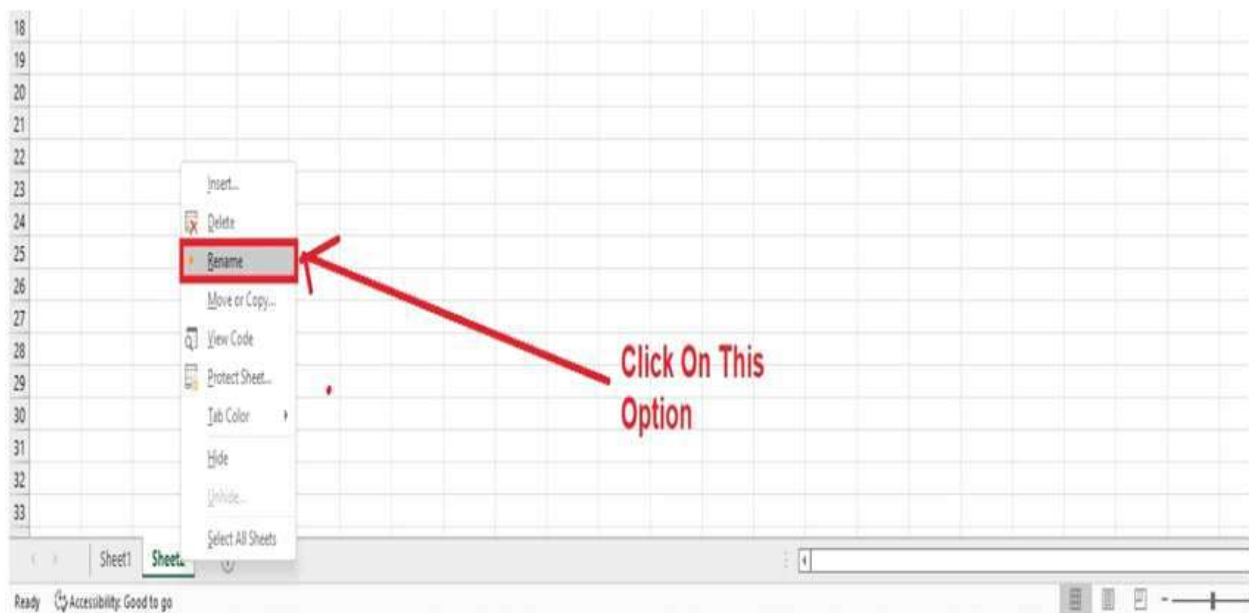


2.2.3 How to rename a worksheet?

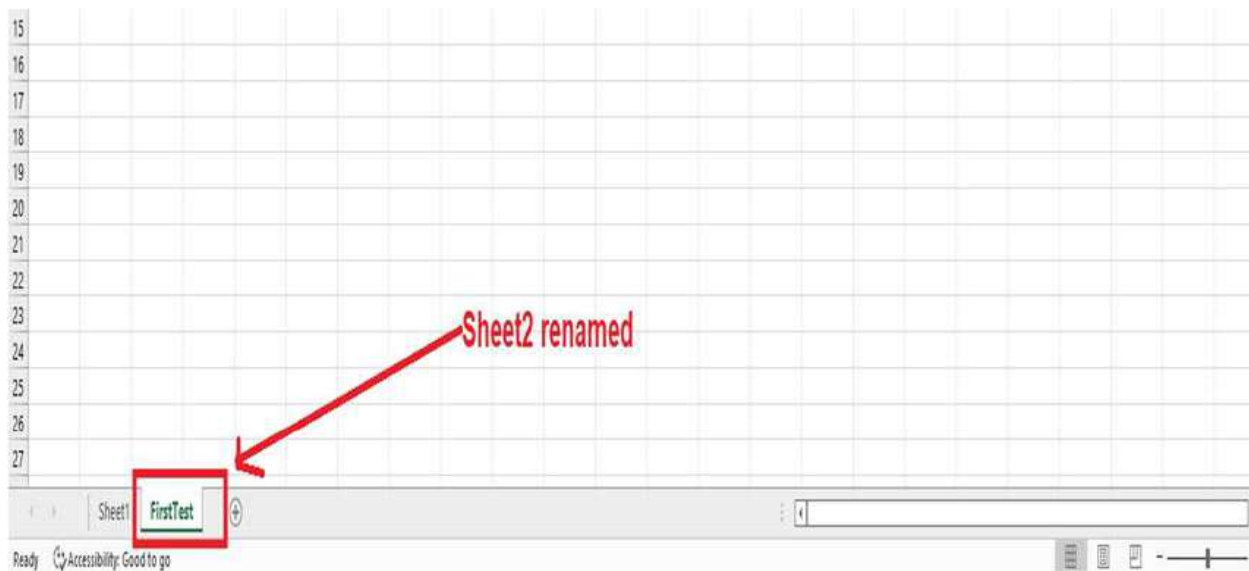


The steps listed below should be followed in order to rename your worksheet with a unique and relevant name.

- Click with your right mouse button on the sheet tab whose name you want to modify.
- You'll see the toolbar window.
- Select Rename from the menu.



- Type the name you like. For instance, we have changed the name of “Sheet1” to “FirstTest” in the screenshot below.



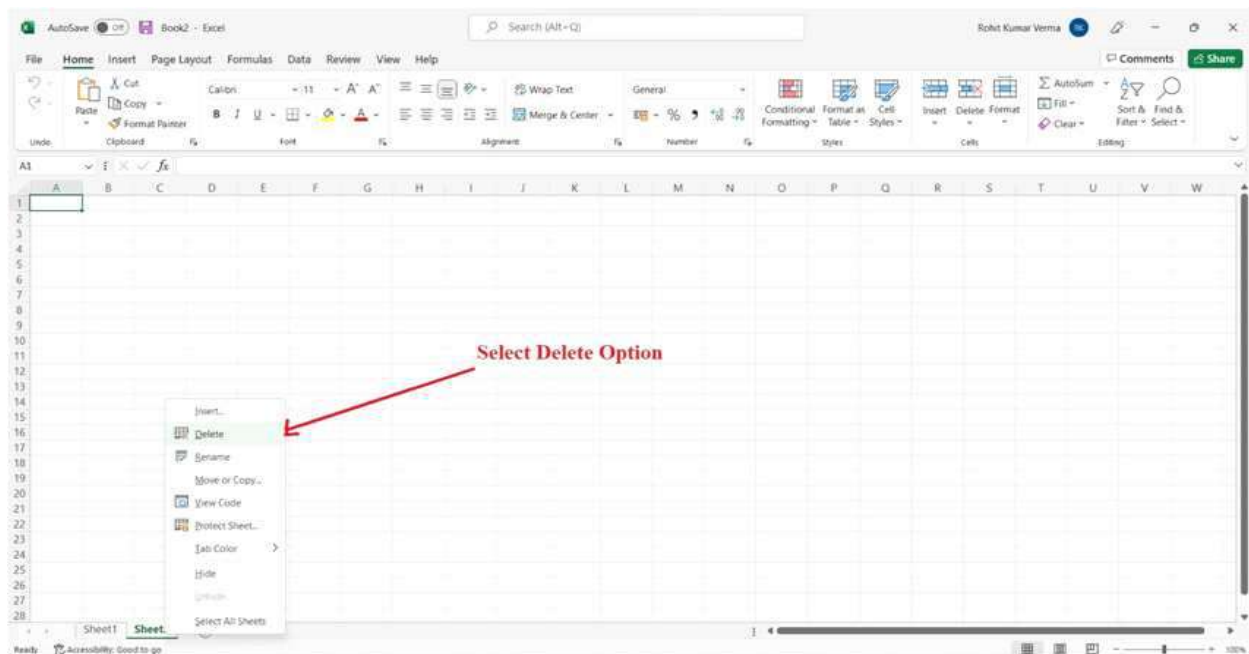
2.2.4 How to move a worksheet?

Excel gives the user complete control over where and how to move their worksheets. Drag the worksheet sheet tab to the desired spot by clicking on it.

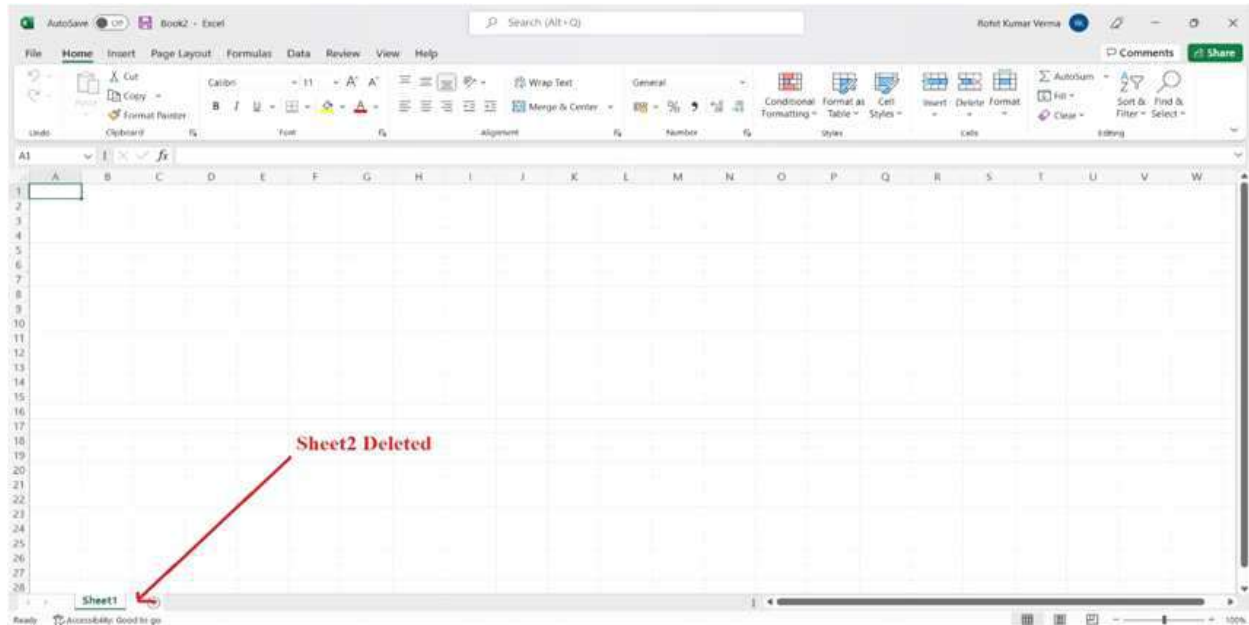
2.2.5 How to delete a worksheet?

The procedures listed below should help you to delete a worksheet:

- Right-click on a sheet tab at the bottom of the page.
- There will be a dialogue box. Select “Delete”



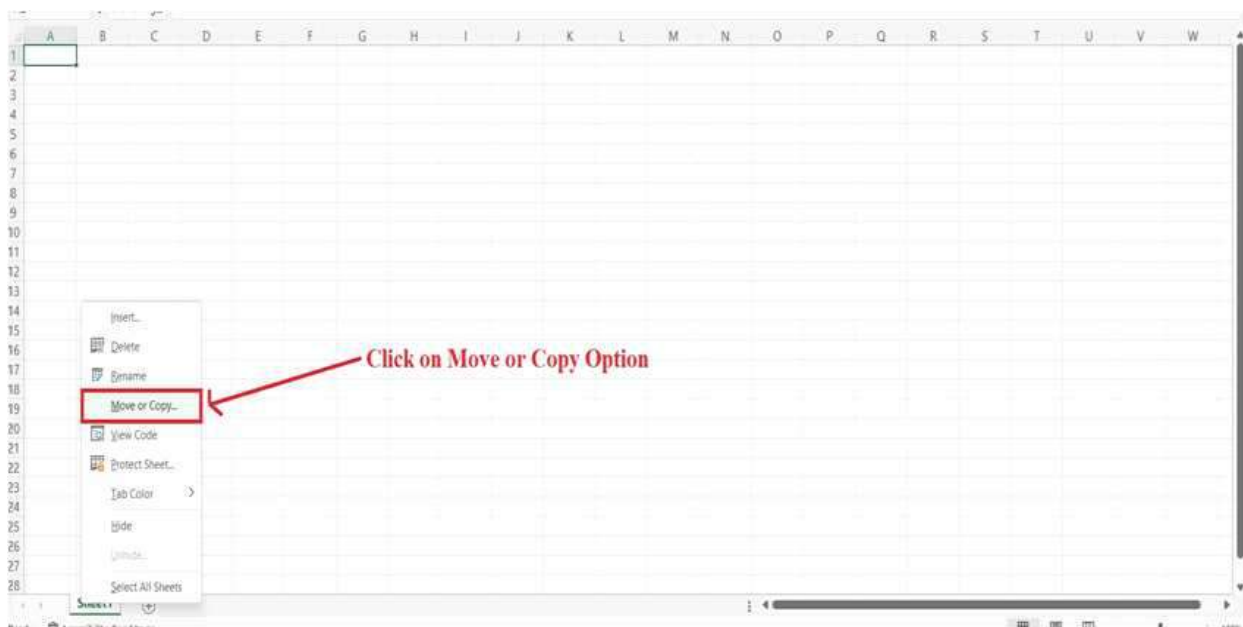
- The sheet will be removed right away as shown below.



2.2.6 How to copy a worksheet?

The procedures listed below should help you to copy a worksheet:

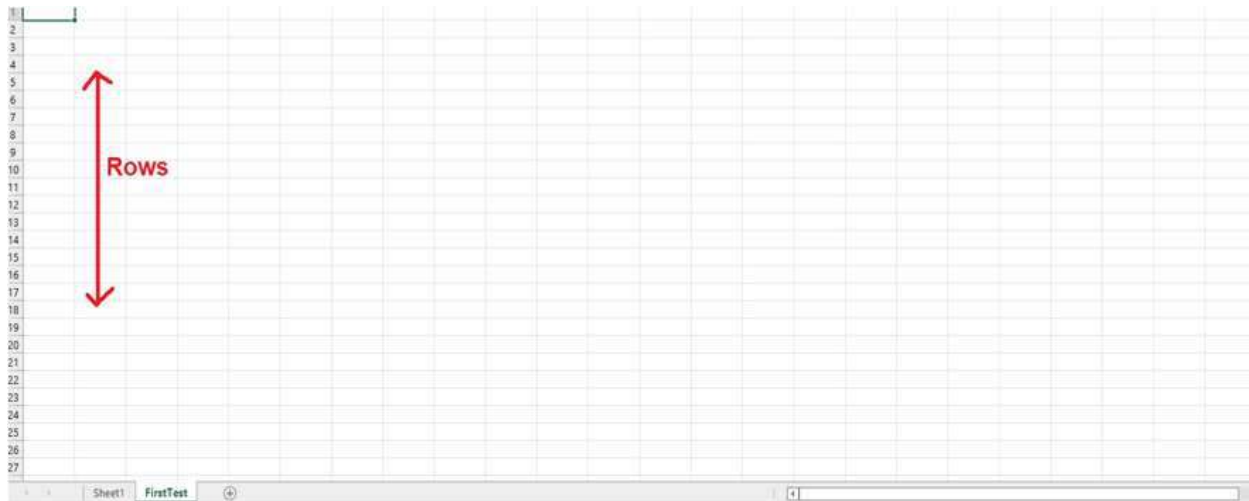
- To copy a sheet tab, use the right-click menu.
- You'll see the toolbar window. Select "Move or Copy" from the menu.



- It will show the "Move or Copy" dialogue box. Make sure to check the "create a copy" box when choosing the "move to end" option in the Before Sheet section.
- Select OK. Your information will be transferred to the new worksheet.

2.3 Introduction to Row

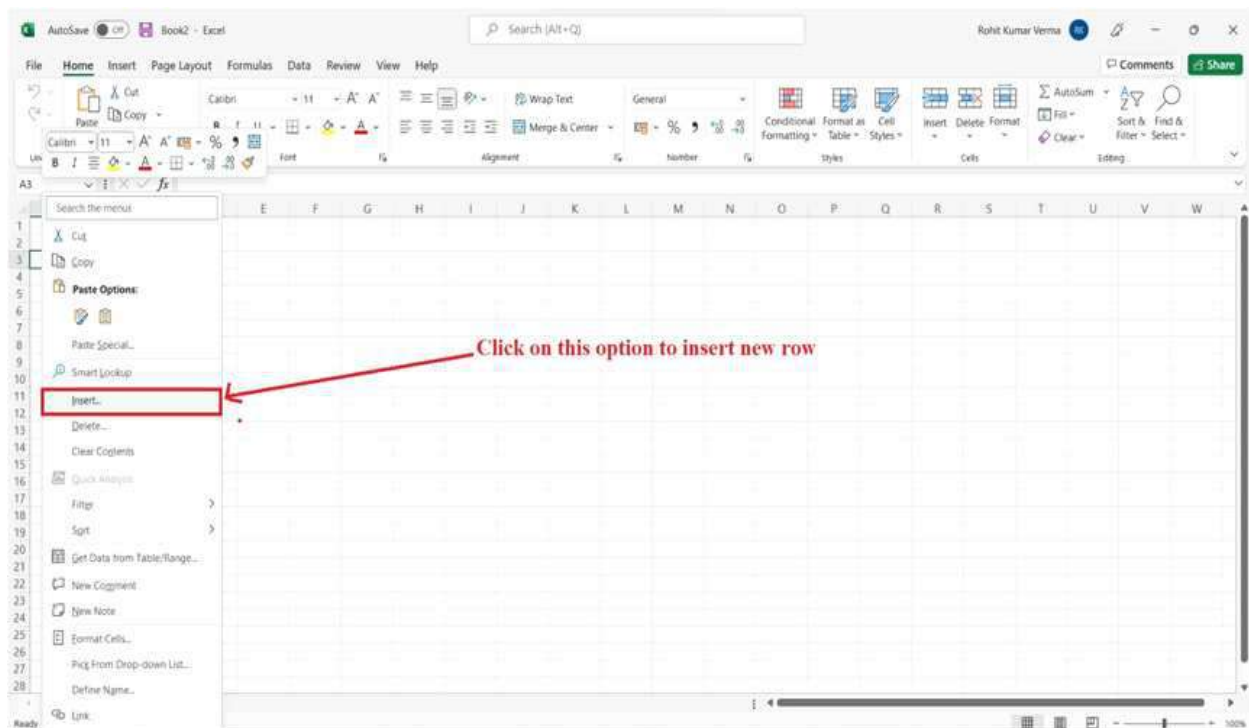
The worksheet has horizontal rows that go from 1 to 1048576. The number on the left side of the row, which indicates where the row came from, is used to identify a row.



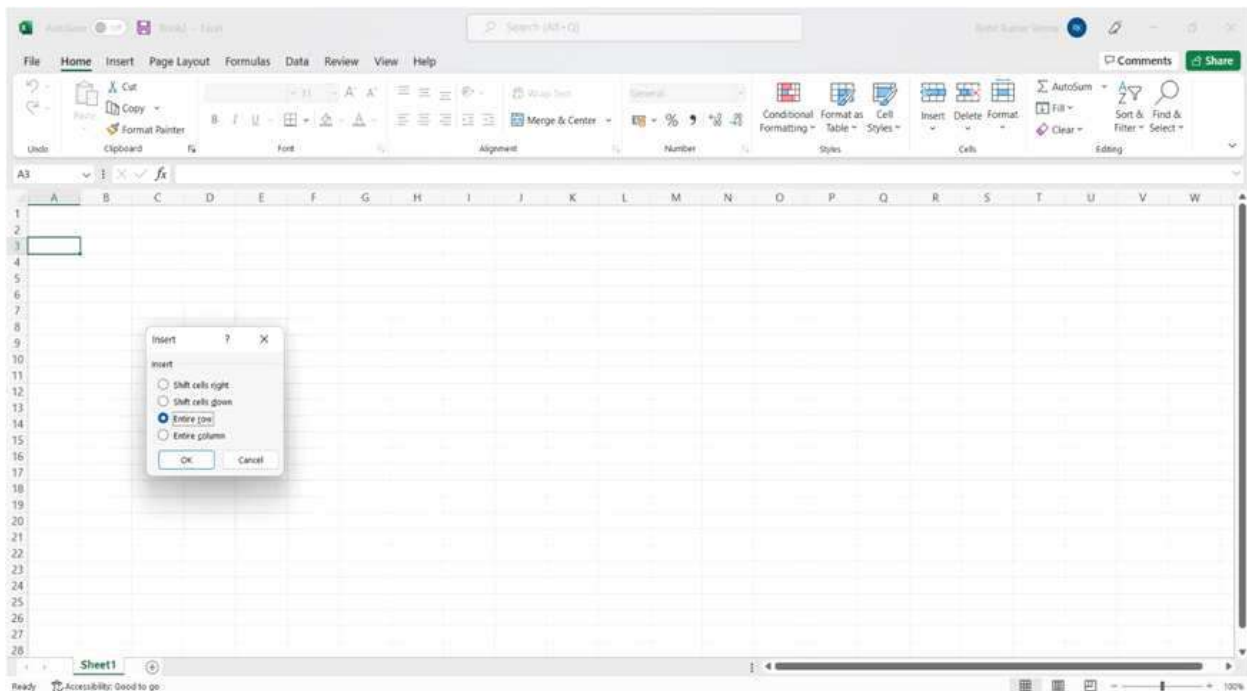
2.3.1 How to insert a row in Excel?

The procedures listed below should help you to insert a row in Excel:

- Choose the cell within the row where you want to introduce a new row by right-clicking it.
- The dialogue box will appear. Pick Insert.



- You'll see the insert dialogue box. Select "Entire row."

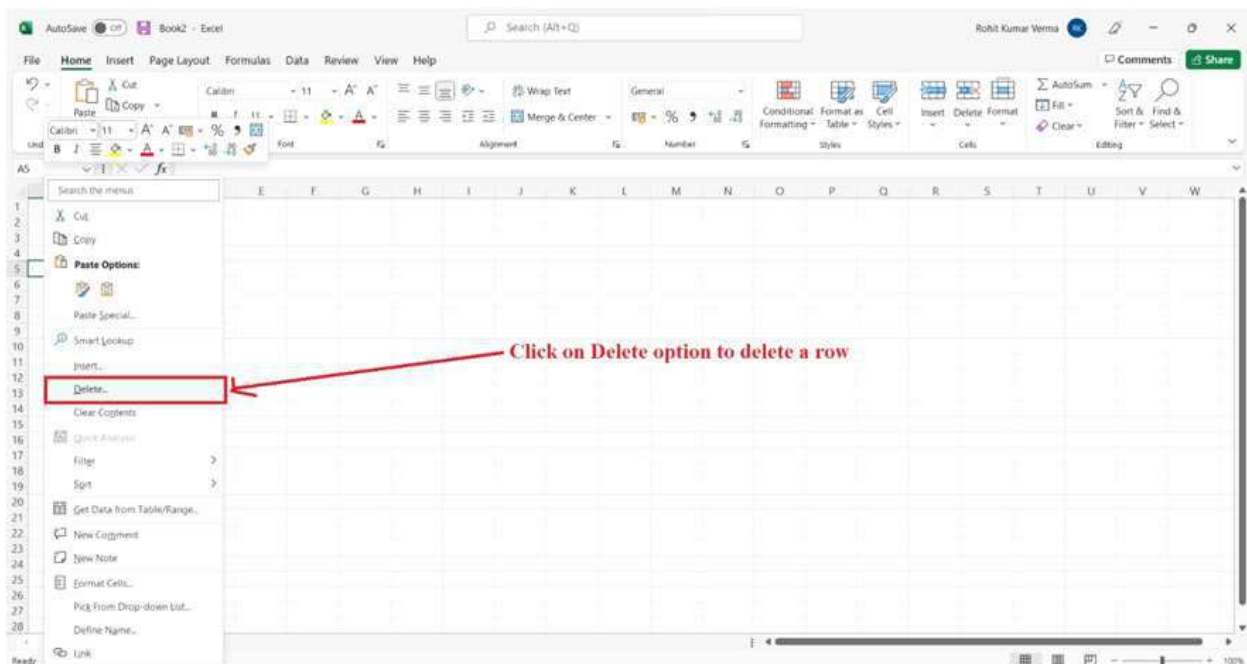


- At the top of the cell you've chosen, it will insert a new row.

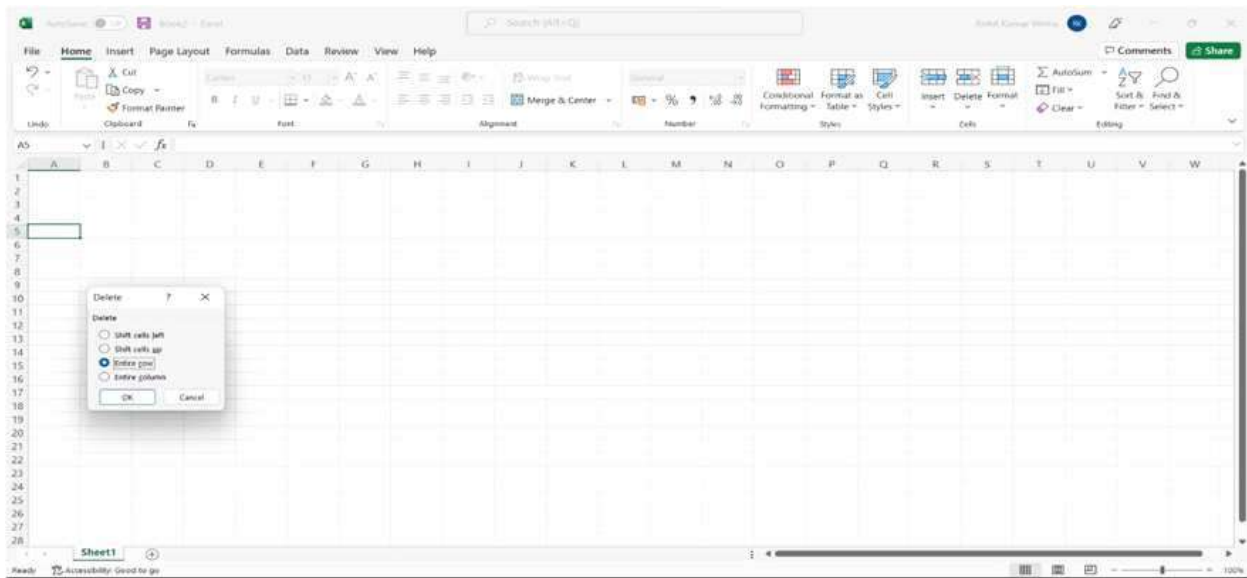
2.3.2 How to delete a row in Excel?

The procedures listed below should help you to delete a row in Excel:

- The cell you want to delete must be chosen from the row. Right-click the cell.
- It will show the dialogue box. Select Delete.



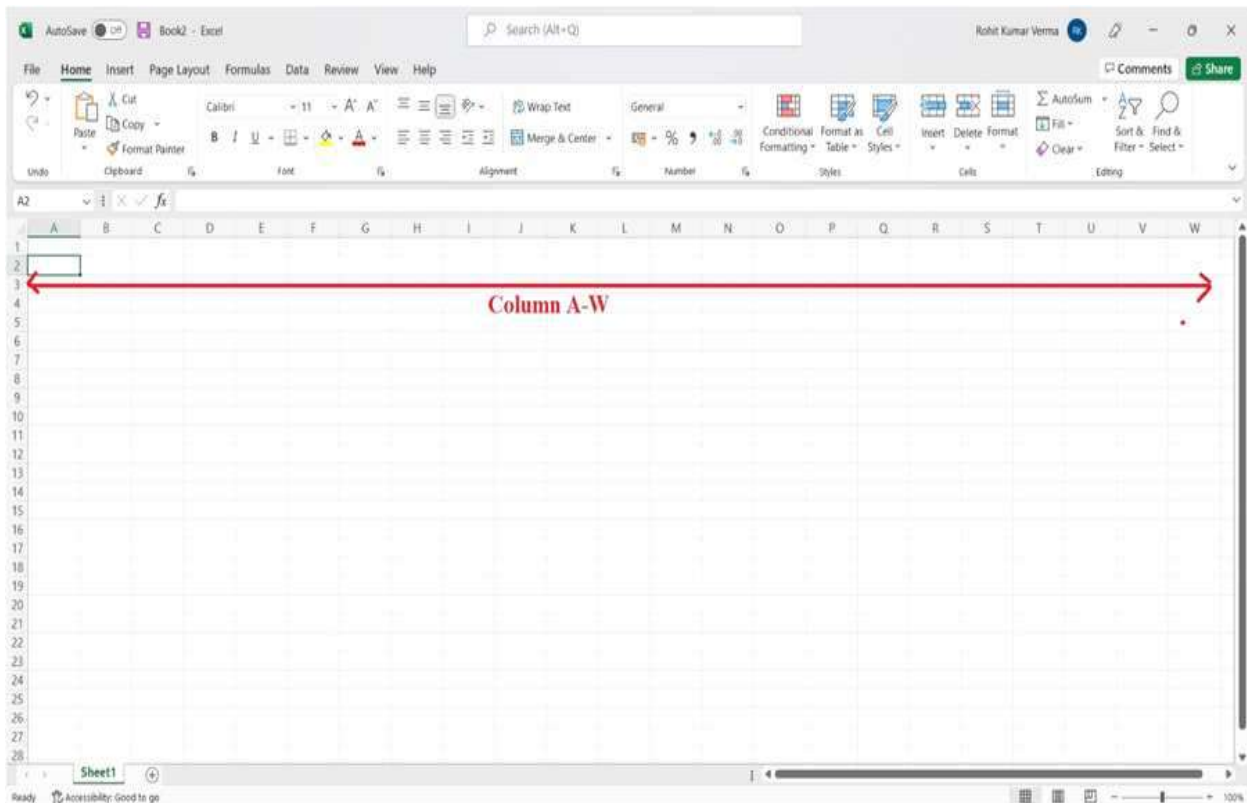
- You'll see the Delete dialogue box. Select “entire row” from the menu.



- It will immediately erase the entire row.

2.4 Introduction to Column

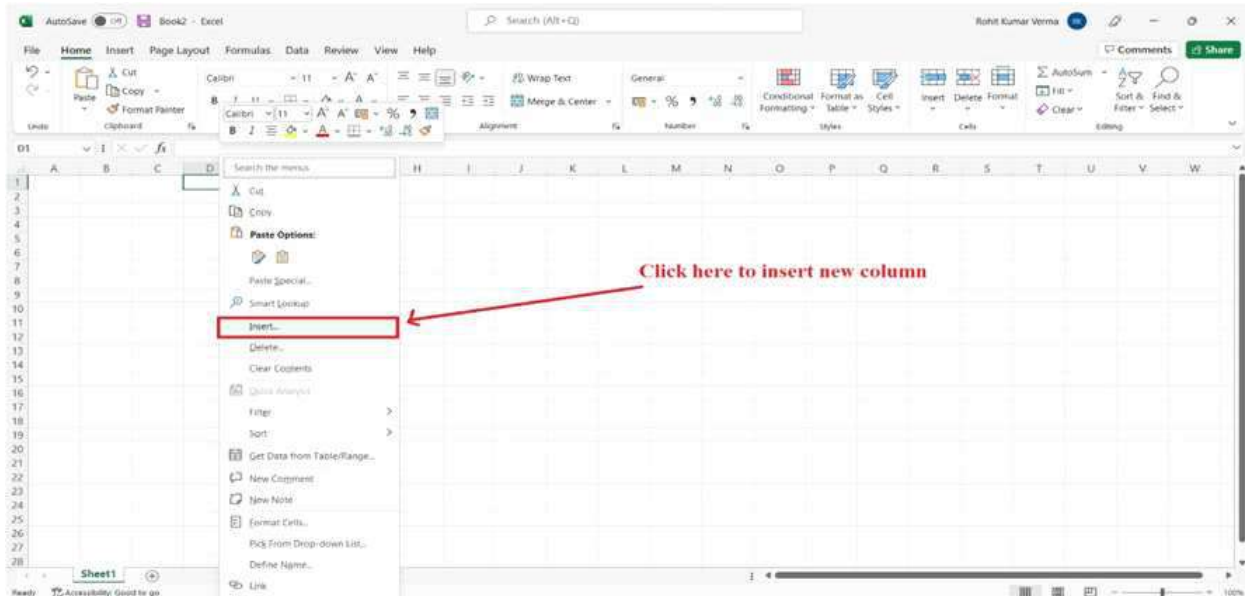
The worksheet is divided vertically into columns from A to XFD (in total 16384 columns). A column's header, which shows where the column comes from, at the top of the column serves as its identifier.



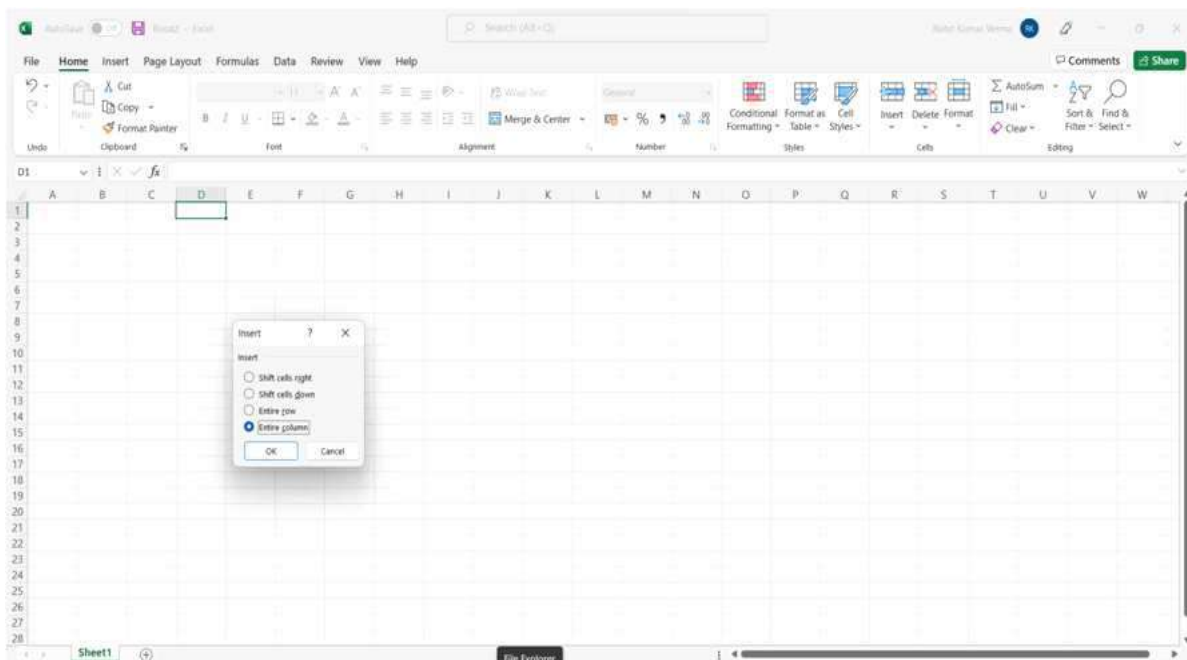
2.4.1 How to insert column in Excel?

The procedures listed below should help you to insert column in Excel:

- Choose the cell within the column you wish to insert a new column in by performing a right-click.
- It will show the dialogue box. Select Insert.



- You'll see the insert dialogue box. Select "Entire column."

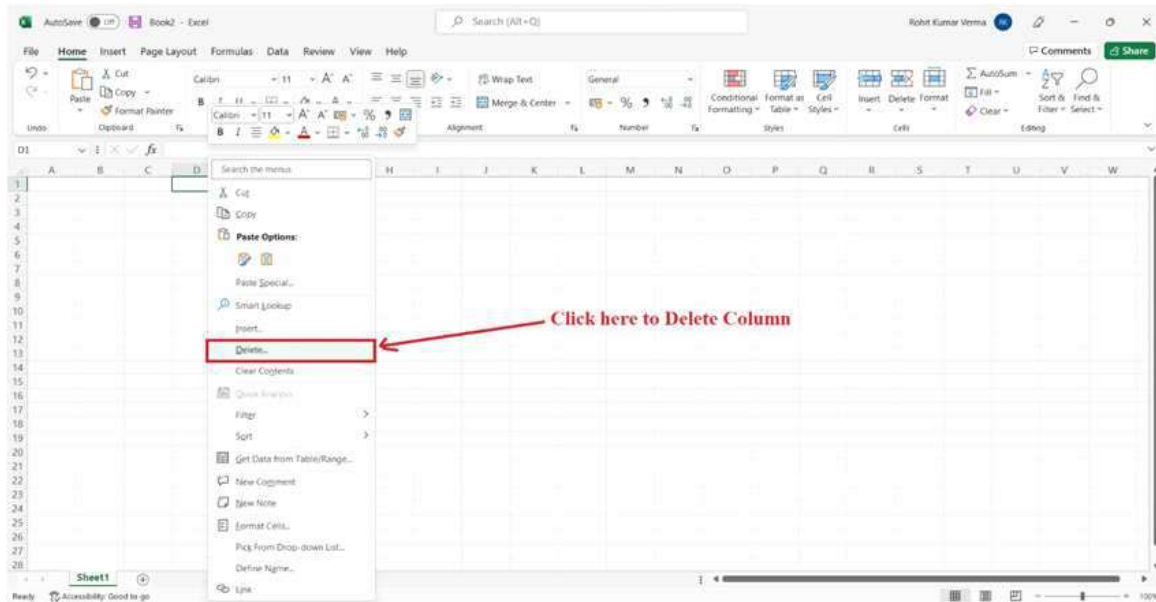


- It will add a new column to the left of the cell you've chosen.

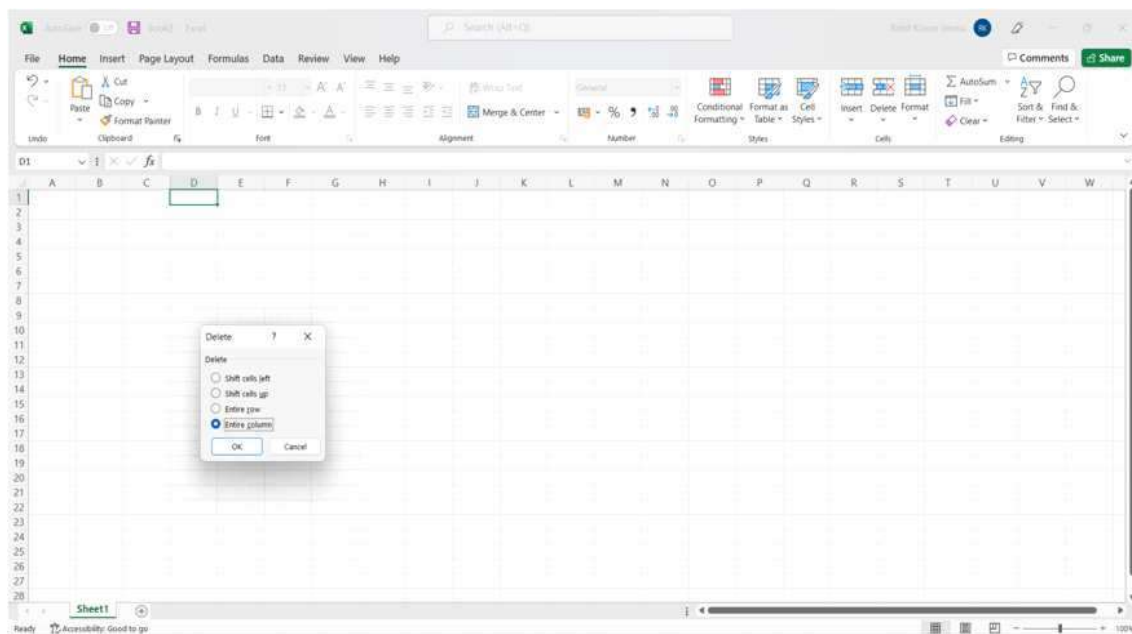
2.4.1 How to delete column in Excel?

The procedures listed below should help you to delete a column in Excel:

- The cell you want to delete must be chosen from the column. Right-click the cell.
- It will show the dialogue box. Select Delete.



- You'll see the delete dialogue box. Select "Entire column."



- It will immediately erase the entire column.

2.5 Summary

In this chapter, we have studied:

- A set of worksheets called an Excel workbook has rows and columns of data storage.
- A worksheet is a single page with a number of cells where the user can input, modify, and store data.
- The worksheet has horizontal rows that go from 1 to 1048576.
- The worksheet is divided vertically into columns from A to XFD (in total 16384 columns).

2.6 Answers to self-check exercises

Q.1 What is the difference between row and column?

Q.2 How workbook is different from worksheet?

Q.3 What are the various ways to create a new workbook?

Q.4 How we can rename a workbook?

Q.5 How we can move a worksheet?

Q.6 Mention various steps to insert a row and a column in Excel?

Q.7 How we can copy a worksheet?

2.7 Suggested Reading

- Ron Mansfield, The Compact Guide to Microsoft office; BPB Publication.
- Dienes, Sheila s., Microsoft office, Professional for windows 95.
- Joan Lambert, Curtis Frye, Microsoft Office Step by Step (Office 2021 and Microsoft 365), Microsoft Press Publication.
- Kenneth N. Berk, Data Analysis with Microsoft Excel.
- Wayne L. Winston, Microsoft Excel 2013: Data Analysis and Business Modeling.
- Danielle Stein, Using Excel for business analysis: a guide to financial modelling fundamentals.
- Olafusi M., Microsoft Excel and Business Data Analysis for The Busy Professional.
- Rajiv Mattus: Learning Excel 97 for windows step by step BPB publication.
- Lonnie. E. Moseley & David M. Boodey: Mastering office 97.
- Micro Soft Office 97: Unleashed: Techmedia.
- John Walken Bach: Excel 97.

Chapter-3

MICROSOFT EXCEL WORKSHEETCELL

STRUCTURE

- 3.0 Learning Objectives
- 3.1 Introduction of Microsoft Excel Worksheet Cell
 - 3.1.1 How to select a Cell?
 - 3.1.2 How to select Cell Range?
 - 3.1.3 How to insert content to a cell?
 - 3.1.4 How to delete cell content?
 - 3.1.5 How to delete Cell?
 - 3.1.6 How to drag and drop Cell?
- 3.2 History of Microsoft Excel
- 3.3 Advantages of Microsoft Excel for Statistical Analysis
- 3.4 Disadvantages of Microsoft Excel for Statistical Analysis
- 3.5 Summary
- 3.6 Answer to Self-Check exercises
- 3.7 Suggested Readings

3.0 LEARNING OBJECTIVES

After going through this lesson, you will be able to:

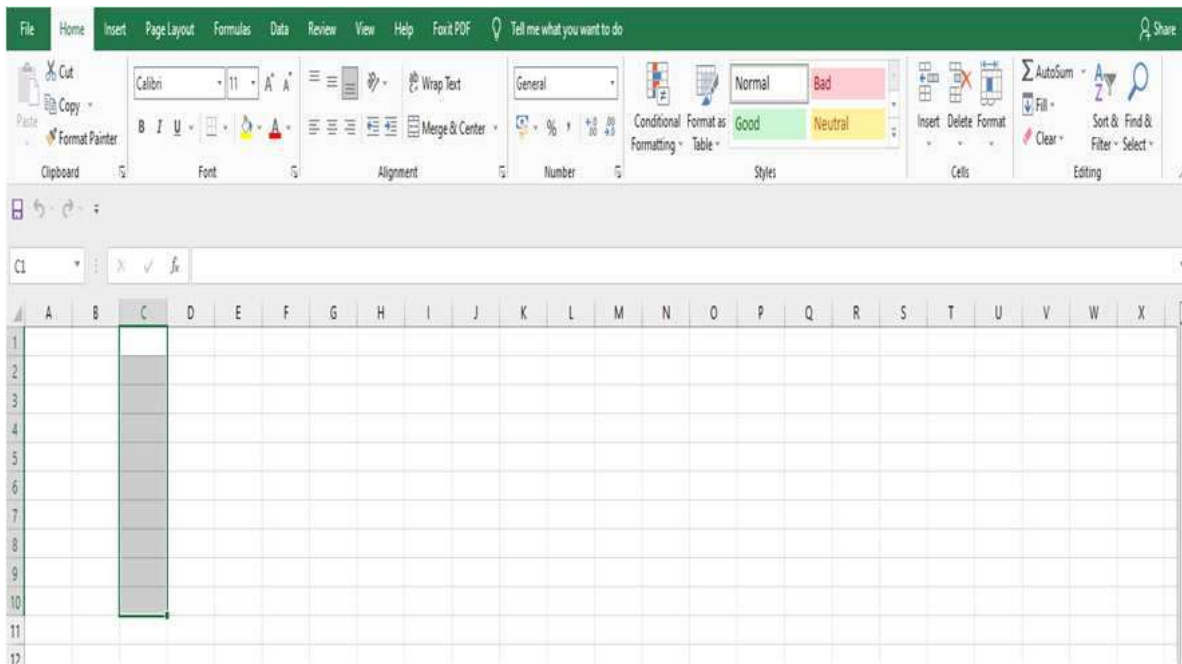
- Understand the basic of Microsoft Excel Worksheet Cell
- Understand the advantages of Microsoft Excel for Statistical Analysis
- Understand the disadvantage of Microsoft Excel for Statistical Analysis

3.1 Introduction of Microsoft Excel Worksheet Cell

The worksheet's cells are the little, rectangular boxes where we enter data. Row and column intersections form cells. The column header and row number serve as identifiers. Using a set of coordinates or positions, such as A1 (where A stands for a column and 1 for a row), C2, or H15, each cell in Excel is uniquely recognized.

Unlike changing the font style, font size, background colour, text alignment, format painter, wrap text, and conditional formatting, Excel cells can be used for a variety of activities.

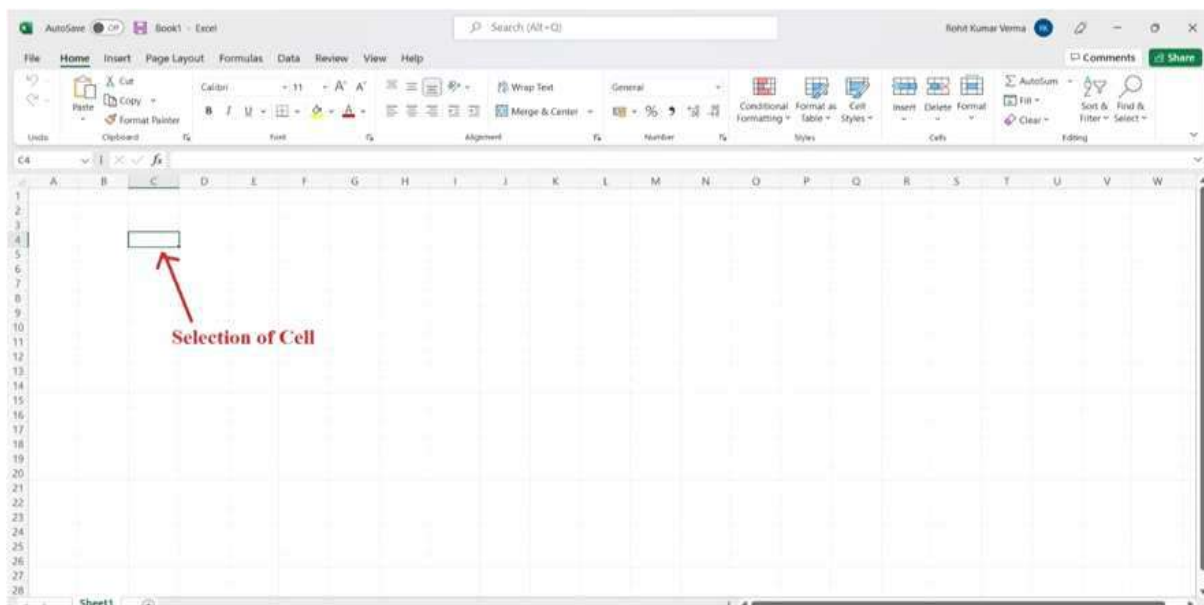
Multiple cells can be selected simultaneously in Excel. Cell range refers to a collection of chosen cells. The user refers to a cell range rather than a single cell address by using the colon-separated cell addresses of the first and final cells they have chosen. As an illustration, C1:C10 would be used to refer to a cell range that includes C2, C3, C4, C5, C6, C7, C8, C9, and C10.



3.1.1 How to select a Cell?

The steps to pick a cell in Excel are listed below:

- Select any cell by clicking it with your pointer.
- Your cell is now highlighted, has a rectangular border around it, and the column and row headings are now highlighted.

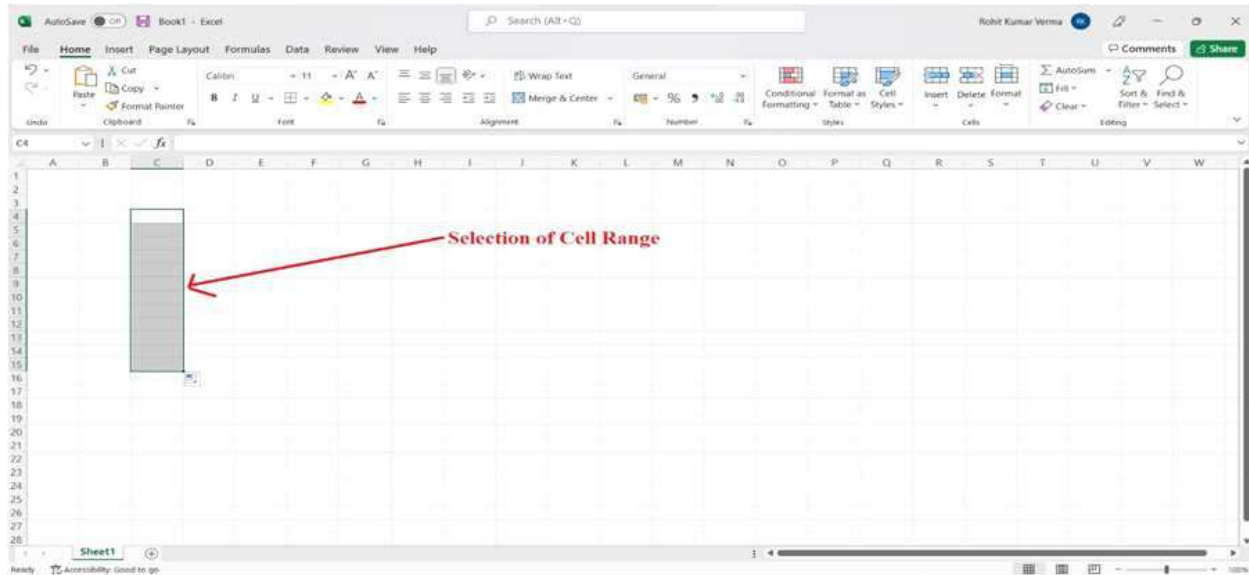


- Until the user moves the cursor to another cell, the selected cell will remain such.

3.1.2 How to select Cell Range?

The steps to select a cell range in Excel are listed below:

- Selecting a cell begins with clicking on the first cell. When you wish to finish the cell range, hold your cursor down and drag it to the final adjacent cells.

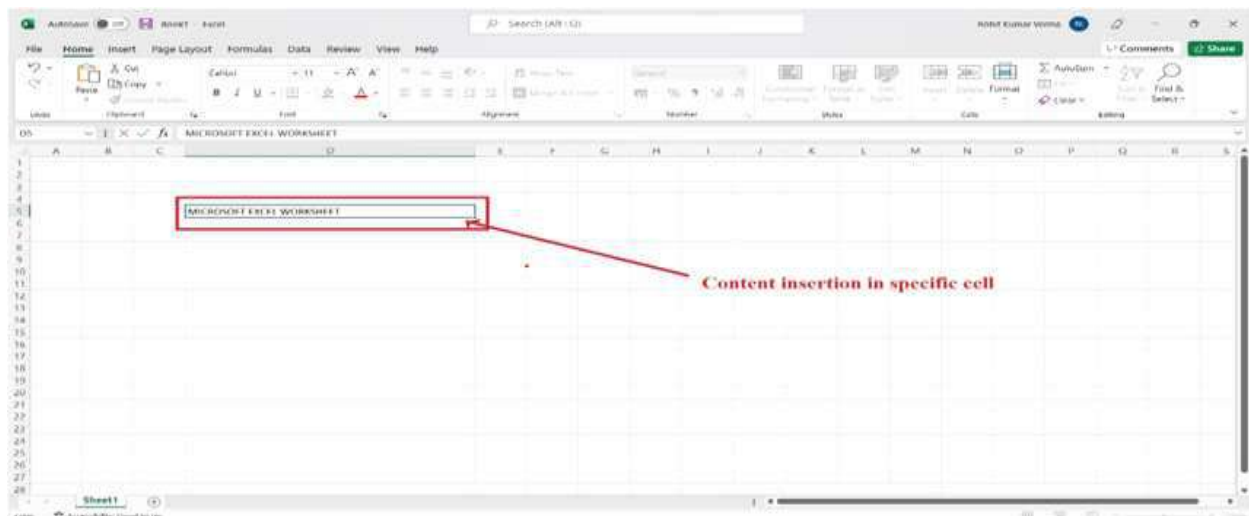


- When you're finished, click and drag the cursor to pick the desired cell range.
- Until the user moves the cursor to another cell, the selected cell will stay selected.

3.1.3 How to insert content to a cell?

The steps to insert content to a cell in Excel are listed below:

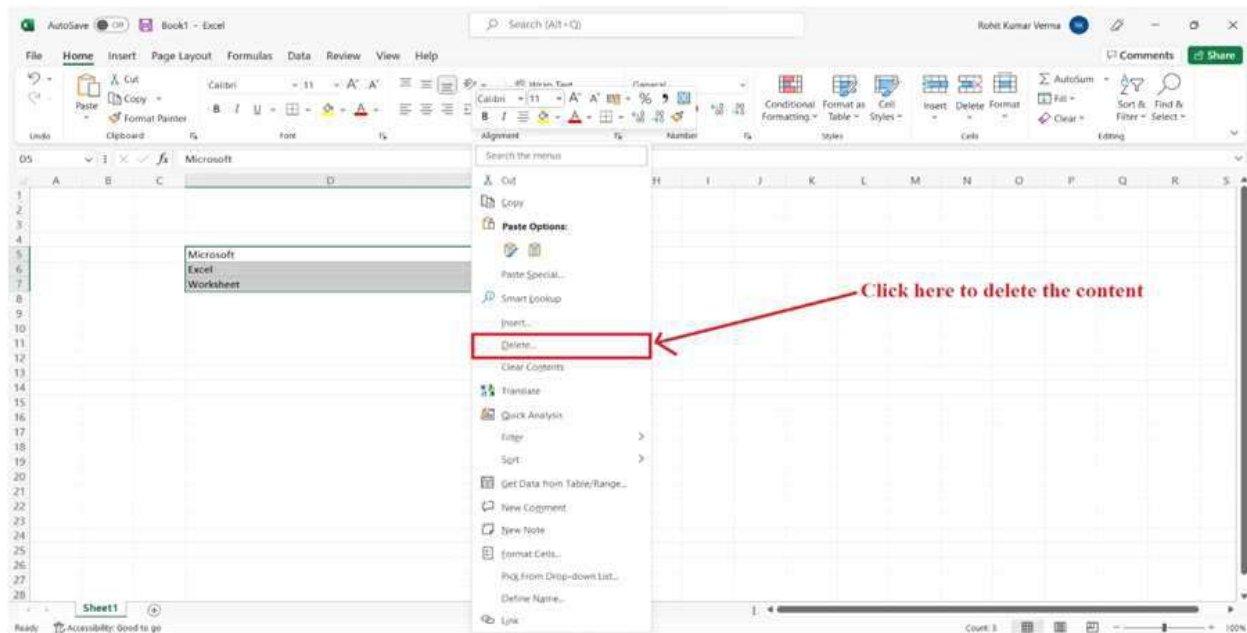
- The cell where you want to insert the content should be clicked and selected.
- Enter the required information into the highlighted cell, then press Enter or shift your pointer to a different cell. Both the formula bar and the cell's content will display the content.



3.1.4 How to delete cell content?

The steps to delete cell content in Excel are listed below:

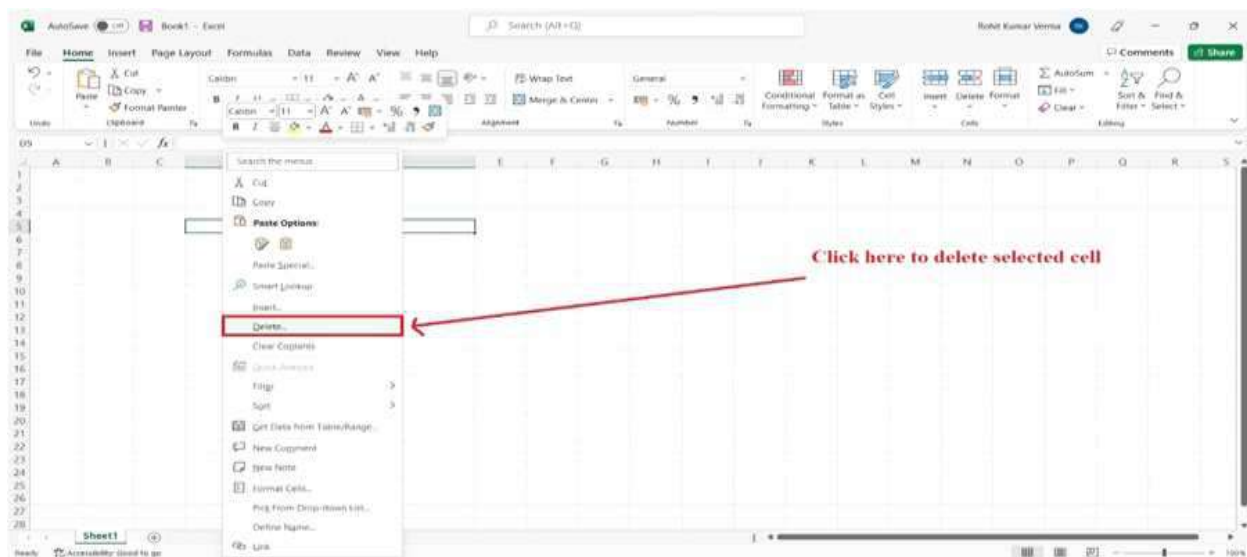
- Select the cell by clicking.
- Press the Delete or Backspace key on your keyboard to remove the cell's content.
- Select the cell range, then press the Erase key to delete several cells.



3.1.5 How to delete cell?

The steps to delete cell in Excel are listed below:

- Select the cell by clicking it (s).
- Simply right-click the cell. It will show the dialogue box. Select Delete.



- The Delete dialogue box shown below will appear. Select “entire row” from the menu.
- The cells underneath the deleted cell will immediately move upward.

3.1.6 How to drag and drop cell?

The steps to drag and drop cell in Excel are listed below:

- Select the cell by clicking it (s).
- When your cursor turns into a black cross with four arrows, move it over the border of the selected cell or cells. Excel Worksheet, Rows, Columns, and Cells
- To paste the content, move the mouse cursor to the desired cell. The mouse pointer will now appear as a dot-filled rectangle.
- The content will be copied from the source cell and pasted in the chosen location once the cursor is released.

3.2 History of Microsoft Excel

Excel has seen numerous changes in terms of function diversity, functionality, and capacity since its creation. In 1985, Microsoft released the initial version of Excel, which was exclusively compatible with Apple Macintosh computers. The description of Microsoft Excel in all versions is provided below:

Sr. No.	Name	Released	Description
1	Version 1	1985	Excel's initial release was limited to Macs only. Many Excel lovers are still unaware of this, which may sound unusual. In 1982, Microsoft actually created a spreadsheet tool called Multiplan, but it was a failure. Up to 2016, Excel versions for various operating systems had several names.
2	Excel 2	1987	To match the Mac version, the first Microsoft Excel release for Windows was labeled “2”. It was essentially a port of the Mac “Excel 2,” and it incorporated a run-time version of Windows.
3	Excel 3	1990	Toolbars, drawing capabilities, outlining, support for add-ins, 3D charts, and numerous other additional features were included in the subsequent edition.

4	Excel 4	1992	The first “popular” version of Excel was version 4. Numerous usability features were implemented, such as AutoFill, which was initially made available in this edition.
5	Excel 5	1993	A significant improvement came with Excel version 5. Workbooks with multiple worksheets, support for VBA, and macros were all included. Excel’s vulnerability to macro virus attacks increased as a result of these additional features, and this problem persisted all the way up to the 2007 update.
6	Excel 95	1995	It was the first significant 32-bit version of Excel and was referred to as Excel 95. A 32-bit version of Excel 5 was also available, but due to distribution issues, it was not extensively used. Excel 95 and Excel 5 are quite comparable in terms of features. You might also be perplexed as to why Excel 6 doesn’t exist. Version numbering was altered, and starting with Excel 7, the same version number was implemented throughout all Microsoft Office programmes.
7	Excel 97	1997	This version included many new features, including UserForms, data validation, and a new interface for VBA developers. Do you recall the annoying Office Assistant Clippy? He was part of this version.
8	Excel 2000	1999	The usage of HTML as a native file format, a “self-repair” function, an improved clipboard, pivot charts, and modeless user forms are some of the new features.

9	Excel 2002	2001	This is the first version of Excel included with Office XP. The extensive list of new features wasn't really useful to the typical user. The new option that enables you to recover your work should Excel crash was one of the most important innovations. Additionally, this edition includes a useful feature called product activation technology, sometimes referred to as copy protection, which restricts the use of the software to a single machine. The implications of this should be taken into account before determining whether to update.
10	Microsoft Office Excel 2003	2003	This version added enhanced XML compatibility, a new "list range" feature, improvements to Smart Tags, and rectified statistical functions. The majority of people won't think the upgrade connected to data is desirable.
11	Microsoft Office Excel 2007	2007	Excel had some significant changes with this Windows release. These included the introduction of the Ribbon interface and the switch from the original .xls file format to the more widely used .xlsx and .xlsm. This modification improved Excel's security (remember the issues with macro viruses in earlier versions?) and made greater row data storage possible (over 1 million). Also substantially enhanced were the charting features. The loss of Clippy from Microsoft Excel was a tragic aspect of the upgrade, which caused some people joy and others pain.

12	Microsoft Office Excel 2010		2010 Sparkline graphics, pivot table slicers, an updated Solver, and a 64-bit version were among the new features in this Excel edition. If you're wondering why Microsoft chose version 14 over 13 instead, it's because 13 is regarded as a bad luck number.
13	Microsoft Excel 2013	2013	This version featured the single-document interface, suggested charts and pivot tables, new charting upgrades, and more than 50 new functions.
14	Microsoft Excel 2016	2016	Despite having separate software versions, Excel for Mac and Windows are now referred to by the same name. If you also subscribe to Office 365, certain Internet upgrades for Excel are available that can greatly alter your user experience. Thus, older versions including ones purchased at retail are at a disadvantage. Histograms, Pareto charts, and PowerPivot, which allows for the import of higher levels of data and has its own language, are some of the newly added features in this version. Histograms are used to depict frequency in data.
15	Microsoft Excel 2019	2019	The new charts, which change how data is presented, are one of the more noticeable new additions of this version. Some of the new data presentation charts that help your data look tidy and organised include map charts and funnel diagrams. Additionally, you have the option to add 3D images to your workbooks.

16	Microsoft Excel 2021	2021	As of right now, this is Microsoft Excel's most recent version. Naturally, it has all the functionality of older Excel versions and even more. This version adds numerous features to earlier versions, including Unhide Multiple Sheets, Navigation Pane, Resize Conditional Formatting Box, and LAMBDA Helper Functions.
----	----------------------	------	--

3.3 Advantages of Microsoft Excel for Statistical Analysis

- **Easy Data Storage:** MS Excel is frequently used to save or analyse data because there is no limit to the amount of information that can be recorded in a spreadsheet. In Excel, information filtering is simple and practical.
- **Simple Data Recovery:** Finding the information might take longer if it were written on paper, but this is not the case with excel spreadsheets. It is simple to locate and retrieve data.
- **Use of Mathematical Formulas:** The MS Excel formulae function has made performing calculations simpler and quicker.
- **Extra Secure:** When compared to data stored on registers or paper, these spreadsheets have a much lower chance of being lost and can be password-secured on a laptop or desktop computer.
- **Data in one location:** When the paperwork was completed, data had to be preserved in various files and registers. Now that multiple worksheets may be added to a single MS Excel file, this has become convenient.
- **Information Visibility Is Sharper and Clearer:** Analysing the data is much simpler when it is recorded as a table. Information is therefore a more readable and intelligible spreadsheet.

3.4 Disadvantages of Microsoft Excel for Statistical Analysis

- **The data source is challenging with Excel:** Excel's difficulty in evaluating real-time unstructured and semi-structured data is one of its biggest drawbacks. On the other hand, a lot of recent and upcoming data analytics technologies are capable of swiftly identifying this kind of data and producing representations. When it comes to data formats like JSON, Excel likewise struggles. JSON is the fastest-growing data type due to the rise of mobile data, apps, and IoT devices. Even the most seasoned Excel users face a hurdle with its intricate structure. However, individuals employing contemporary analytics tools may quickly decode and examine this kind of data in an Excel-like setting.

- **Excel cannot create interactive dashboards:** Utilizing a data visualization tool, you can mix various chart types to produce dynamic dashboards that list all of your KPIs in one location. After that, you can delve more deeply into each chart, examining its level of detail and looking for trends. Excel allows you to create dashboards, but they are static, only display high-level trends, and act as an endpoint rather than the beginning of the additional investigation. When compared to dynamic dashboards, which have the capability of automatic updates, this is a drawback.
- **Excel doesn't provide real-time updates:** Spreadsheets in Excel are used to store stale data. Data gets outdated the instant you import a CSV file and build out a worksheet because it isn't connected to a live data warehouse. Users are compelled by this to endure tedious manual procedures in order to repeatedly submit the same information. A cloud-based data visualization tool may easily connect to dozens of data sources using a cloud data warehouse like Snowflake, whereas Excel can connect to external data sources via plugins. In this way, you can rapidly get the most recent data, display it quickly, distribute it around your organisation, and use it to instantly make smarter business decisions.
- **Excel Is Not a Collaboration Tool:** Most Excel files live locally. While they do have connectivity to the cloud, the worksheets are primarily intended for solitary usage and lack sophisticated collaboration features. Every time someone wants to build on the work of another person, this leads to infinite duplication of effort and chaos from an organisational standpoint. Analyses carried out in a vacuum are a waste of time. Excel can eventually put businesses at a disadvantage since it makes it difficult for teams to properly communicate information, collaborate, and find solutions. Teams may eliminate data silos, work more effectively together, and elevate each other's work by using a data visualisation tool. By first deriving insights from the primary and then deriving insights from those insights, this is how you construct compound interest with data.
- **Sprawling Spreadsheets Poses a Security Risk:** It is risky to download data to a spreadsheet and store it locally. Excel spreadsheets don't require professional SQL coding knowledge, unlike BI tools, which is why so many corporate customers still use them in spite of the dangers. Unfortunately, the risks are numerous, and it also has many drawbacks. There is a lack of security and compliance monitoring when data is downloaded to Excel since the data team is unable to monitor how employees utilise or distribute the data. Vulnerabilities, lost data, and misuse result from this.

3.5 Summary

In this chapter, we have studied:

- The worksheet's cells are the little, rectangular boxes where we enter data.
- Excel has seen numerous changes in terms of function diversity, functionality, and capacity since its creation.

- The MS Excel formulae function has made performing calculations simpler and quicker.
- MS Excel is frequently used to save or analyse data because there is no limit to the amount of information that can be recorded in a spreadsheet.

3.6 Answers to self-check exercises

- Q.1 What do you mean by Cell?
- Q.2 How to select a Cell?
- Q.3 How to drag and drop cell?
- Q.4 How to delete cell content?
- Q.5 How to insert content to a cell?
- Q.6 How to select Cell Range?
- Q.7 Explain history of Microsoft Excel.
- Q.8 What are various advantages of Microsoft Excel for Statistical Analysis?
- Q.9 Explain various disadvantages of Microsoft Excel for Statistical Analysis?

3.7 Suggested Reading

- Ron Mansfield, The Compact Guide to Microsoft office; BPB Publication.
- Dienes, Sheila s., Microsoft office, Professional for windows 95.
- Joan Lambert, Curtis Frye, Microsoft Office Step by Step (Office 2021 and Microsoft 365), Microsoft Press Publication.
- Kenneth N. Berk, Data Analysis with Microsoft Excel.
- Wayne L. Winston, Microsoft Excel 2013: Data Analysis and Business Modeling.
- Danielle Stein, Using Excel for business analysis: a guide to financial modelling fundamentals.
- Olafusi M., Microsoft Excel and Business Data Analysis for The Busy Professional.
- Rajiv Mattus: Learning Excel 97 for windows step by step BPB publication.
- Lonnie. E. Moseley & David M. Boodey: Mastering office 97.
- Micro Soft Office 97: Unleashed: Techmedia.
- John Walken Bach: Excel 97.

Chapter-4

MICROSOFT EXCEL TITLE BAR, MENU BAR AND TOOLBAR

STRUCTURE

- 4.0 Learning Objectives
- 4.1 Introduction of Microsoft Excel Worksheet Title Bar
- 4.2 Introduction of Microsoft Excel Worksheet Menu Bar
- 4.3 Introduction of Microsoft Excel Worksheet Toolbar
- 4.4 Introduction of Formula and Function
 - 4.4.1 How to enter formula in Excel?
- 4.5 Summary
- 4.6 Answer to Self-Check exercises
- 4.7 Suggested Readings

4.0 LEARNING OBJECTIVES

After going through this lesson, you will be able to:

- Understand the basic of Microsoft Excel Worksheet Title Bar
- Understand the basic of Microsoft Excel Worksheet Menu Bar
- Understand the basic of Microsoft Excel Worksheet Toolbar
- Understand the basic of Formula and Function

4.1 Introduction of Microsoft Excel Worksheet Title Bar

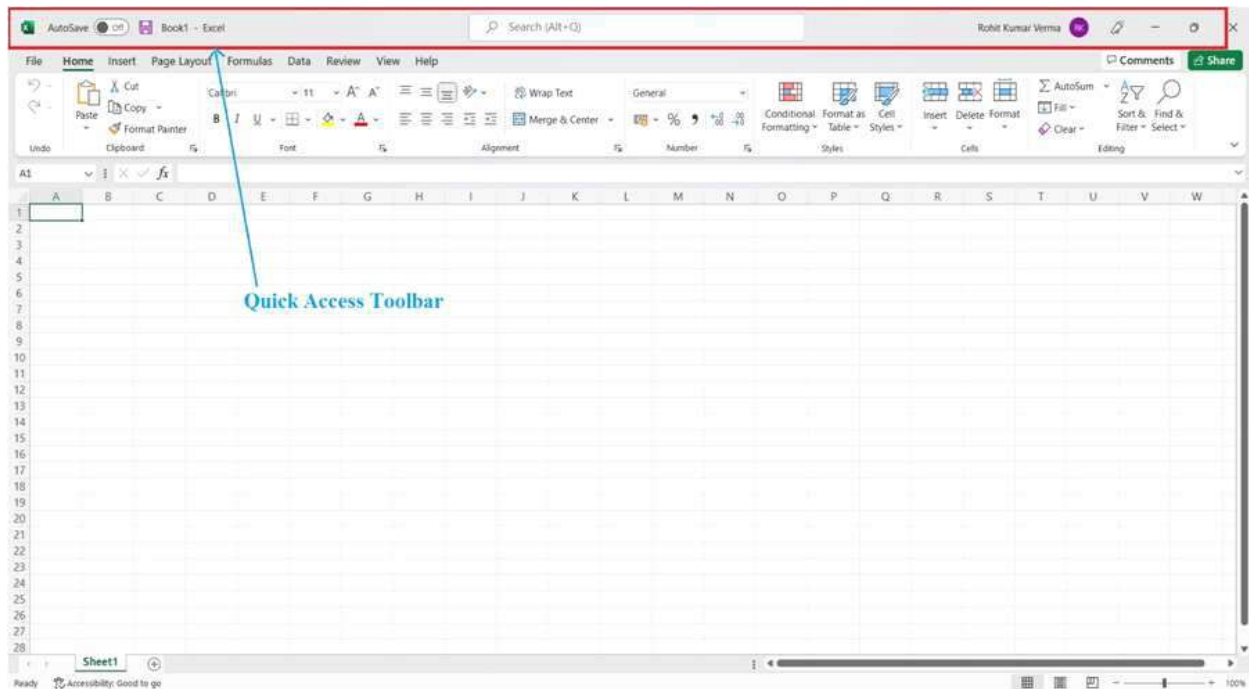
The excel file's title is held by the bar known as the Title Bar. But it also has a lot of other crucial features in addition to presenting the title.

The title bar has three sections:

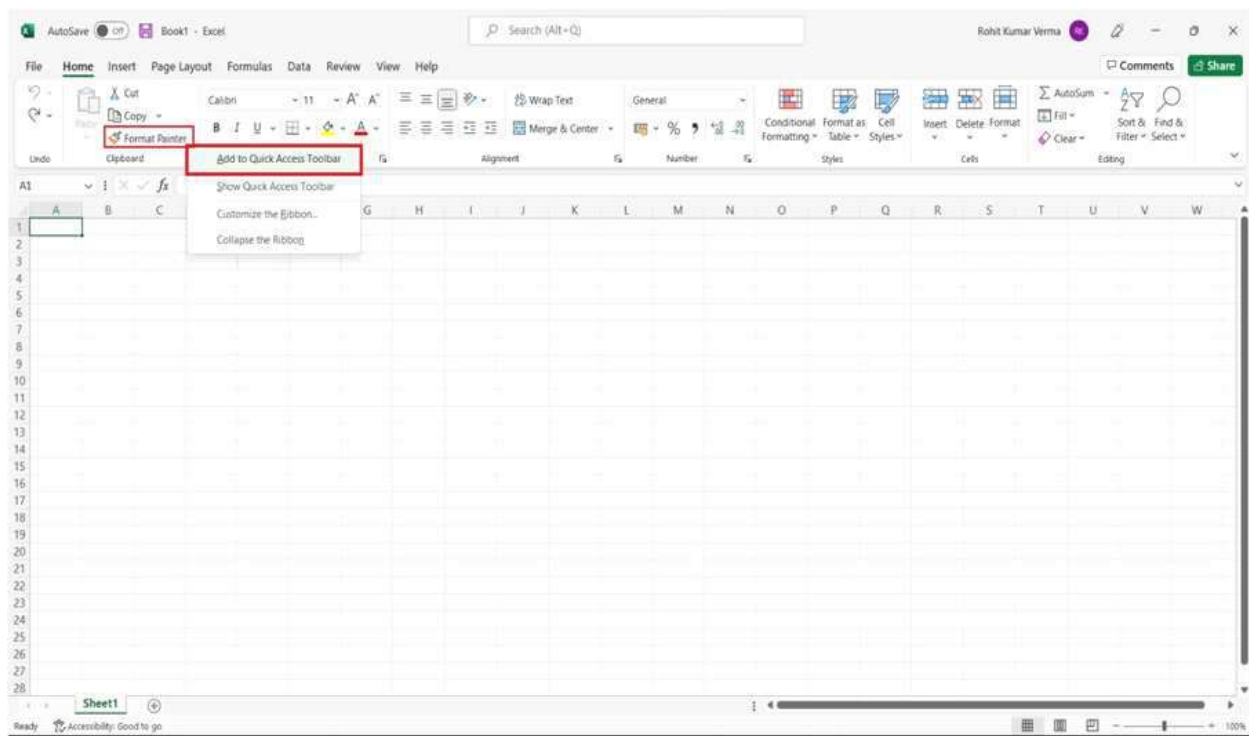
- Quick Access Toolbar
- File Name
- The Minimize, Maximize/Restore, and Close buttons

Quick Access Toolbar

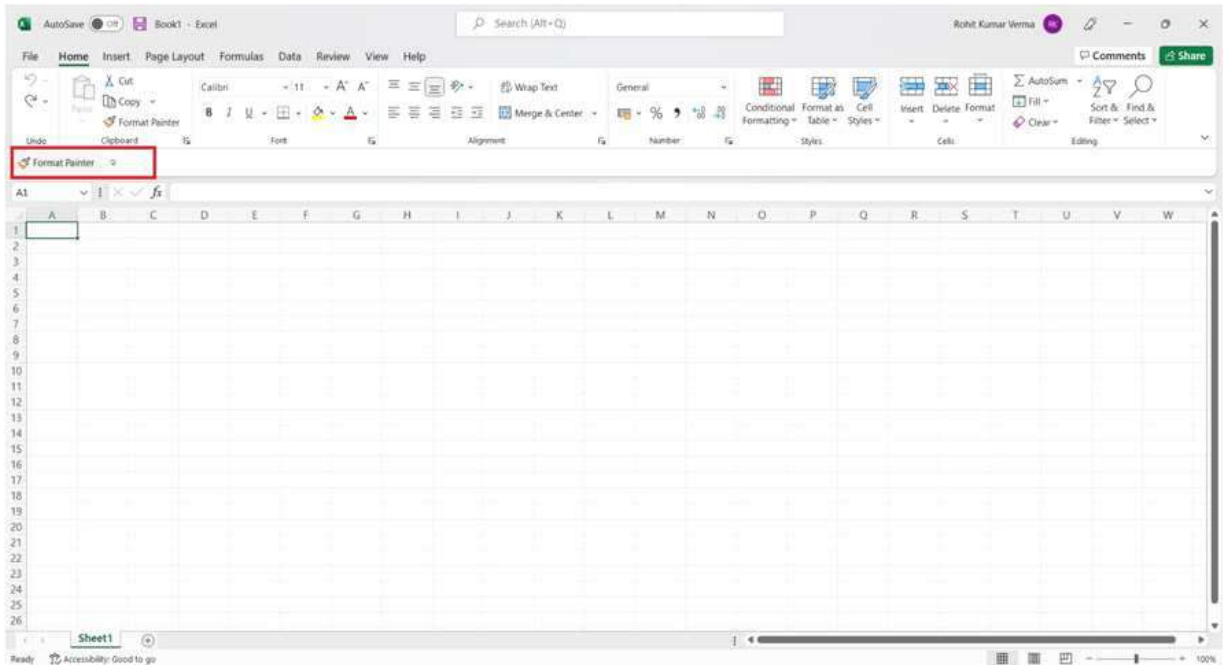
We can add commands to the Quick Access Toolbar that we commonly use. You can add the desired commands here by following these procedures.



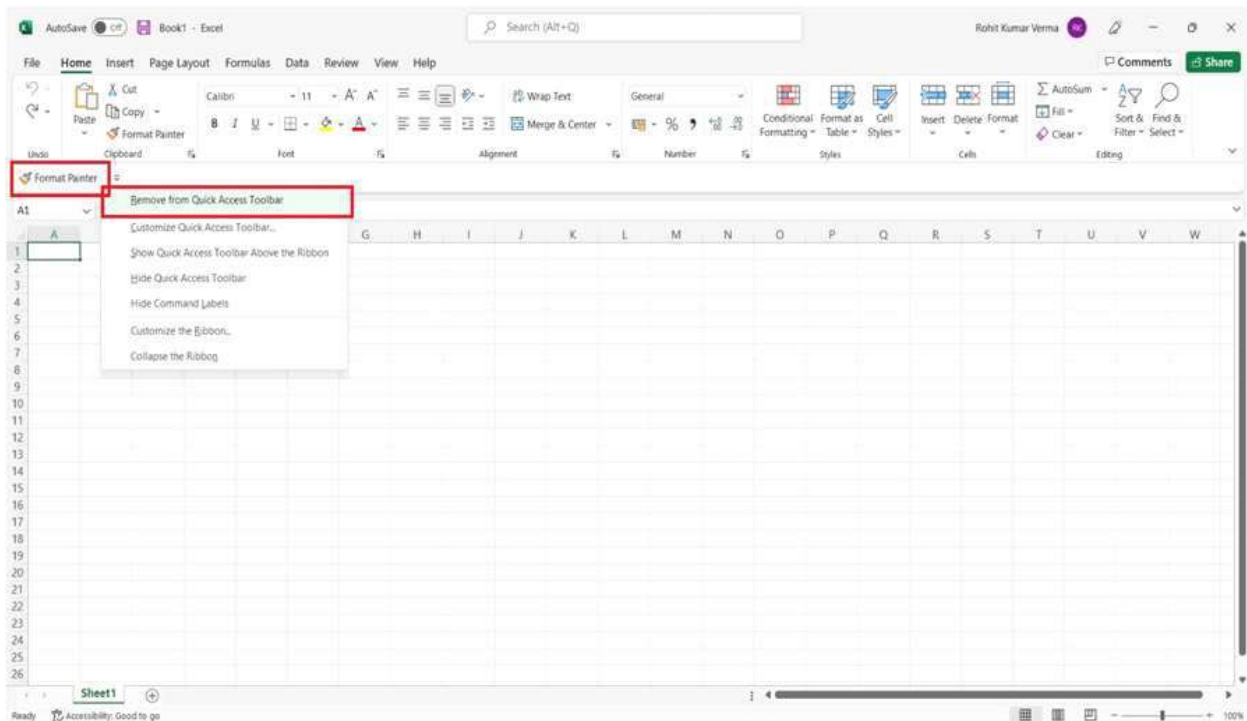
- Right-click the command you want to use. (In the example below, we have “Format Painter” chosen as a desired command.)
- Choose “Add to Quick Access Toolbar” from the menu.



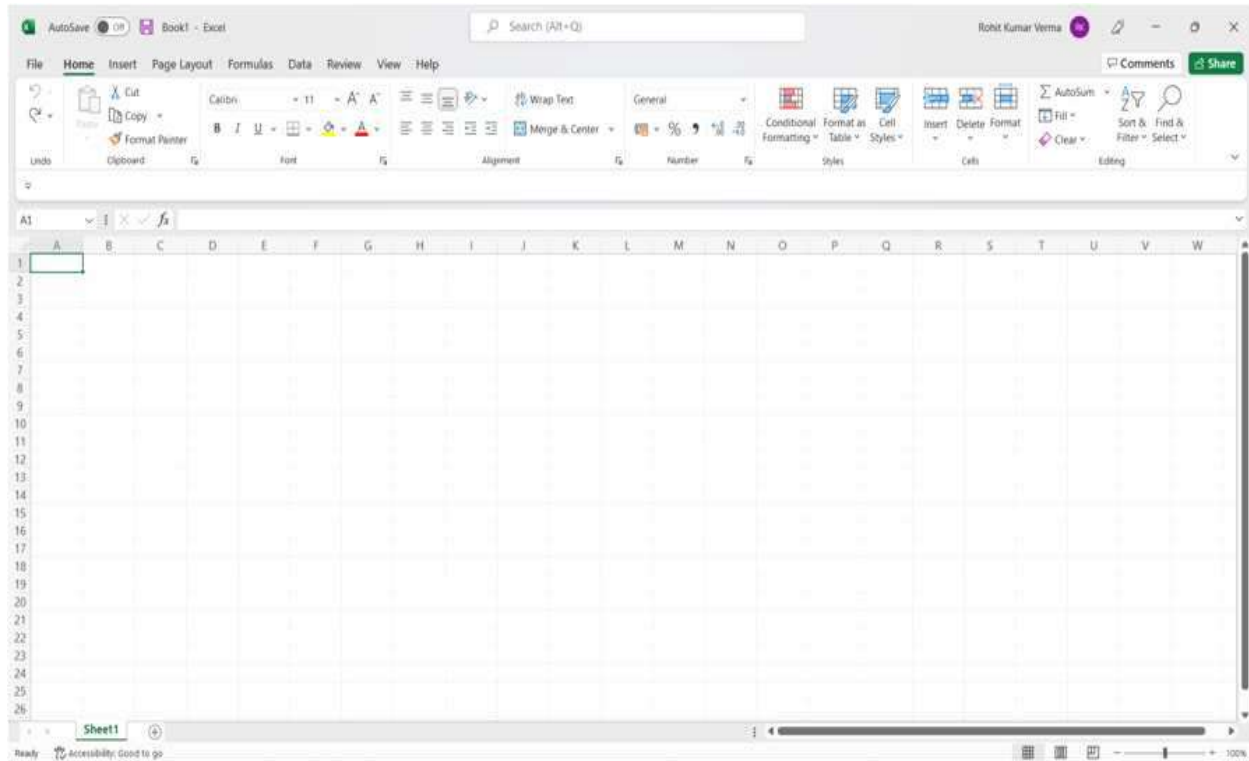
- The “Format Painter” command has now been added to the Quick Access Toolbar, as you can see.



- Simply right-click on the relevant command and choose “Remove from Quick Access Toolbar” from the menu that appears.

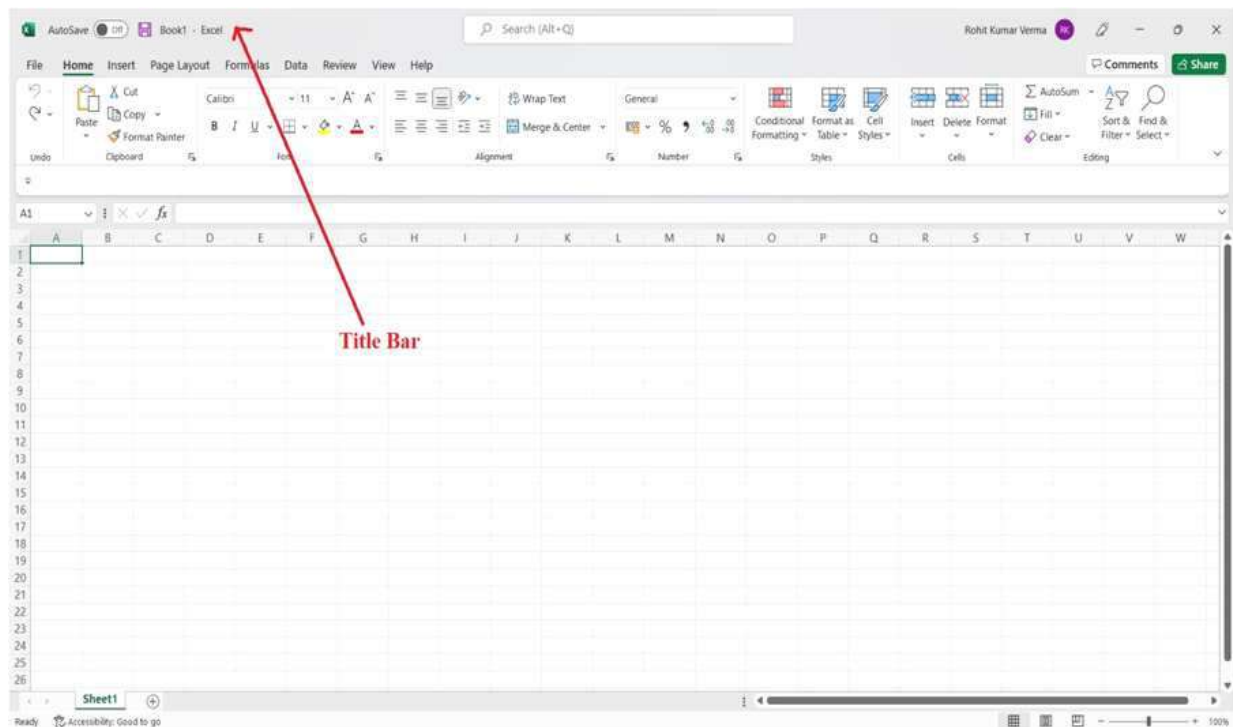


- You will now see that the “Format Painter” command has been deleted from the Quick Access Toolbar.



File Name

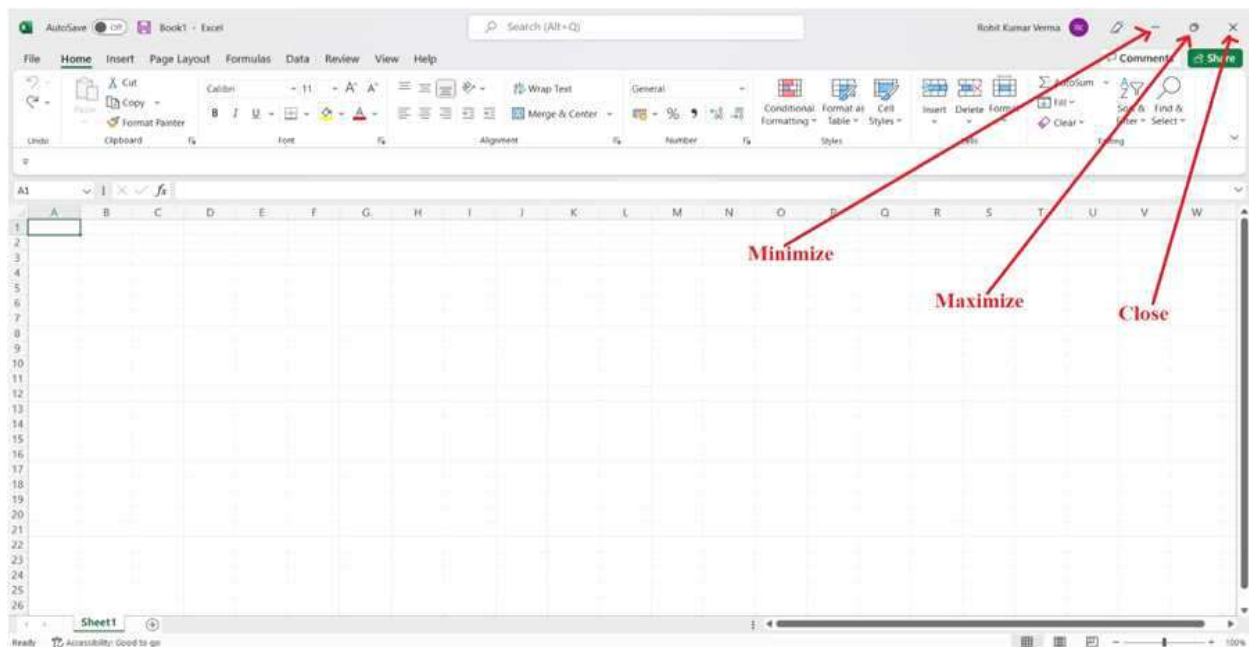
It displays the file name we gave to our excel file. The excel file name is “Book 1” in the screenshot below, and “- Excel” signifies that this is an excel file.



Minimize, Maximize/Restore, Close Buttons

On the right side of the Title Bar are these buttons:

- Minimize: This button has a minus-like appearance. This button will relocate the Excel window to the Taskbar (Taskbar is located at the bottom of the screen).
- Maximize/Restore: When we click on this, the Excel window will fill the entire screen. The Excel window will appear in a lesser size if we click this button after it has already been stretched to fill the entire screen, allowing us to resize the window.
- Close: Select this button to shut off the Excel document.

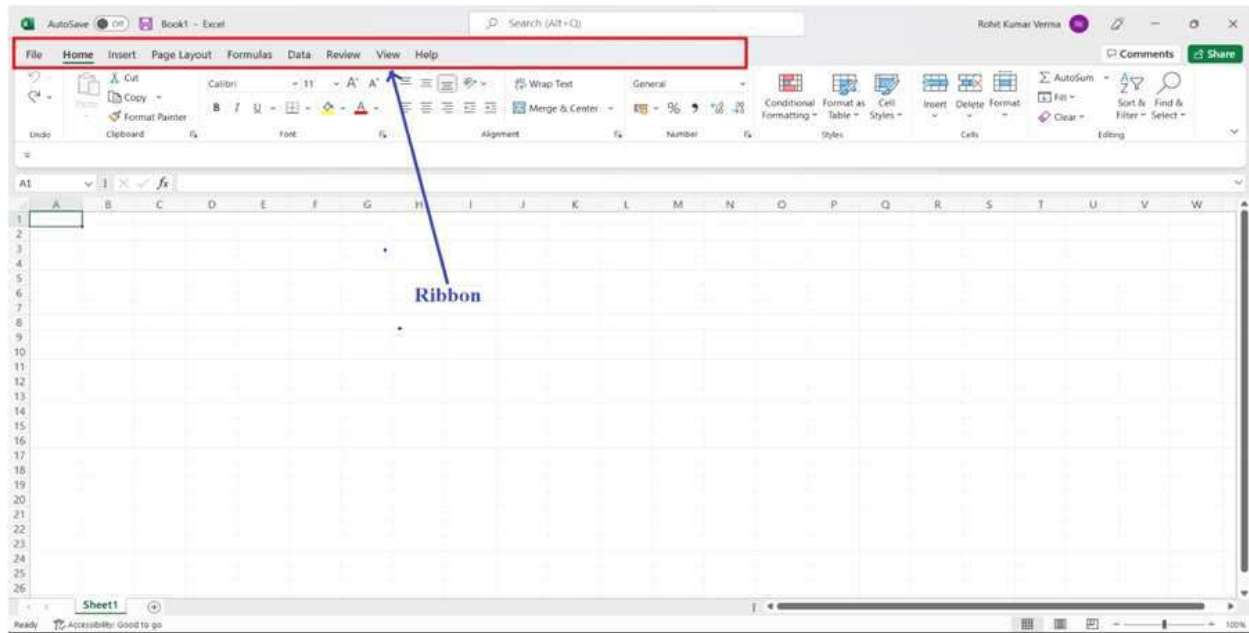


4.2 Introduction of Microsoft Excel Worksheet Menu Bar

You may access several commands required for tasks like opening and closing files, printing documents, formatting data, and other actions using the Menu Bar at the top of the screen. The options in the Menu Bar and how they work are captured on screen in the images below.



Nowadays, the name of Excel's primary menu bar is "Ribbon." The ribbon's items come with collections of icons that serve the same purposes.



These consist of:

File, Home, Insert, Formulas, Page Layout, View, Data

File

The item on the Excel ribbon farthest to the left is the File menu. You may manage your files by using the open, save, close, and print options available on the File ribbon items. Along with tools that let you modify Excel itself; you may import data into Excel from outside sources using the File menu.

Home

The second option in the Excel menu bar is the home menu. Formatting options for font, colour, conditional formatting, filter, number type, and more are available in the home ribbon items. Each of these functions' aids in the execution of several efficient calculations.

Insert

The Insert menu, as its name implies, enables you to add a variety of choices and objects to an Excel spreadsheet. A pivot table, a photo, clip art, shapes, screen captures, charts and graphs, text boxes, header and footers, symbols, equations, and more can all be inserted.

Page Layout

Once more, the name alludes to the group of options available on the Page Layout menu. You'll find a variety of options for setting up pages for reading and printing, including options for page size, margins, colours, and fonts. On the Page Layout menu, you may also change the width and height of individual cells.

Formulas

You can find all the options for number-crunching in the Formulas menu. There are several formulae in Excel, including ones for algebra and trigonometry as well as for financial, logical, text, date and time, lookup and reference.

Data

Numerous crucial Excel features, such as imports and links to databases, are now available under the Data menu. On the Data menu, you can also access the sort, filter, duplicate-removal, data validation, consolidation, group, ungroup, and subtotal options.

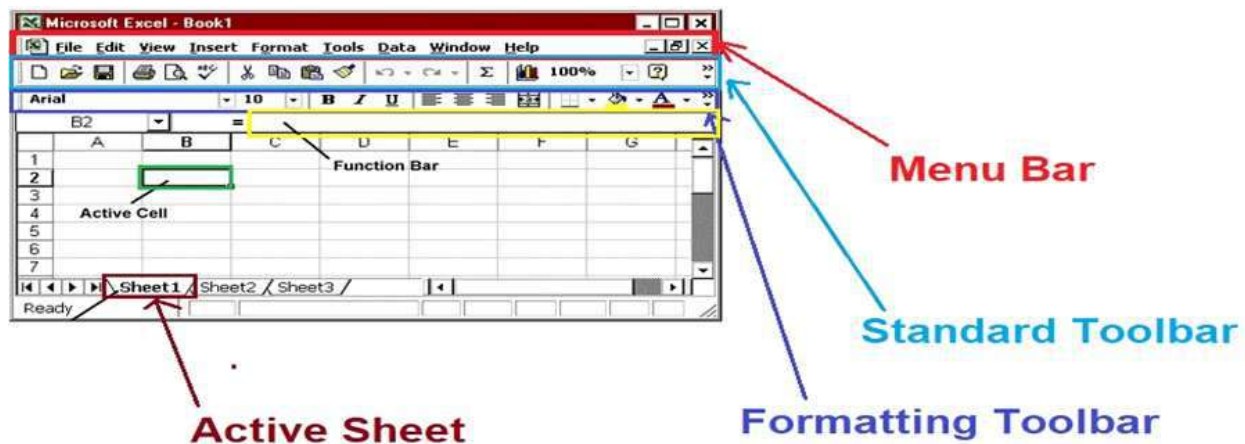
Review

The majority of the tasks can be found in the Review menu. Using features from the Review menu, you may track changes, check spelling, add comments in cells for your colleagues, and even restrict authorization.

View

You can alter how spreadsheets display on your screen by using the View menu. Grid lines between cells can be displayed, the formula bar and headings can be switched, and more options are available. You have the ability to see and record macros from this menu as well, which saves you from having to repeat actions you already know how to accomplish over and over again.

4.3 Introduction of Microsoft Excel Worksheet Toolbar



The Excel toolbar is a place where you can put various commands or tools. It can be seen by default in the Excel window's upper right corner, above the ribbon with various tools. Toolbars can now be customized based on how frequently users use particular tools, which improves customer friendliness.

There are two toolbars that offer easy access to many of the Excel features that are utilized the most. The Standard Toolbar is the first bar, followed by the Formatting Toolbar. A yellow box will display next to a toolbar icon when the mouse pointer is placed over it without clicking.

Using the Tools menu, you may modify how the Standard and Formatting toolbars are displayed as seen below:

- Click Tools on the Menu Bar.
- Customize can be selected (clicked) from the drop-down menu.
- To move it to the front, click the Options tab.
- The box next to “Standard and Formatting Toolbar share one row” can be checked or unchecked.
- If the checkbox is unchecked, just one row will be shared by the toolbars.
- The toolbars will be displayed in separate rows if the checkbox is left empty.
- To get back to your workbook window, select Close.

From left to right in the standard toolbar, the buttons are: New Worksheet, Open Worksheet, Save, Print, Print Preview, Spell Check, Cut, Copy, Paste, Format Paint, Undo, Redo, Insert Hyperlink, Web Toolbar, AutoSum, Function Wizard, Sort Ascending, Sort Descending, Chart Wizard, Mapping, Drawing, Zoom, and Help. Additionally, each of these options can be selected from a pull-down menu.



Standard Toolbar

The buttons in the formatting toolbar are, in order, Bold, Italics, Underline, Margins, Font Type, Font Size, and (left, center, right, merge ¢er,) Borders, fill colour, font colour, increase decimal, decrease decimal, decrease indent, increase indent, currency style, percent style, comma style, and all of the above. Additionally, each of these options can be selected from a pull-down menu.



**Formatting
Toolbar**

Toolbars are really helpful. They provide quick access to commands without requiring users to navigate through menu options. Excel comes with a lot of toolbars. Click “Get” on the Menu Bar and choose “Toolbars” to view a list of all the toolbars that are currently accessible. The toolbars are visible in a submenu that emerges. Simply click on a toolbar in the submenu to choose it. Your choice will now be seen from your worksheet due to the check mark that will now appear in front of it.

Function or Formula Bar: The active cell’s or block’s contents are shown in the formula bar. Text, numbers, formulas, and functions are all included.

Active Cell

The chosen cell at the moment is the active cell. The active cell is surrounded by a large border block known as the Cell Pointer. The active cell is filled with data.

Active Sheet

A file frequently contains multiple worksheets. To switch between sheets, click the tab located on each sheet. To make it simpler to recall what each sheet includes, you can rename them. To accomplish this, double-click Sheet1, for example, type a new name, and then press the ENTER key.

Worksheet Window

The Excel worksheet is visible in the worksheet window. There are rows and columns in it. The alphabetic letters, such as A, B, C, etc., are used to identify the columns. On the left side, the rows are numbered. Cells are formed when rows and columns cross. The column name and row number can be used to refer to any cell.

4.4 Introduction of Formula and Function

Formula

Calculating values in a cell or a group of cells is done by a formula, which is an expression.

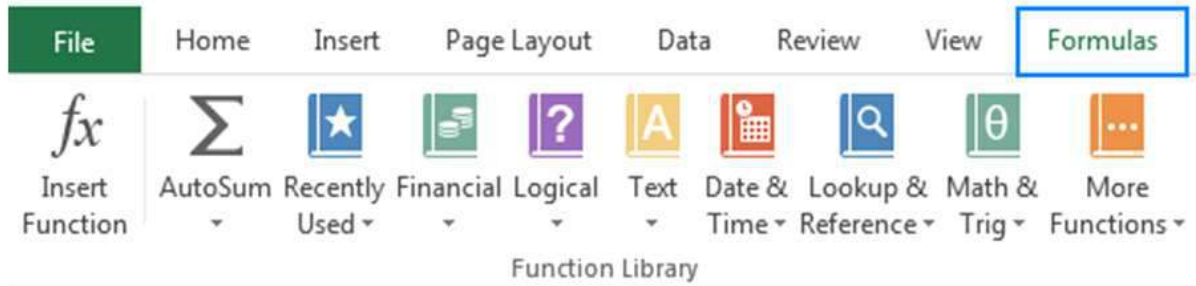
For instance, the values in cells A2 through A4 can be added up using the formula =A2+A2+A3+A4.

Function

Function is a built-in, predefined formula in Excel. Based on the supplied values, referred to as arguments or parameters, functions carry out specific calculations in a precise order.

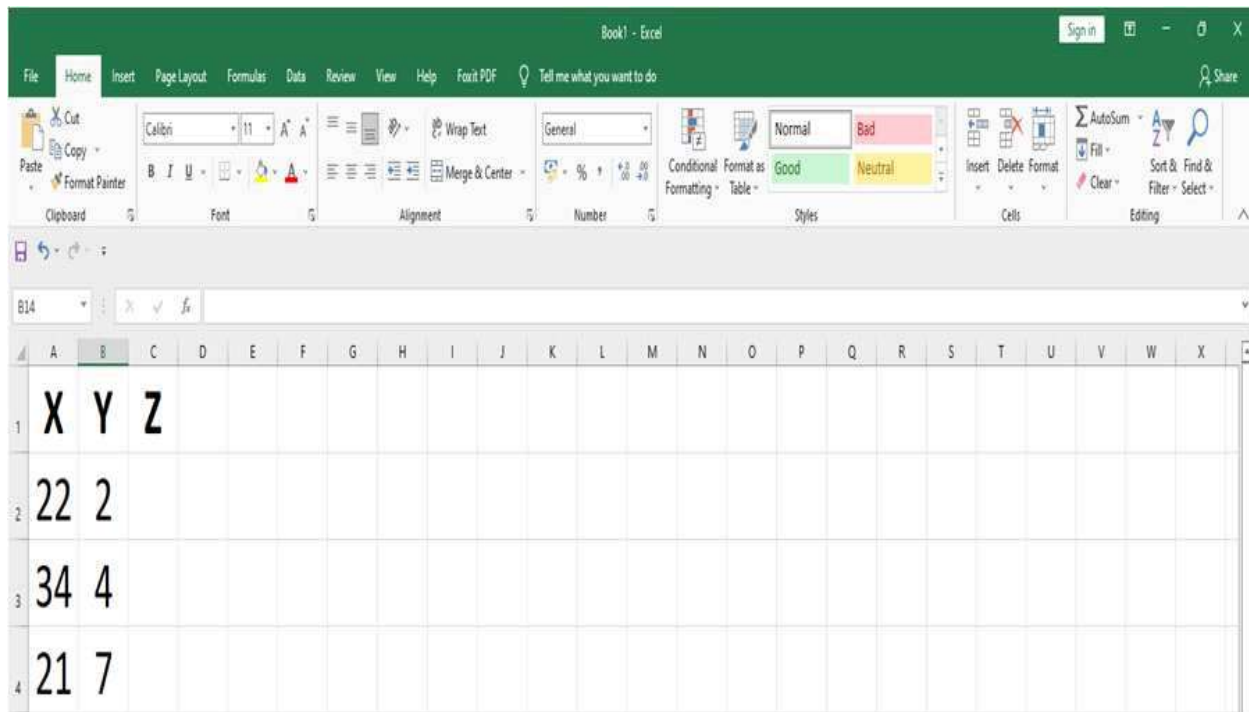
To sum a range of cells, for instance, use the SUM function rather than specifying each value to be added as in the formula above: =SUM(A2:A4)

All of the Excel functions are listed in the Function Library under the Formulas tab:

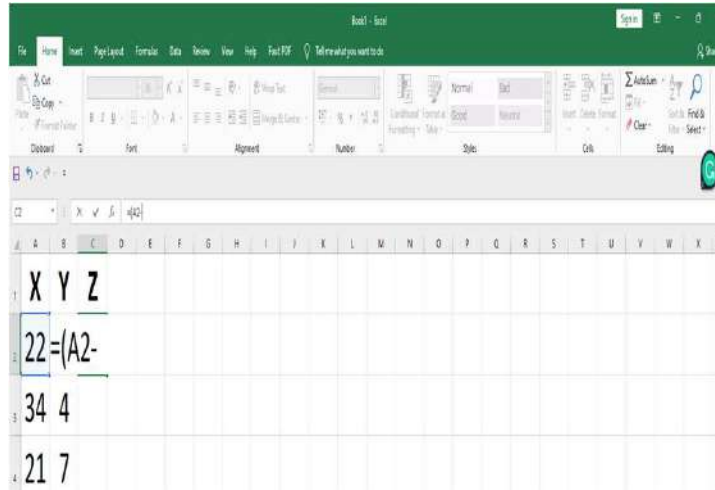


4.4.1 How to enter formula in Excel?

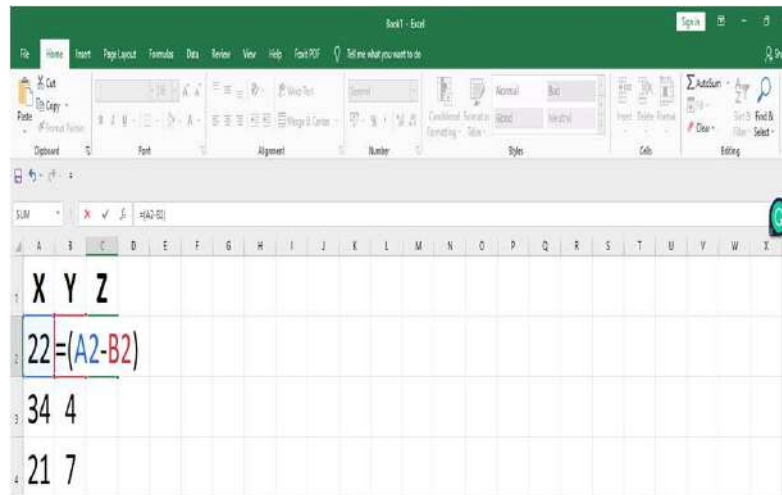
- Click the cell in a spreadsheet with numerical columns where you wish the formula results to show.
- Add the equal sign =.
- To include the first cell in your computation, click it



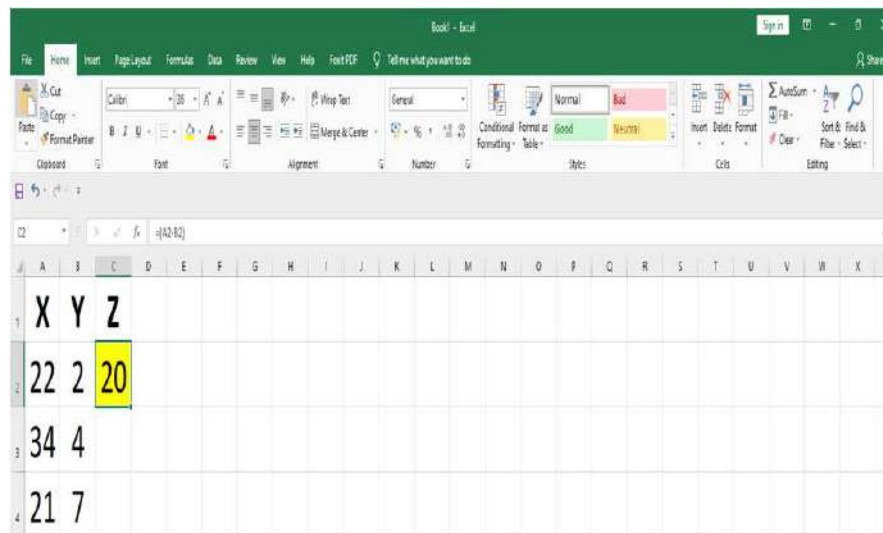
- Type an operator. An operator is a specific type of calculation that the formula carries out. For instance, the * operator multiplies numbers. To subtract in this instance, use the - operator. Your formula should now resemble the following:



- To add the following cell to your computation, click it. Your formula should now seem as follows:



- Press RETURN.



4.5 Summary

In this chapter, we have studied:

- The Excel toolbar is a place where you can put various commands or tools.
- The Insert menu enables you to add a variety of choices and objects to an Excel spreadsheet.
- Calculating values in a cell or a group of cells is done by a formula, which is an expression.

4.6 Answers to self-check exercises

Q.1 What do you mean by Function?

Q.2 Explain Title bar briefly?

Q.3 What are various options available under Menu bar?

Q.4 How to enter formula in Excel?

Q.5 Explain Toolbar briefly?

4.7 Suggested Reading

- Ron Mansfield, The Compact Guide to Microsoft office; BPB Publication
- Dienes, Sheila s., Microsoft office, Professional for windows 95
- Joan Lambert, Curtis Frye, Microsoft Office Step by Step (Office 2021 and Microsoft 365), Microsoft Press Publication
- Kenneth N. Berk, Data Analysis with Microsoft Excel
- Wayne L. Winston, Microsoft Excel 2013: Data Analysis and Business Modeling
- Danielle Stein, Using Excel for business analysis: a guide to financial modelling fundamentals
- Olafusi M., Microsoft Excel and Business Data Analysis for The Busy Professional
- Rajiv Mattus: Learning Excel 97 for windows step by step BPB publication.
- Lonnie. E. Moseley & David M. Boodey: Mastering office 97.
- Micro Soft Office 97: Unleashed: Techmedia.
- John Walken Bach: Excel 97.

Chapter-5

CELL FORMATTING, CUT, COPY AND PASTE OPERATION

STRUCTURE

- 5.0 Learning Objectives
- 5.1 Introduction of Microsoft Excel Worksheet Cell
 - 5.1.1 How to change format of Cell?
 - 5.1.2 How to enter data into a Cell?
 - 5.1.3 How to clear data from a Cell?
- 5.2 Introduction of Various types of Operation
 - 5.2.1 Cut
 - 5.2.2 Copy
 - 5.2.3 Paste
 - 5.2.4 Paste Special
 - 5.2.5 Paste Value
 - 5.2.6 Paste Formula
 - 5.2.7 Paste Transpose of Data Set
- 5.3 Summary
- 5.4 Answer to Self-Check exercises
- 5.5 Suggested Readings

5.0 LEARNING OBJECTIVES

After going through this lesson, you will be able to:

- Understand the basic of Cell Formatting
- Understand the basic of Cut, Copy, Paste Operation
- Understand the basic of Paste Special, Paste Transpose

5.1 Introduction of Microsoft Excel Worksheet Cell

The worksheet's cells are the little, rectangular boxes where we enter data. Row and column intersections form cells. The column header and row number serve as identifiers. Using a set of coordinates or positions, such as A1 (where A stands for a column and 1 for a row), C2, or H15, each cell in Excel is uniquely recognized.

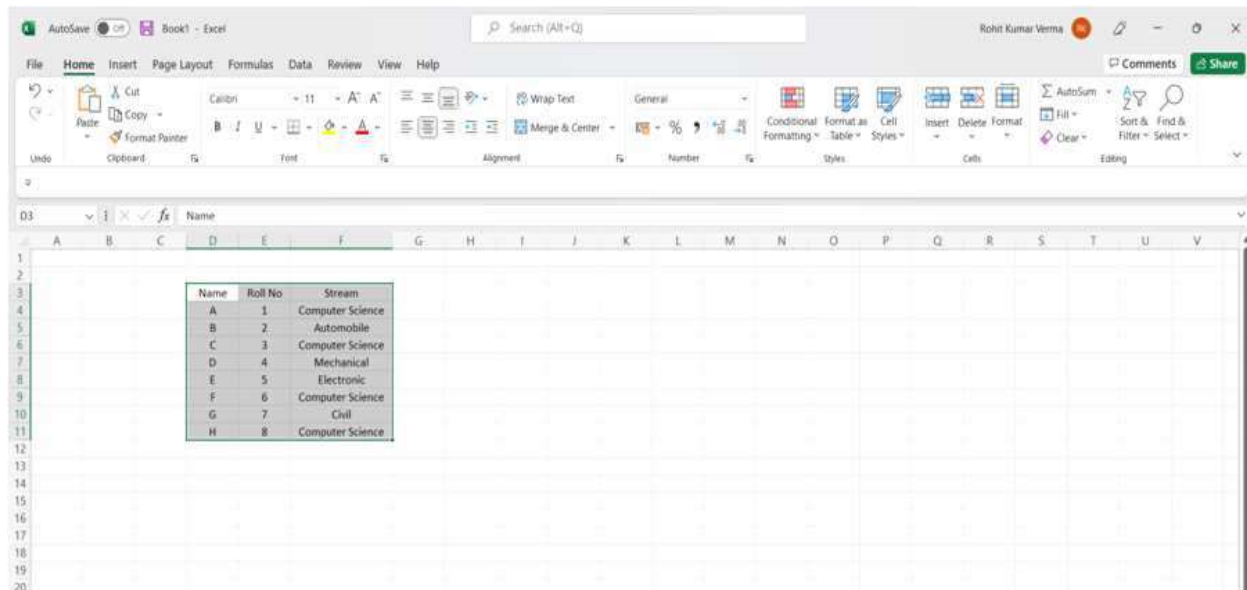
5.1.1 How to change format of a Cell?

A cell's contents, including its data, and a set of related cells can all be formatted. One way to conceptualise this is as if the data were the picture inside the frame and the cells were the picture's frame.

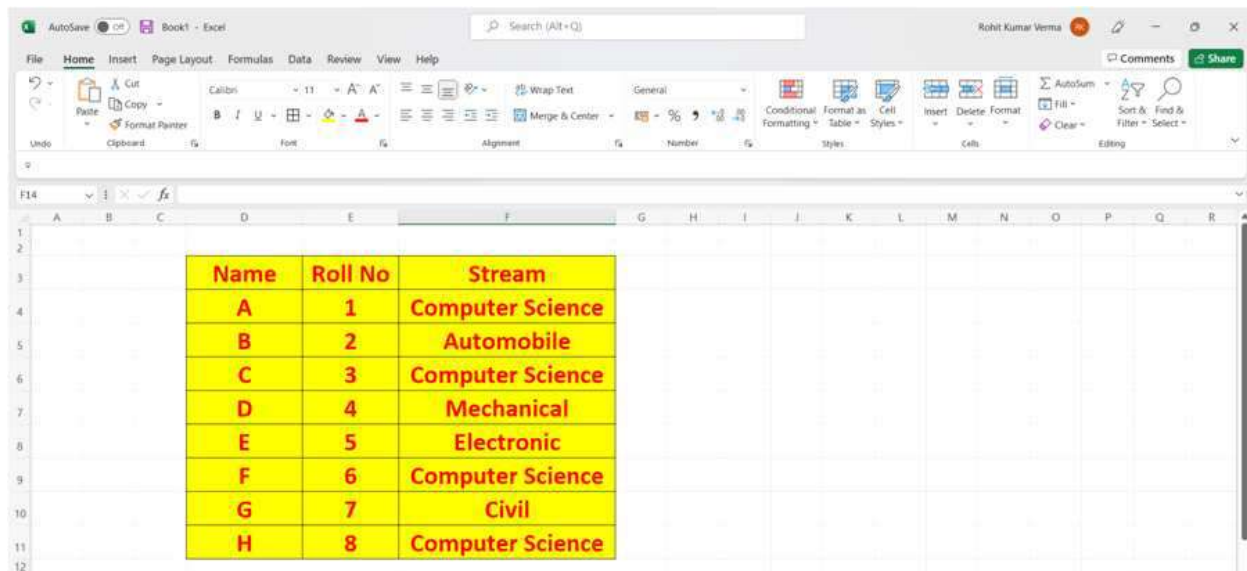
Format Cells

The steps to format cells in Excel are listed below:

- Select the cells.



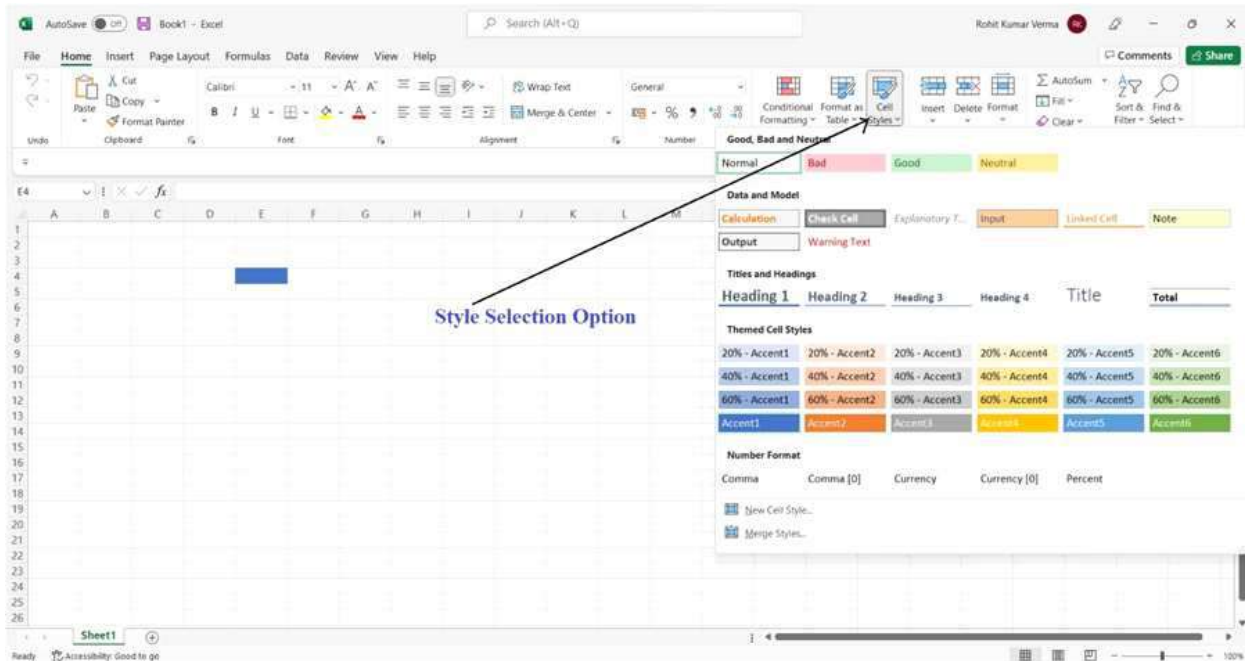
- To make adjustments such as Bold, Font Color, or Font Size, use the ribbon.



How to apply style to a Cell in Excel?

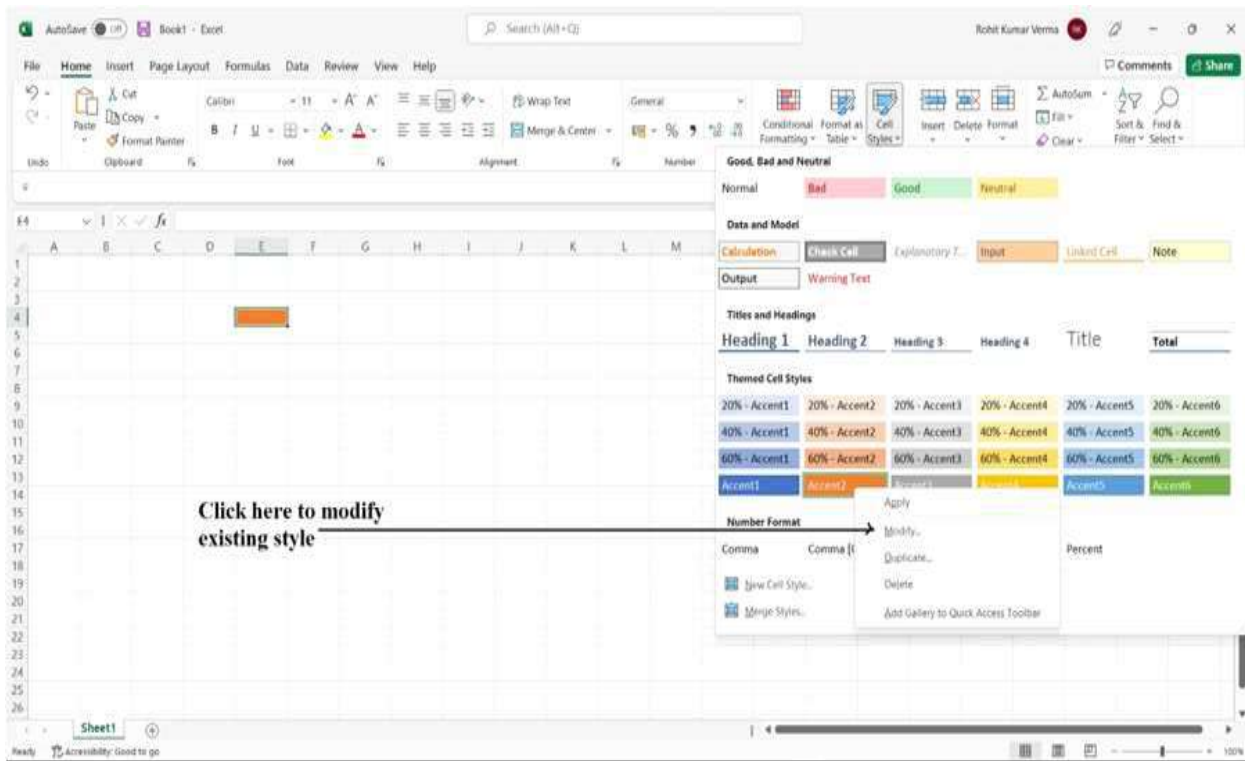
The steps to apply style to a cell in Excel are listed below:

- Choose the cells.
- Pick a style by selecting Home > Cell Style.



How to modify style of a Cell in Excel?

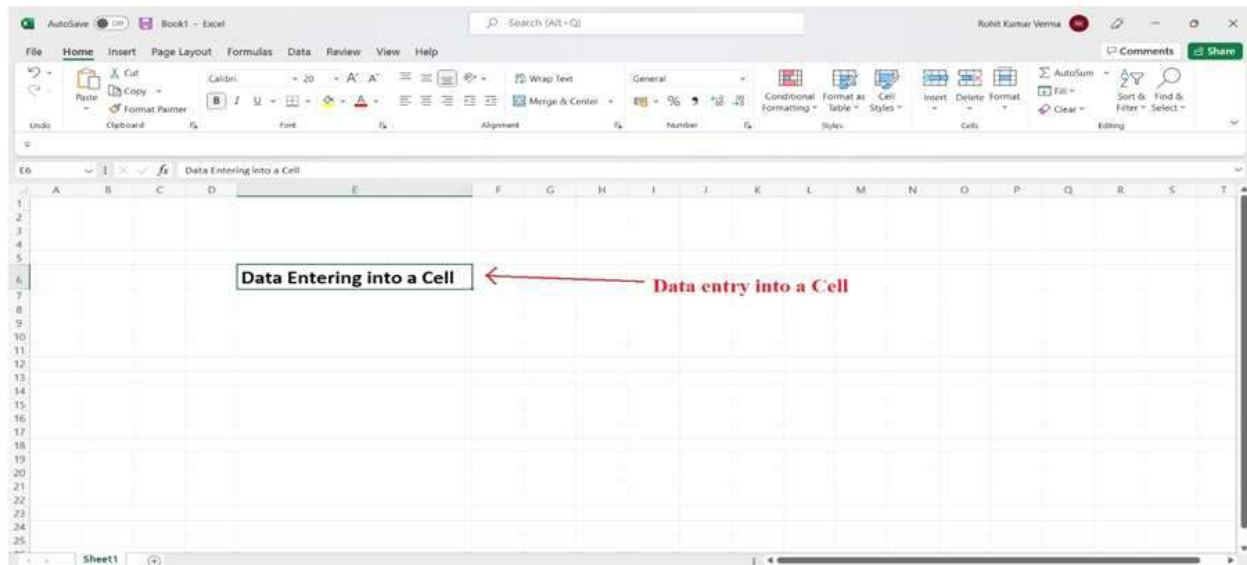
- Choose the cells with the Excel Style.
- Right-click the style that was just applied under Home > Cell Styles.
- To make the changes you wish, choose Modify > Format.



5.1.2 How to enter data into a Cell?

The steps to enter data into a cell in Excel are listed below:

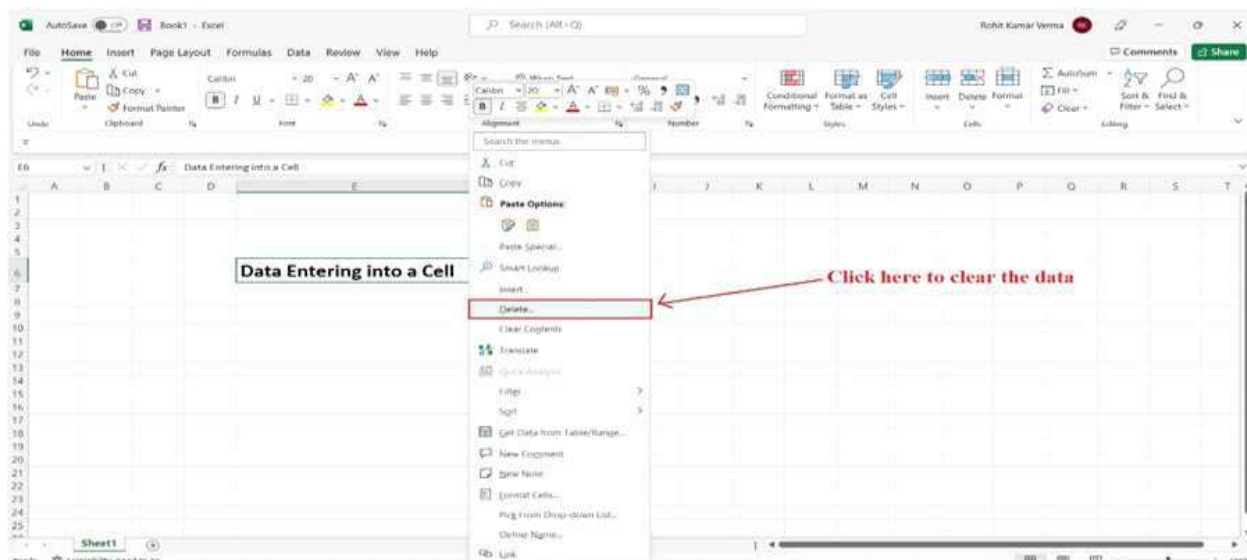
- The cell where you want to enter the data should be clicked and selected.
- Enter the required information into the highlighted cell, then press Enter or shift your pointer to a different cell. Both the formula bar and the cell's content will display the content.



5.1.3 How to clear data from a Cell?

The steps to clear Cell data in Excel are listed below:

- Select the cell by clicking.
- Press the Delete or Backspace key on your keyboard to remove the cell's content.
- Select the cell range, then press the Erase key to delete several cells.



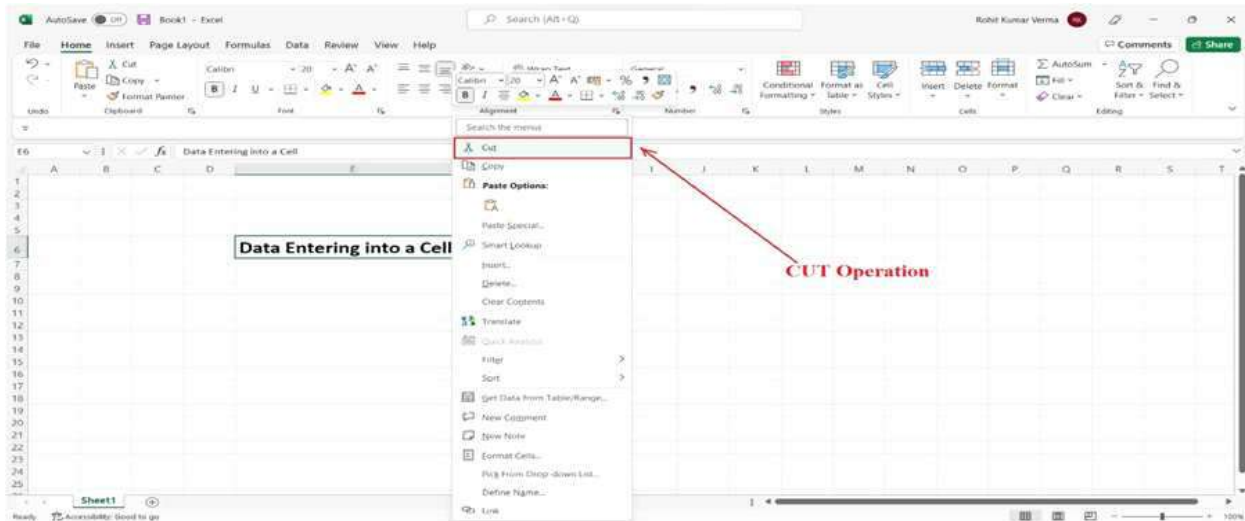
5.2 Introduction of Various types of Operation

5.2.1 CUT Operation

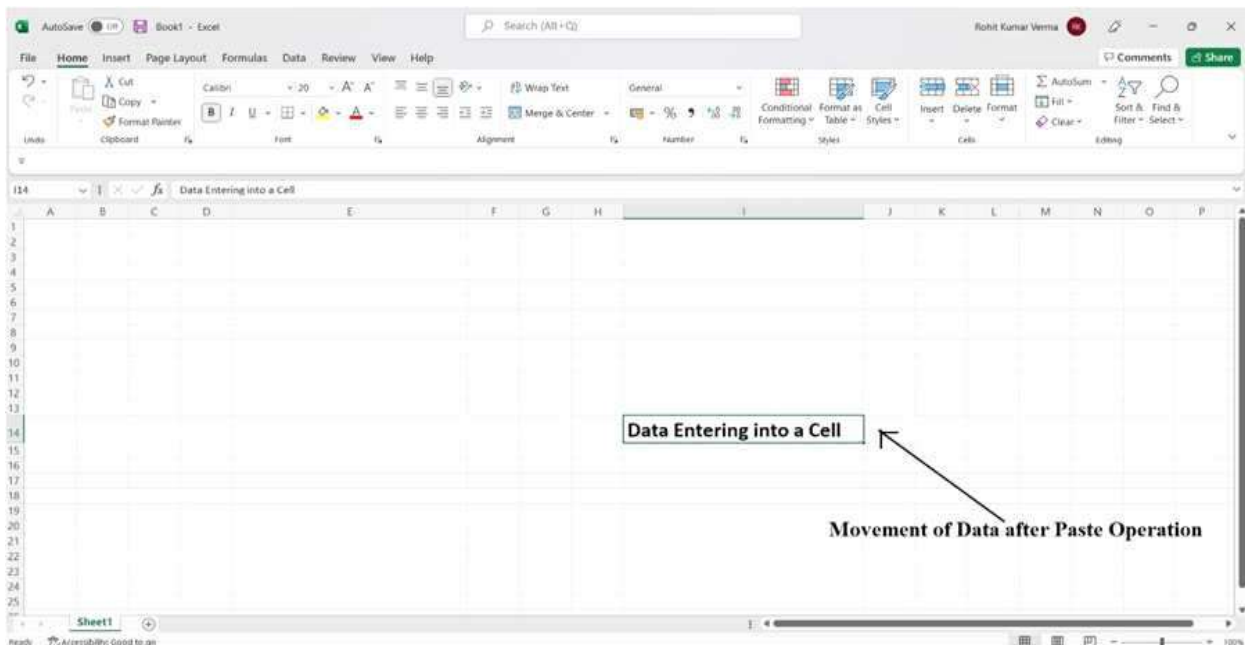
Transferring data from one place to another is known as cutting data in Excel. This can be done by moving the data to a new position in the same worksheet, another worksheet in the same workbook, a different Excel workbook, or even a different application.

Move cells by using Cut and Paste

- Choose a cell or a range of cells.
- Press Ctrl + X, choose Home > Cut, or both.



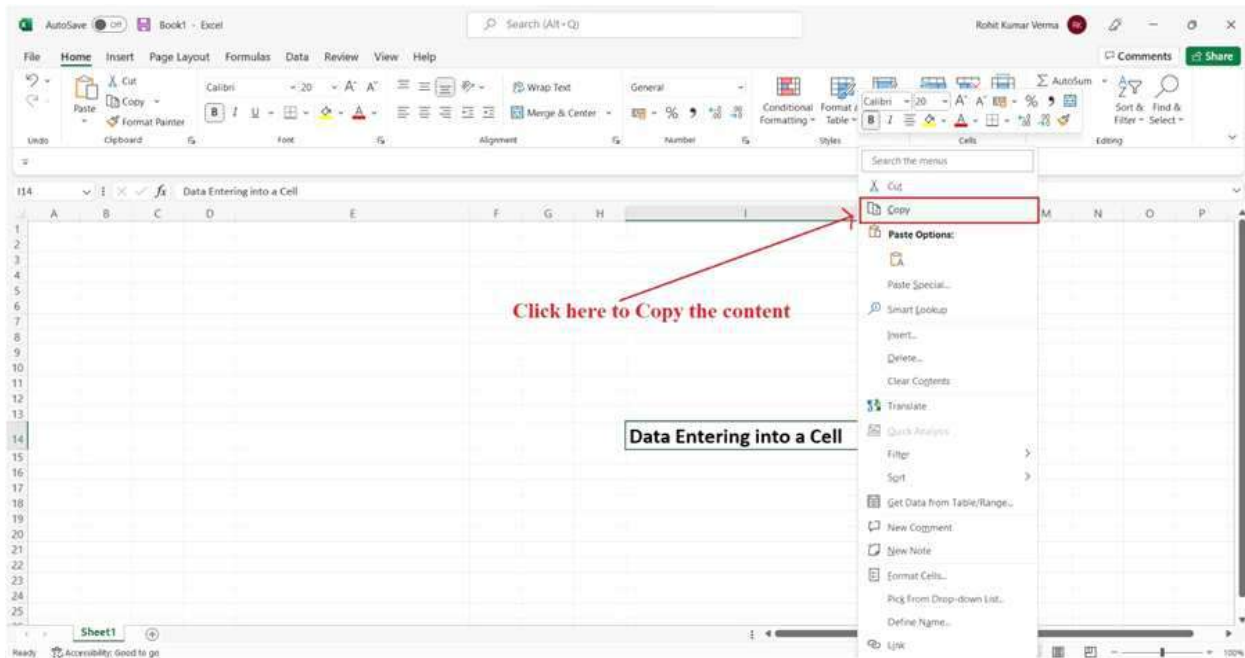
- The cell you want to move the data to is selected.
- Ctrl + V or Home > Paste are two options.



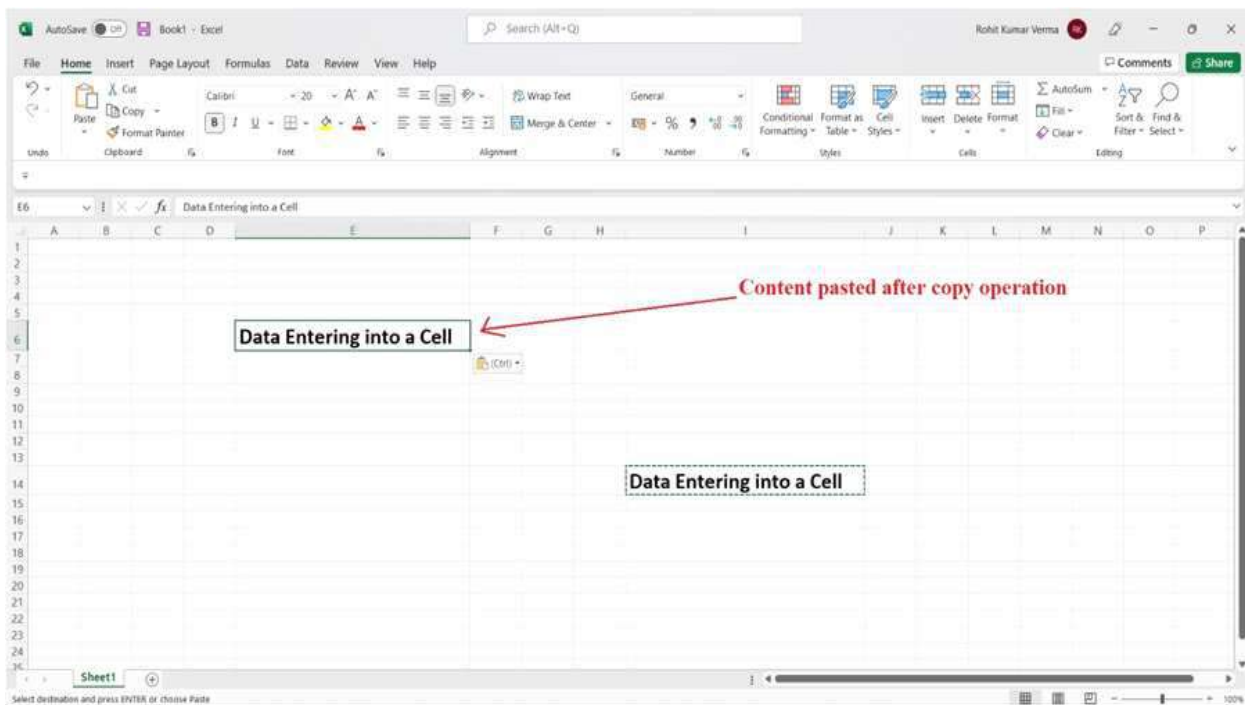
5.2.2 COPY Operation

The steps to copy Cell data from one location to another location in Excel are listed below:

- Choose a cell or a range of cells.
- Press Ctrl + C or choose Copy.



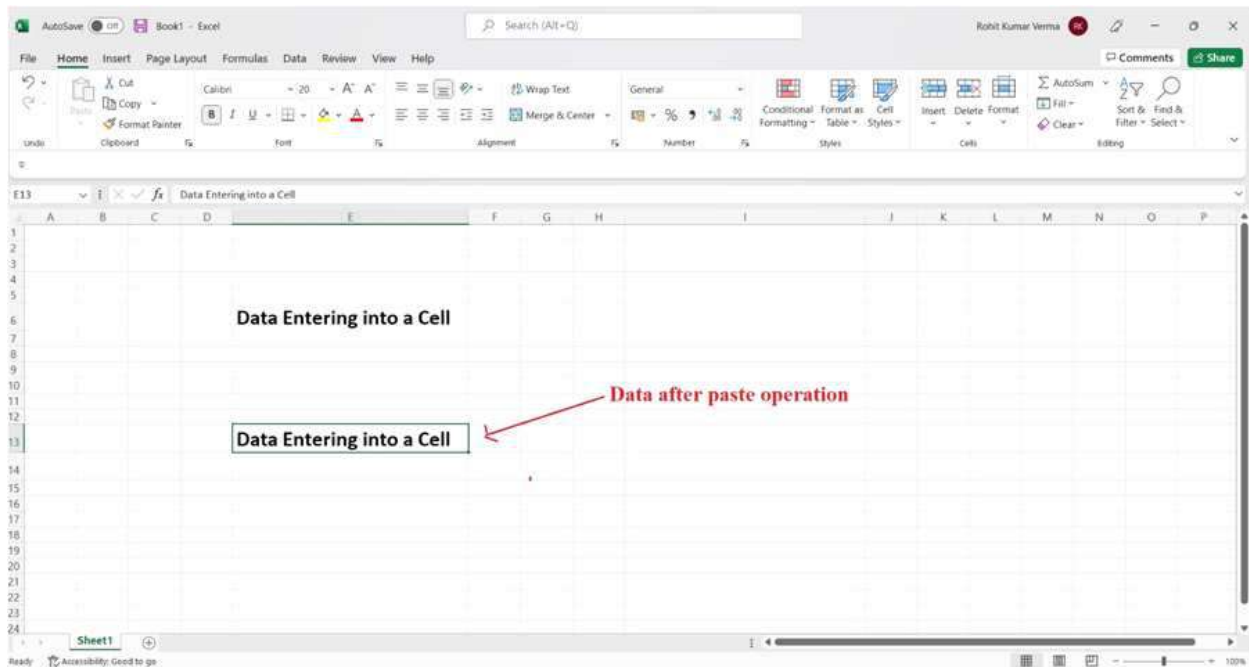
- Choose Paste or click Ctrl + V.



5.2.3 Paste Operation

The Paste option pastes everything.

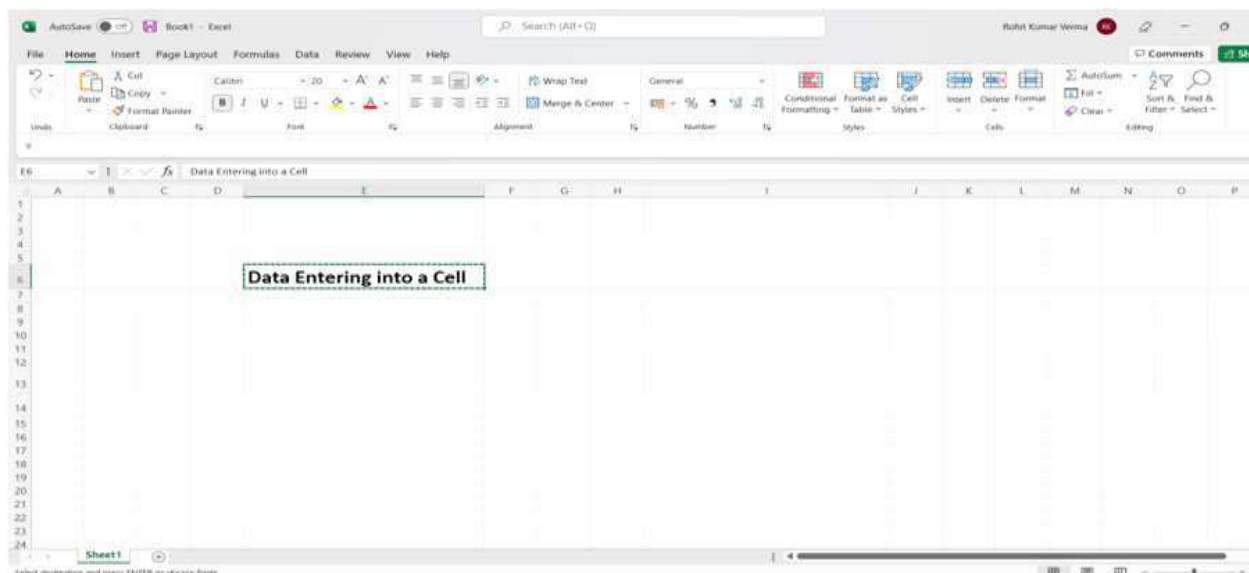
- Select cell E6, right click, and then click Copy (or press CTRL + c).
- Next, select cell E13, right click, and then click Paste under 'Paste Options:' (or press CTRL + v).



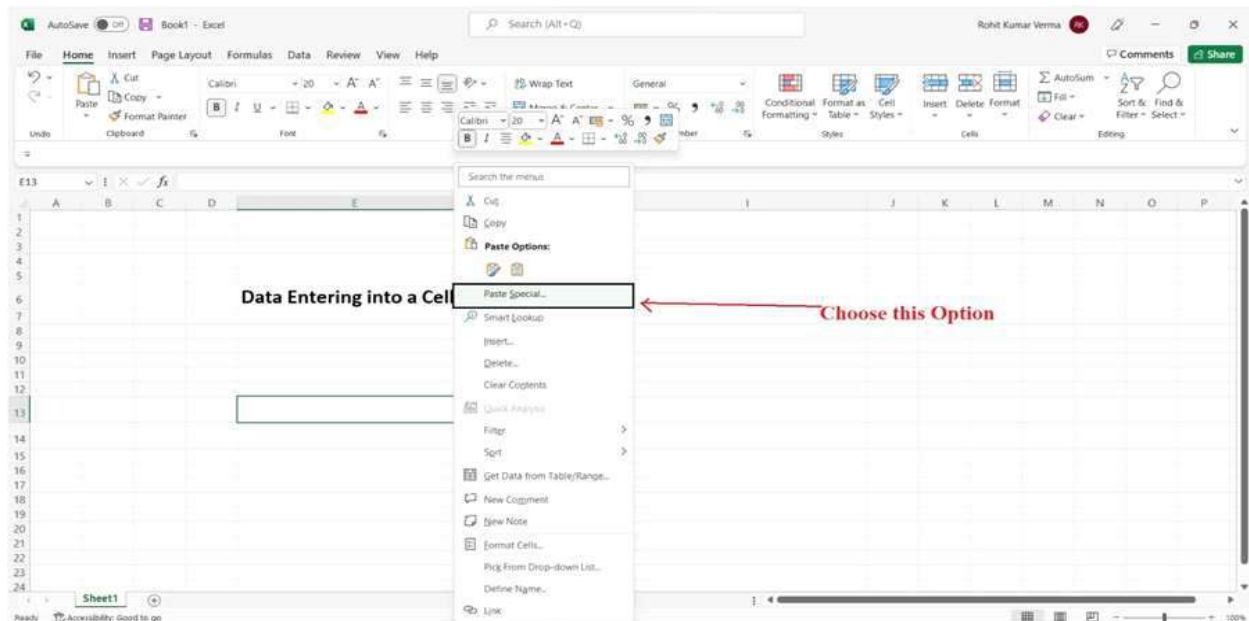
5.2.4 Paste Special

In Excel, we use the Paste Special option when we wish to paste only specific portions of the data rather than the entire set of data.

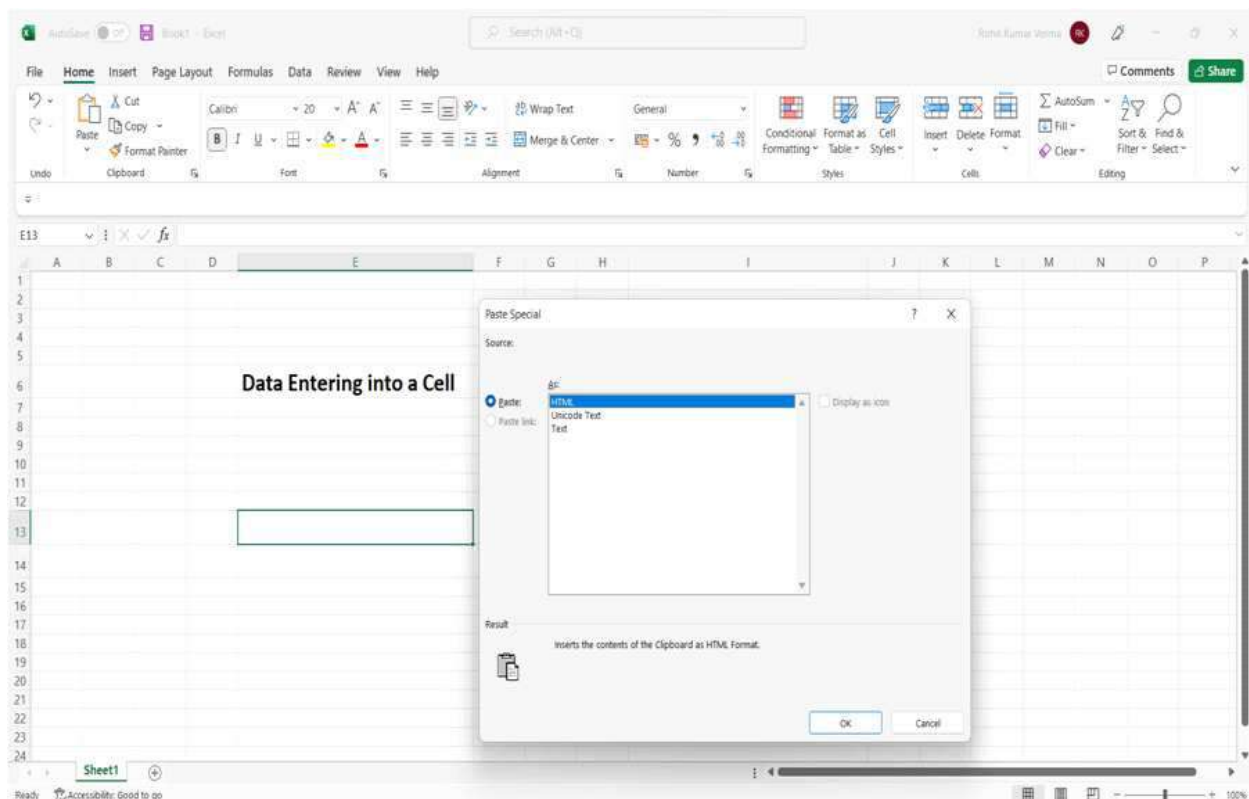
- Select cell E6, right click, and then click Copy (or press CTRL + c).



- Next, select cell E13, right click, and then click Paste Special.



- The Paste Special dialog box appears.

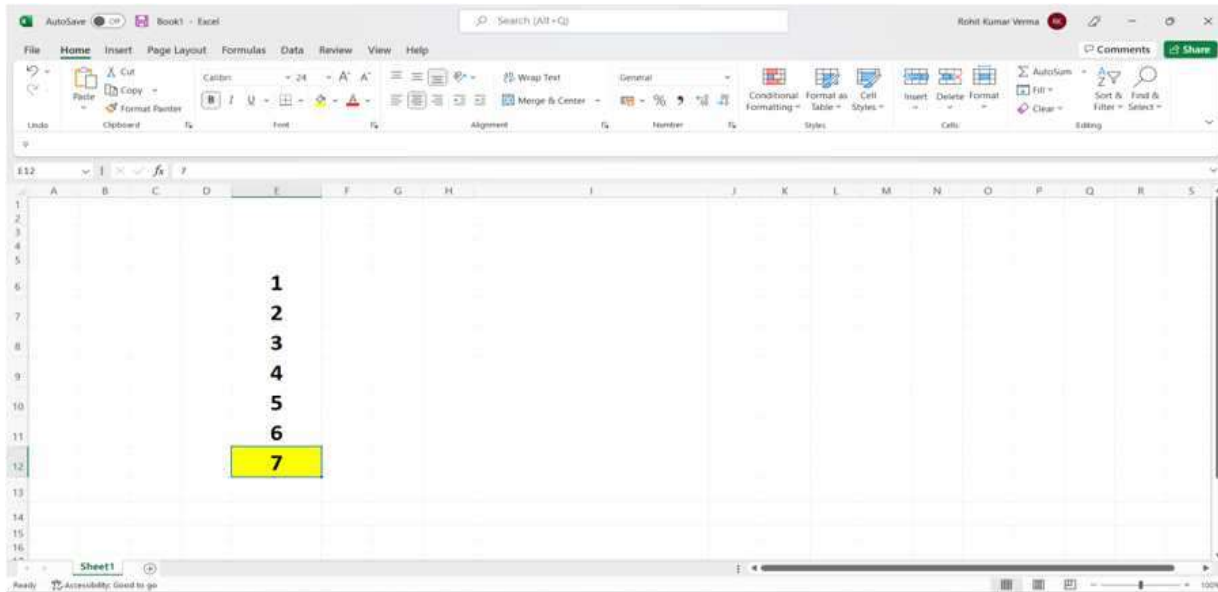


- The above-described pasting choices are also available here. Additionally, you can paste everything except borders, column widths, formulas and number formats, values and number formats, validation criteria only, and comments.

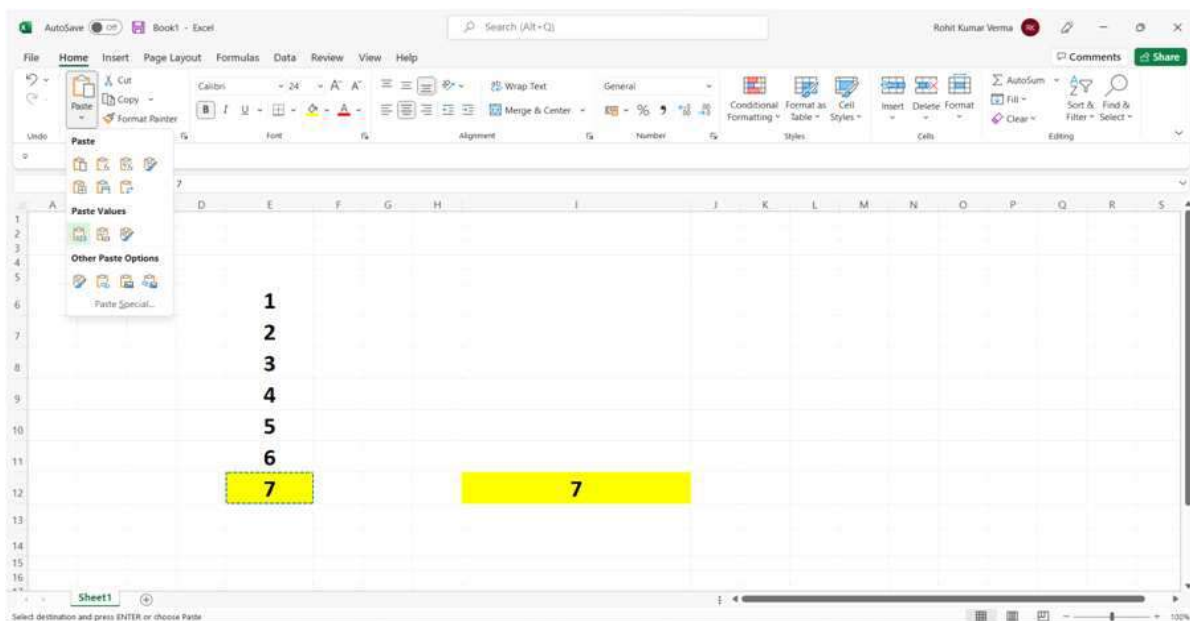
5.2.5 Paste Value

The Values option pastes the result of the formula.

- Select cell E6, right click, and then click Copy (or press CTRL + c).



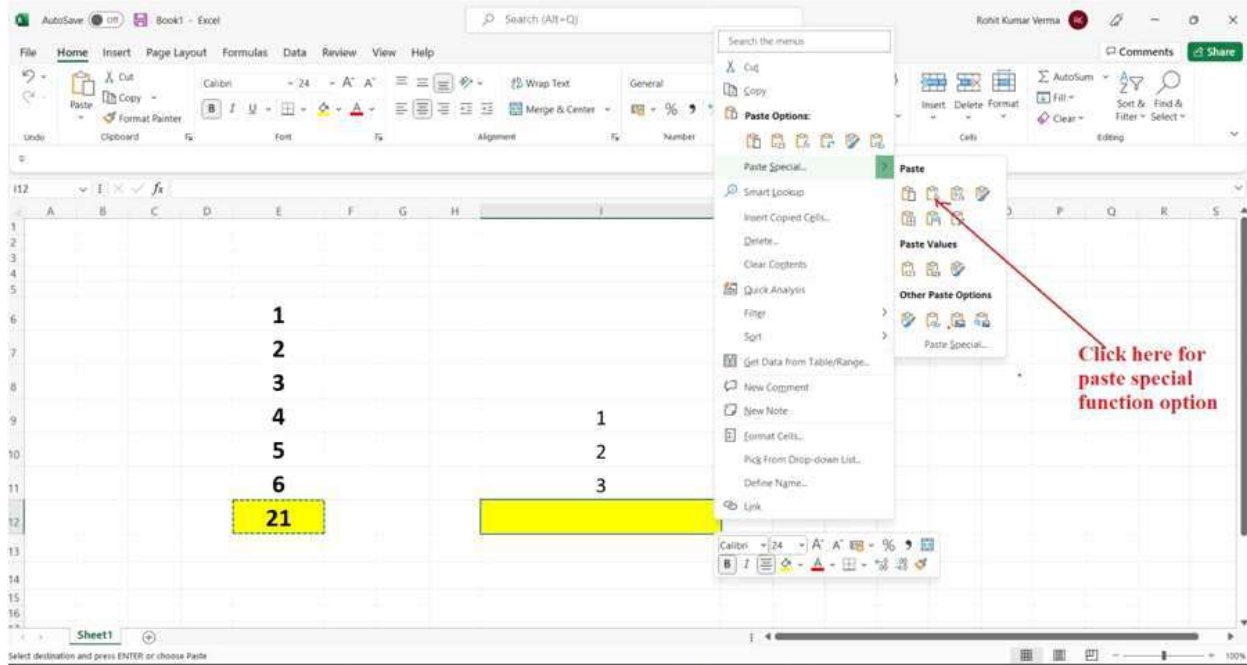
- Next, select cell E12, right click, and then click Values under 'Paste Options:'



5.2.6 Paste Formula

The Formulas option only pastes the formula.

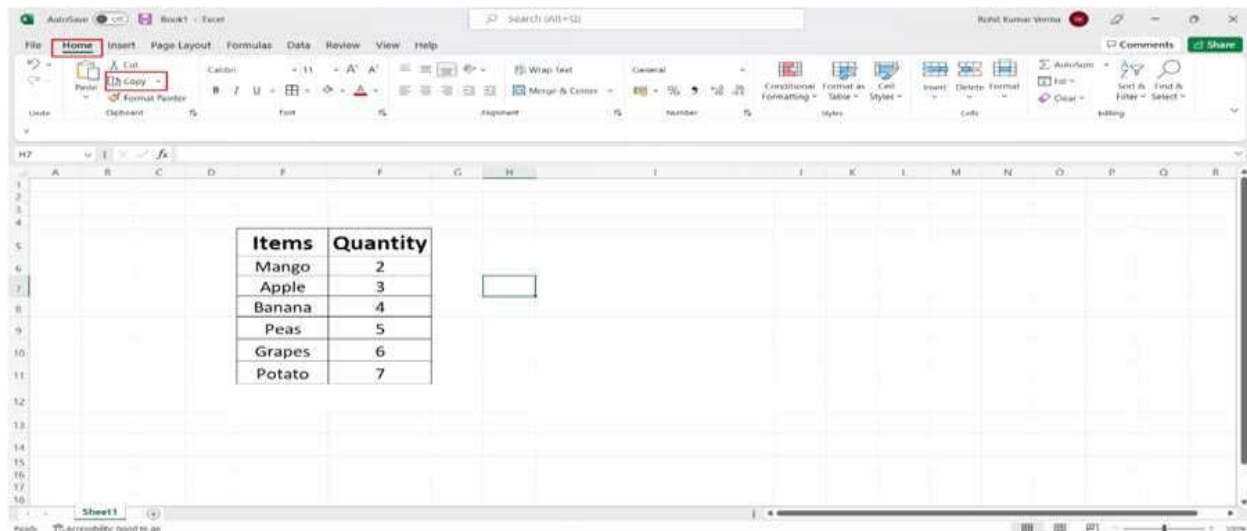
- Select cell E12, right click, and then click Copy (or press CTRL + c).
- Next, select cell F5, right click, and then click Formulas under 'Paste Options:'



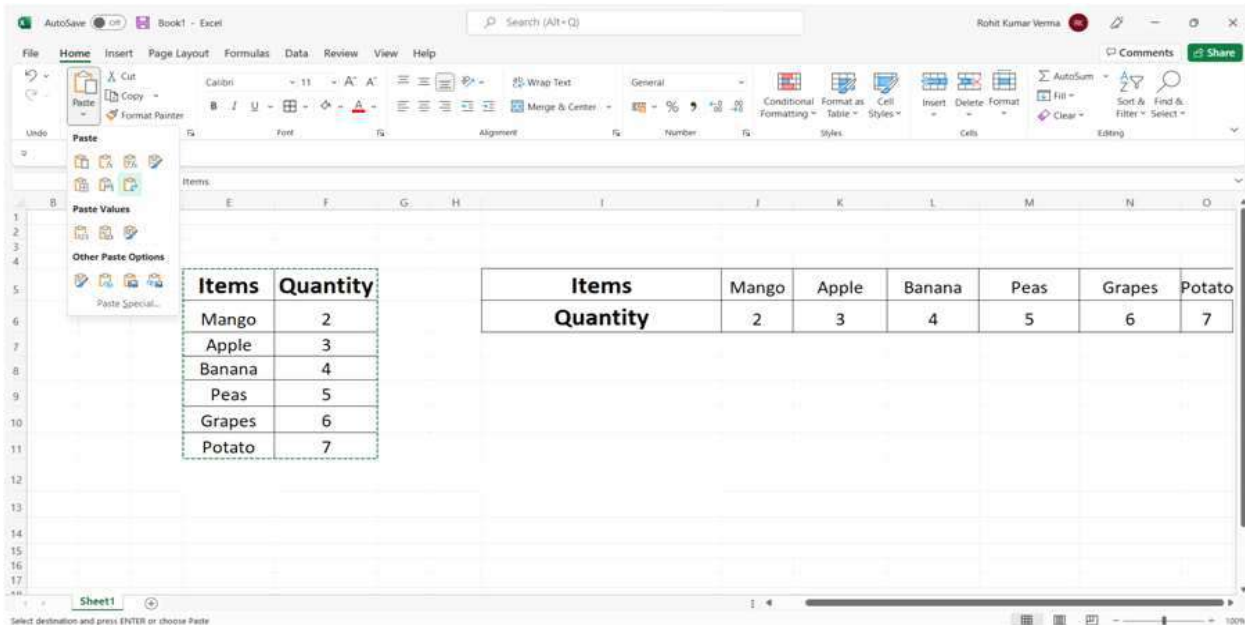
5.2.7 Paste Transpose of Data Set

The steps to paste transpose of data set in Excel are listed below:

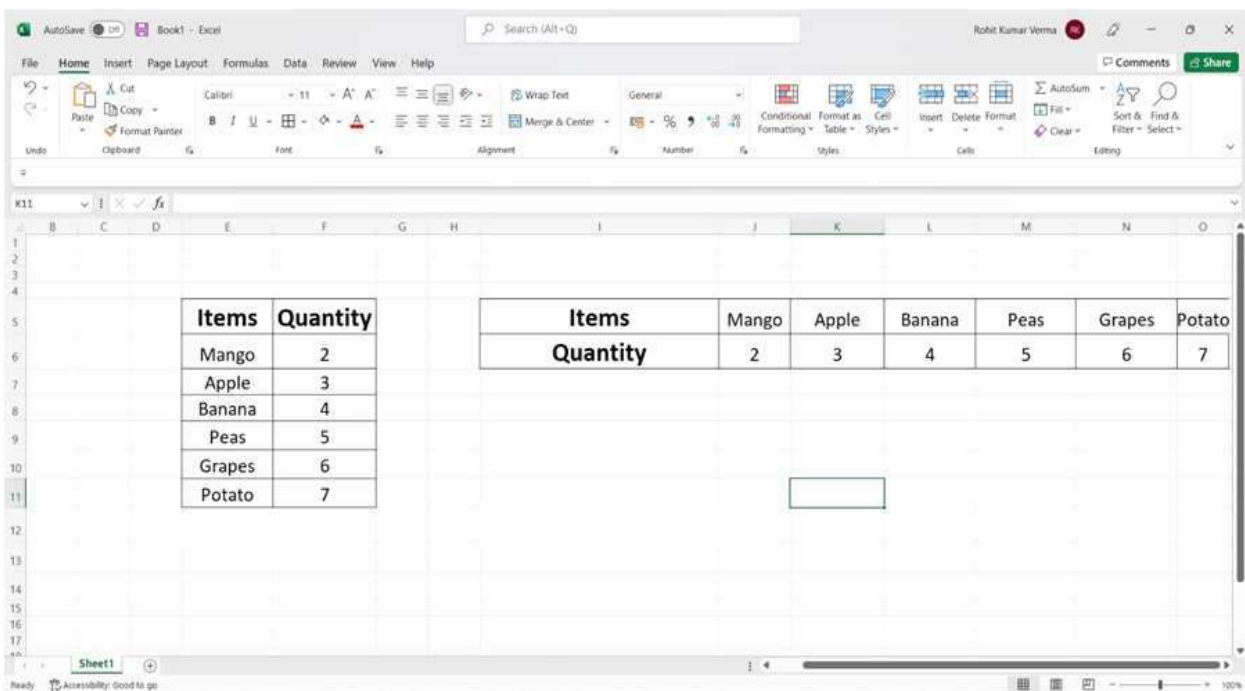
- Select the data we have to transpose. Here I've selected range B4:C9.
- Then go to Home.
- Now, select Copy from the Clipboard group of commands.



- Now, go to Cell I5 to Paste and Transpose.
- Go to the main Home tab.
- Then select Paste from the commands.
- From the drop-down of Paste option, select Transpose(T).



➤ After selecting Transpose(T), we will get our transposed data.



5.3 Summary

In this chapter, we have studied:

- The Excel toolbar is a place where you can put various commands or tools.
- The Insert menu enables you to add a variety of choices and objects to an Excel spreadsheet.
- Calculating values in a cell or a group of cells is done by a formula, which is an expression.

5.4 Answer to Self-Check exercises

- Q.1 What do you mean by Cell Formatting?
- Q.2 Explain CUT, COPY, Paste Operation briefly?
- Q.3 Explain Paste Value, Paste Function briefly?
- Q.4 How to Paste Special Operation in Excel?
- Q.5 How to enter data into a Cell?
- Q.6 How to clear data from a Cell?

5.5 Suggested Readings

- Ron Mansfield, The Compact Guide to Microsoft office; BPB Publication
- Dienes, Sheila s., Microsoft office, Professional for windows 95
- Joan Lambert, Curtis Frye, Microsoft Office Step by Step (Office 2021 and Microsoft 365), Microsoft Press Publication
- Kenneth N. Berk, Data Analysis with Microsoft Excel
- Wayne L. Winston, Microsoft Excel 2013: Data Analysis and Business Modeling
- Danielle Stein, Using Excel for business analysis: a guide to financial modelling fundamentals
- Olafusi M., Microsoft Excel and Business Data Analysis for The Busy Professional
- Rajiv Mattus: Learning Excel 97 for windows step by step BPB publication.
- Lonnie. E. Moseley & David M. Boodey: Mastering office 97.
- Micro Soft Office 97: Unleashed: Techmedia.
- John Walken Bach: Excel 97.

Chapter-6

DATA MANAGEMENT

STRUCTURE

6.0 Learning Objectives

6.1 Introduction

6.1.1 Types of Data Management

6.1.2 Importance of Data Management

6.2 Data Transformation

6.3 Arithmetic Operation

6.3.1 Addition

6.3.2 Subtraction

6.3.3 Multiplication

6.3.4 Division

6.3.5 Log Transformation

6.3.6 Exponential Transformation

6.3.7 Square and Square Root

6.3.8 Index Function

6.4 Summary

6.5 Answer to Self-Check exercises

6.6 Suggested Readings

6.0 Learning Objectives

After going through this lesson, you will be able to:

- Understand about the data management
- Understand the process of data transformation
- Understand the various functions and formulas for data transformation

6.1 Introduction

Data management is a systematic way of collecting, organizing, protecting, and storing an Establishment's data so it can be analyzed to make better business decisions. As organizations create and consume data at an unprecedented rate, data management solutions are becoming essential to making sense of the vast amounts of data. Today's leading data management software ensures that reliable and up-to-date data is always used to make decisions. The software helps with everything from data preparation to cataloging, searching, and management, so people can quickly find the data they need for analysis.

Becoming an Excel expert is more than learning functions and combining formulas. An important part of using Excel to its full potential is knowing the art of data management. This covers everything from setting up spreadsheets to data entry and good maintenance practices. In today's digital world, organizations have access to more data than ever before. This data forms the foundation for making important business decisions. Data management methods improve visibility, reliability, security, and scalability.

6.1.1 Types of Data Management

Data management plays a vital role in making essential functions easier and less time-intensive. Some of the commonly used data management techniques are:

- **Data preparation** is used to clean and transform raw data into the right shape and format for analysis, including making corrections and combining data sets.
- **Data pipelines** enable the automated transfer of data from one system to another.
- **ETLs** are three operations combined namely, Extract, Transform and Load. These are made to take the data from one system, transform it, and load it into the system's data storage.
- **Data catalogs** help deal with the metadata to create a complete picture of the data, providing brief information about its changes, locations, and quality while also making the data easy to find.
- **Data warehouses** are places to merge and unite various data sources. It provides a clear route for data analysis.
- **Data governance** sets standards, processes, and policies to preserve data security and integrity.
- **Data architecture** supplies a formal approach for creating and managing data flow.
- **Data security** defends data from unauthorized access and corruption.
- **Data modeling** documents the direction of the transfer of data through an application or organization.

6.1.2 Importance of Data Management

Data management is an important first step to employing effective data analysis at scale. Effective data management helps the person to find and access trusted data for their queries. A few benefits of effective data management include:

Visibility

An increase in the visibility of an organization's data assets makes it easier for people to easily and confidently find the right data for their analysis. Data visibility allows your establishment to be more organized and productive, allowing employees to find the necessary data they need to better do their jobs.

Reliability

Data management helps minimize potential errors by making data more reliable for decision-making. With reliable, up-to-date data, the organization can answer more effectively to market changes and customer needs.

Security

Data management protects the data of your organization and its employees from losses, thefts, and breaches with various authentication and encryption tools. Strong data security ensures that critical information is backed up and retrievable in case the primary source becomes unavailable.

Scalability

Data management allows the effective scalability of data in organizations. When actions are easy to repeat, your establishment can avoid the unnecessary costs of duplication, such as employees conducting the same operation over and over again.

6.2 Data Transformation

Data transformation means changing the data in some way to meet your data analysis needs. For example, you can filter rows, delete columns or change the data type. Each of these actions transforms the data. Data shaping or data processing is a term used to describe transformations (and aggregations) to a set or sets of data.

Transformation of data gives several benefits:

- Transformation of data to make it better organized. Transformed data is easier to apply for both humans and computers.
- formatting and validation of data in a proper way improves the data quality and protects applications from possible landmines such as NaN (Not-a-Number), null values, unexpected duplicates, incorrect indexing, and incompatible formats.
- Data transformation eases compatibility between various applications, systems, and types of data. Data used for various purposes may need to be transformed in different ways.

However, there are challenges to transforming data effectively:

- Data transformation can be expensive. Costs depend on the specific infrastructure, software and tool used to process the data. Costs may include costs associated with licensing, computing resources and hiring the necessary personnel.
- Data transformation processes can be resource intensive. Making changes to the local data store after loading or converting data before it is injected into applications can create a computational load that slows down other operations. If you use a cloud-based data warehouse, you can perform transformations after uploading, as the platform can scale to meet demand.
- Lack of expertise and carelessness can cause problems during the changeover. Data analysts without proper knowledge are less likely to notice typos or incorrect data because they do not know the exact and allowed values. For example, a person working with medical information who is unfamiliar with the relevant terms may leave unmarked disease names that should be associated with a single value, or notice typographical errors.

- Businesses can make transformations that do not meet their needs. A company can convert data to a specific format for one application only, to return the data to the previous format for another application.

6.3 Arithmetic Operation

An arithmetic operation is carried out by combining two operands with a single arithmetic operator such as '+', '-', or '*' etc. An arithmetic operator accepts two operands and performs computation on them. Most of the programming languages have a collection of these operators that may be used inside expressions and equations to carry out various kinds of sequential calculations. Everyone utilizes them in day-to-day math.

6.3.1 Addition

Addition is a mathematical process of adding numbers or integers together. The addition process is denoted by the '+' sign. It combines two or more numbers into a single term. In addition, the order does not matter. It means that the addition process is commutative. It can involve any type of number whether it be a real or complex number, fraction, or decimal.

Example: $5.12 + 3.88 = 9$

The process of adding more than two numbers, values, or terms is also known as a summation of terms and can involve n number of values.

There are several ways to perform addition operations in Excel. Below is an explanation of a few of these operations:

1. Sum a column in Excel with one click
 - Select the column alphabet for which you wish to know the sum by clicking on it.
 - You can see the total of the cells in the Excel Status bar (Fig. 6.1).
 - To calculate the sum, you can choose a subset of the cells rather than the complete column. The Excel status bar will directly show the sum of the selected cells. In addition to the total Sum of the selected cells, it also shows the number of cells selected with data (not empty cells) and the average of the data in the selected cells.

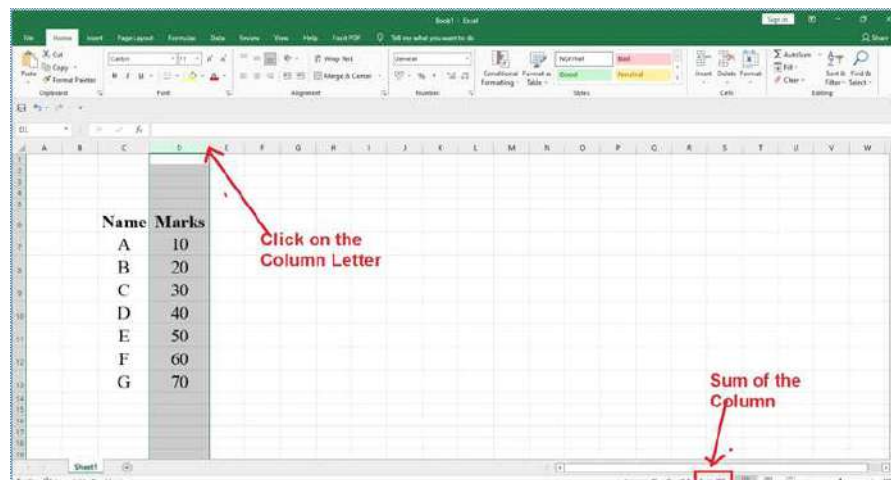
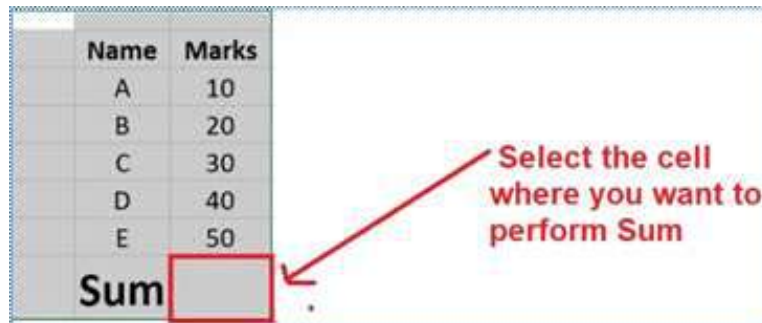


Fig. 6.1

2. Calculate the sum in Excel with AutoSum

- Select the blank cell below the column where you wish to add the numbers by clicking on it (Fig. 6.2).



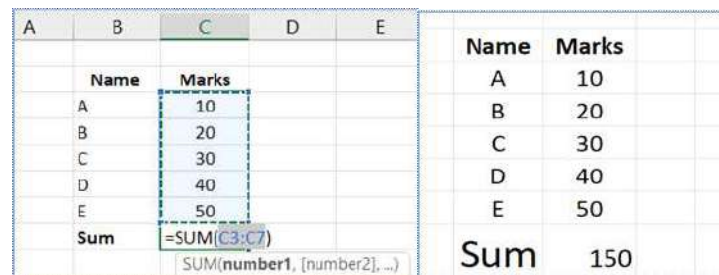
Name	Marks
A	10
B	20
C	30
D	40
E	50
Sum	

Fig. 6.2

- Select the AutoSum option in the Editing group under the Home menu.



- Excel will choose the range and insert it into the SUM function automatically. However, you can also change the range if necessary (Fig. 6.3).



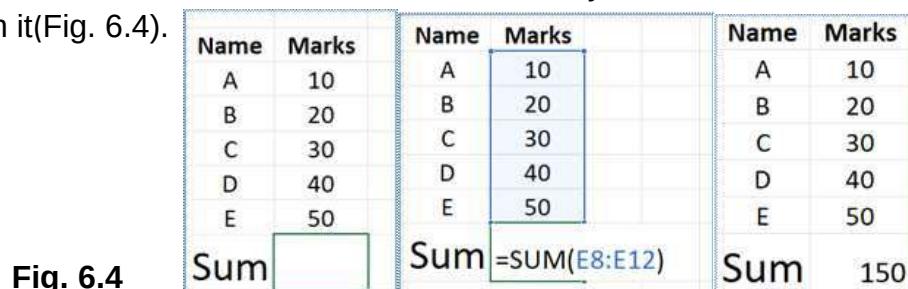
Name	Marks
A	10
B	20
C	30
D	40
E	50
Sum	=SUM(C3:C7)

Fig. 6.3

- The moment you press the Enter key on your keyboard, Excel will display the total of the values in the range you have chosen (Fig. 6.3).

3. Addition using the SUM function

- Select the blank cell below the column where you wish to add the numbers by clicking on it (Fig. 6.4).



Name	Marks
A	10
B	20
C	30
D	40
E	50
Sum	

Name	Marks
A	10
B	20
C	30
D	40
E	50
Sum	=SUM(E8:E12)

Name	Marks
A	10
B	20
C	30
D	40
E	50
Sum	150

Fig. 6.4

- Begin the formula with an equal to (=), then add SUM (). By placing the cursor on one cell and moving it to the last, you can choose the range of the cells in the arguments. You can choose to provide the number of cell references directly, separated by commas.
- As soon as you press the Enter key, Excel will calculate the sum of the values.

6.3.2 Subtraction

Subtraction gives the difference between two numbers. Subtraction is denoted by the '-' sign. It is the inverse process of addition. The addition of the term with the negative term is known as subtraction. This process is mostly used to find how many are left when some things are taken away.

Example: $16 - 7 = 9$

There is no predefined function for Subtraction in Excel. But you can still perform the subtraction using a formula. As you are already familiar with the implementation of formulas in Excel, let us take an example to perform subtraction.

- Select the cell that will display the outcome. In this example (Fig 6.5), it's cell C5. Type the "=" sign in the cell. Then choose the first cell, C4 succeeded by the "-" symbol followed by the second number in Cell C3. Once you press the enter key, the two values will be subtracted.

	A	B	C		
1					
2		Name	Marks	Name	Marks
3		A	10	A	10
4		B	20	B	20
5		Difference	= (C4 - C3)	Difference	10
6					

Fig 6.5

In case one needs to apply the subtraction operation on the set of values in a particular column, all you need to do is drag down the formula till the last cell in the result column.

6.3.3 Multiplication

Multiplication is a process of repeated addition. It is denoted by the '×' or '*' symbol. It also combines with two or more values to result in a single value. The multiplication process involves multiplicand and multiplier. The multiplication of the multiplicand and the multiplier gives the product.

Example: $6 \times 7 = 42$

Here, "6" is the multiplier, "7" is the multiplicand, and the result "42" is called the product.

To use the multiplication operation in Excel one can simply use the * (asterisk) arithmetic operator.

For example, if you type =5*20 in a cell, the cell displays the result, 100.

Multiply an entire column of numbers by a constant value

Let us say, one needs to multiply each cell in a column of seven numbers by a number that is contained in another cell. In this example (fig. 6.6), the number you want to multiply by is 2, contained in cell B2.

	A	B	C	D
1	Data	Constant	Formula	Result
2	10	2	=A2*\$B\$2.	20
3	22		=A3*\$B\$2.	44
4	33		=A4*\$B\$2.	66
5	34		=A5*\$B\$2.	68
6				

Fig. 6.6

- Type =A2*\$B\$2 in a new column in your spreadsheet (the above example uses column D for the result). Make sure to include a \$ symbol before B and before 2 in the formula, and press ENTER.
- Drag the formula down to the other cells in the column.
- The formula column is showing the formula used for the multiplication

6.3.4 Division

The division is usually denoted by '÷' or '/' and is the inverse of multiplication. It includes two values dividend and divisor, where the dividend is divided by the divisor to give a single-term value. When the dividend is greater than the divisor, the result obtained is greater than 1, or else it would be less than 1.

There is no Divide function in Excel. To perform Division in Excel use the '/' (forward slash) arithmetic operator.

For example, if you type =20/10 in a cell, the cell displays 2.

One thing to make sure of is to type an equal sign (=) in the cell before typing the numbers and the / operator; else, Excel will interpret what one types as a date. Such that, for typing 7/30, Excel may display 30-Jul in the cell. Or, for 12/36, Excel will first convert that value to 12/1/1936 and display 1-Dec in the cell.

6.3.5 Log Transformation

It transforms the value x to $\text{Log}(x)$. The log function is mostly used in data analysis to transform the skewed dataset distribution to less skewed. This helps in finding a pattern in the dataset. Finding a pattern in the dataset help in making an effective decision, which is the end task in the data analysis.

There are multiple ways to implement the Log transformation in Excel. Two of those are discussed in this chapter.

1. Use of LOG10 function

This function returns the Logarithm value of a number and the base will be always 10. The function to transform the base to another number is explained later.

- Select the cell for function application, E2 in our example (Fig 6.7).
- Type the formula '=LOG10(D2)' in the cell and hit enter. You can apply the formula for multiple cells in the column by dragging the rectangular symbol in the bottom right corner of the selected cell.

fx =LOG10(D2)		
D	E	F
Number	Log	
20	=LOG10(D2)	
44	LOG10(number)	
66		
68		

Fig. 6.7

2. Use of LOG function

This function is similar to the LOG10 function explained above. There is an option to add a different base in this function. When we do not enter any value for the base, Excel will assume the base is 10. The value of the base cannot be – negative, 1, or 0.

fx =LOG(D2,2)		
D	E	F
Number	Log	
20	=LOG(D2,2)	
44	LOG(number, [base])	
66		
68		

Fig 6.8

The steps to implement the LOG function are the same as the LOG10 function. Just change the function to 'LOG(number,[base])', where number is the cell number, D2 in this case, and [base] is the base value of the Log function. In this example base is 2.

6.3.6 Exponential Transformation

Excel provides users with two methods to calculate exponents. The first option is to use the exponent character, the “^” symbol. The second option is to use a built-in POWER() function. It takes two values, number, and power.

The use of the exponent character '^' is demonstrated in fig 6.9. For multiple-cell implementation of the formula, drag the formula down to the other cells in the column.

D	E
Number	Exponent
20	=D2^2
44	
66	
68	

Fig 6.9

The use of the POWER function is demonstrated in fig 6.10. The way to use this function is the same as the other functions. Start with the '=' and type POWER followed by number and power. The number value can be replaced by the cell reference.

D	E	F
Number	Exponent	
20	=POWER(D2,2)	
44		
66		
68		

Fig 6.10

For multiple-cell implementation of the function, drag the function down to the other cells in the column.

6.3.7 Square and Square Root

Squaring a number:

Excel provides two ways in which one can square a number:

1. Using a Formula
2. Using a Function

Both ways are very quick and quite easy to implement.

1. Using a Formula

The Square of a number simply means multiplying a number by itself. In other words, raising it to the power of 2.

To square a number one can write the formula in two different ways:

- Using the multiplication operator to multiply it by itself.
- Using the caret operator to raise the number to the power of 2.

2. Using a Function

As discussed in the previous topic the POWER function is used to find the exponential of a number. So, the POWER function with power value 2 can be used for squaring a number.

Square Root

Excel provides the user with the SQRT function to calculate the square root of any number. The syntax for the function is:

SQRT(*number*)

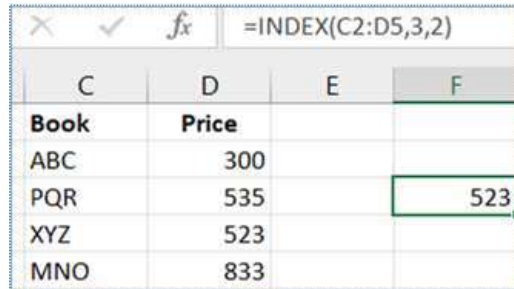
Here the *number* is the number or cell reference which contains the number for which you want to find the square root. For example, SQRT(400) will give the user value of 20.

6.3.8 INDEX Function

The INDEX function returns a value or the reference to a value from within a table or range. In case one needs to return the value of a specified cell or array of cells then this INDEX function is used. It gives the value of an element in a given location, selected by the row and column number indexes. Syntax is:

INDEX(*array*, *row_num*, [*column_num*])

- array –This value is required. It is a range of cells or an array constant. This gives the range of the cells for indexing.
- row_num– This is the row position in the reference or array.
- col_num - [optional] This is the column position in the reference or array.



	C	D	E	F
Book		Price		
ABC		300		
PQR		535		523
XYZ		523		
MNO		833		

Fig 6.11

Fig 6.11 demonstrates the use of the INDEX function. Here it is used to find the Price of the book named XYZ which is 523. You can check the values of parameters in the formula bar. The range of cells is C2 to D5. Indexing starts at ABC so the index of XYZ is 3. Similarly for the column value. This concept can be implemented in a large dataset.

6.4 Summary

In this chapter, we have studied:

- Data management, its types and benefits of data management.
- Data transformation and different types of transformation.
- Implementation of different Arithmetic operators in Excel.
- Data transformation functions in Excel such as LOG, SQRT, POWER and INDEX, etc.

6.5 Answer to Self-Check exercises

- Q.1 Define data management and explain its features.
- Q.2 What are the various arithmetic operators of Microsoft Excel?
- Q.3 Explain the different ways of squaring a number in Excel.
- Q.4 What is Log transformation and how we can calculate the Log of a number with different base values?

6.6 Suggested Readings

- Ron Mansfield, The Compact Guide to Microsoft office; BPB Publication
- Dienes, Sheila s., Microsoft office, Professional for windows 95
- Joan Lambert, Curtis Frye, Microsoft Office Step by Step (Office 2021 and Microsoft 365), Microsoft Press Publication
- Kenneth N. Berk, Data Analysis with Microsoft Excel
- Wayne L. Winston, Microsoft Excel 2013: Data Analysis and Business Modeling

- Danielle Stein, Using Excel for business analysis: a guide to financial modelling fundamentals
- Olafusi M., Microsoft Excel and Business Data Analysis for The Busy Professional
- Rajiv Mattus: Learning Excel 97 for windows step by step BPB publication.
- Lonnie. E. Moseley & David M. Boodey: Mastering office 97.
- Micro Soft Office 97: Unleashed: Techmedia.
- John Walken Bach: Excel 97.

Chapter-7

CONDITIONAL STATEMENT

STRUCTURE

- 7.0 Learning Objectives
- 7.1 Introduction
- 7.2 IF Statement
 - 7.2.1 Use of IF statement
 - 7.2.2 Nested IF Statement
 - 7.2.3 SUM, SUMIF, SUMIFS Functions
 - 7.2.4 COUNT, COUNTIF, COUNTIFS Functions
 - 7.2.5 AVERAGEIF Function
 - 7.2.6 MAX and MIN Function
 - 7.2.7 CONCATENATE Function
- 7.3 Summary
- 7.4 Answer to Self-Check exercises
- 7.5 Suggested Readings

Learning Objectives

After going through this lesson, you will be able to:

- Understand the conditional Statements
- Understand the use of IF and its variants
- Understand the use of CONCATENATE, AVERAGEIF

7.1 Introduction

There are many scenarios in real life where you need to work on some conditions. So, while dealing with real-life data in excel one can use the functionality of condition statements or predefined functions to solve problems. Excel provides a wide range of functions for condition checking. In this chapter, the most common Conditional statements and functions are discussed with examples.

7.2 IF Statement

The Excel IF Statement is used to test a given condition and returns one value for a TRUE result and another value for a FALSE result. For example, if marks are more than 40%, then return a "PASS" as the result – Otherwise, return a "FAIL" for the result. One can implement the IF function in different ways, such as to evaluate a single function, or includes several IF functions in a single formula. The use of Multiple IF statements in Excel is known as nested IF statements.

7.2.1 Use of IF statement

The IF function is commonly implemented to evaluate and analyze data by evaluating specific conditions.

This function can be used to evaluate text, values, and even errors. It's not just about checking if one thing is equal to the other and returning a result. According to our standards, we can also use arithmetic operators and perform additional calculations. One can also combine multiple IF functions together for multiple comparisons.

The syntax for the IF function:

=IF(logical_test, value_if_true, value_if_false)

The function uses the following arguments:

- Logical_test (required argument) – This is the condition that is to be tested and evaluated as either TRUE or FALSE.
- Value_if_true (optional argument) – It is the value to be returned if the logical_test evaluates to TRUE.
- Value_if_false (optional argument) – It is the value to be returned if the logical_test evaluates to FALSE.

When using the IF function to construct a test, we can use the following logical operators:

= (equal to)

> (greater than)

>= (greater than or equal to)

< (less than)

<= (less than or equal to)

<> (not equal to)

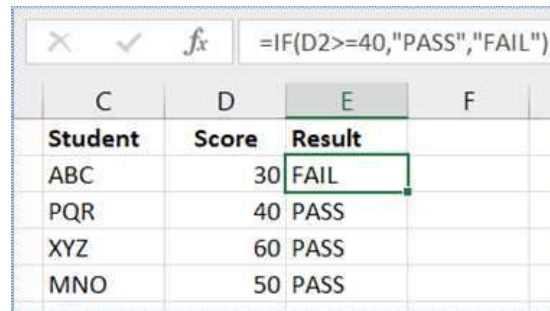
Example: In Fig 7.1 three columns are present namely, Student, Score, and Result. The Result column provides the values PASS or FAIL after evaluating the value in column D (Score).

The IF function is used here to evaluate the score so that the result will be PASS for a Score greater than or equal to 40 and FAIL otherwise.

C	D	E	F	G
Student	Score	Result		
ABC	30	=IF(D2>=40,"PASS","FAIL")		
PQR	40	IF(logical_test, [value_if_true]		
XYZ	60			
MNO	50			

Fig 7.1

To implement the condition to the whole E column just drag the formula downwards. The final result is shown in fig 7.2 after implementation.



The formula bar shows: `=IF(D2>=40,"PASS","FAIL")`

	C	D	E	F
Student	Score	Result		
ABC	30	FAIL		
PQR	40	PASS		
XYZ	60	PASS		
MNO	50	PASS		

Fig 7.2

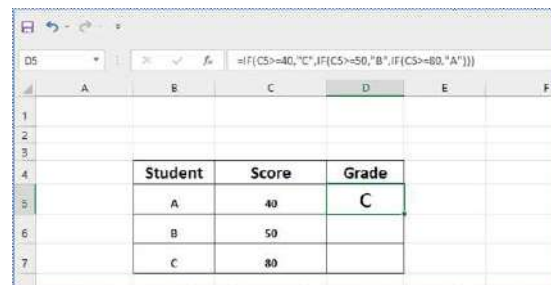
Note that The IF function is not case-sensitive.

7.2.2 Nested IF Statement

Excel provides the functionality of Nested IF. A Nested IF is referred to a case in which at least one IF condition is implemented inside another IF for evaluating the condition in more detail and providing a more precise result. It helps narrow down the result and helps in evaluating more conditions within one IF.

Example: Here we are Grading the students based on their scores in the examination. See Fig 7.3.

The Grade (column D) gives the final value by evaluating the Score (Column C). The nested IF is used to evaluate the score and divide the final result into three grades, A, B and C. Fig 7.3 shows the result for student A.

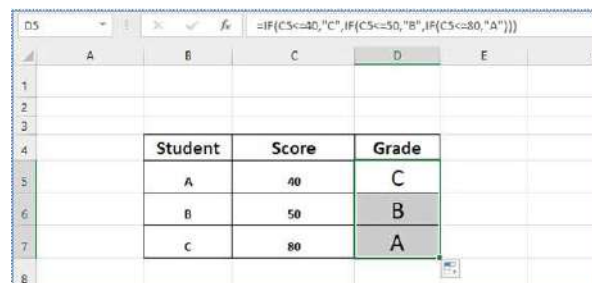


The formula bar shows: `=IF(C5>=40,"C",IF(C5>=50,"B",IF(C5>=80,"A")))`

	A	B	C	D	E	F
1						
2						
3						
4		Student	Score	Grade		
5		A	40	C		
6		B	50			
7		C	80			

Fig 7.3

After implementing the nested IF to the rest of the cells in the table the final result will look like Fig 7.4.



The formula bar shows: `=IF(C5<=40,"C",IF(C5<=50,"B",IF(C5<=80,"A")))`

	A	B	C	D	E	F
1						
2						
3						
4		Student	Score	Grade		
5		A	40	C		
6		B	50	B		
7		C	80	A		

Fig 7.4

While implementing the nested IF, one IF needs to be carefully nested inside the other so that the proper logic gets evaluated.

7.2.3 SUM, SUMIF, SUMIFS Functions

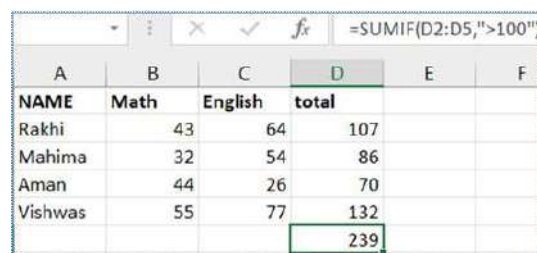
SUM

We can use the SUM function for the addition of a range of cells in MS Excel. SUM function can either be used by directly writing in the cell starting with equal to sign or by clicking on the AutoSum command from the 'Editing' group under the 'HOME' tab on the Ribbon.

The example of the Sum function is already discussed in chapter 6.

SUMIF

SUMIF function is also used to add the range of cells based on one condition or criteria. In the below example (fig 7.5), we are adding the value of the D column only when the cell value is more than 100.



The screenshot shows the Excel formula bar with the formula `=SUMIF(D2:D5,">100")`. Below it is a table with columns A to F. Column A contains names, B contains Math marks, C contains English marks, and D contains total marks. The table data is as follows:

A	B	C	D	E	F
NAME	Math	English	total		
Rakhi	43	64	107		
Mahima	32	54	86		
Aman	44	26	70		
Vishwas	55	77	132		
			239		

Fig 7.5

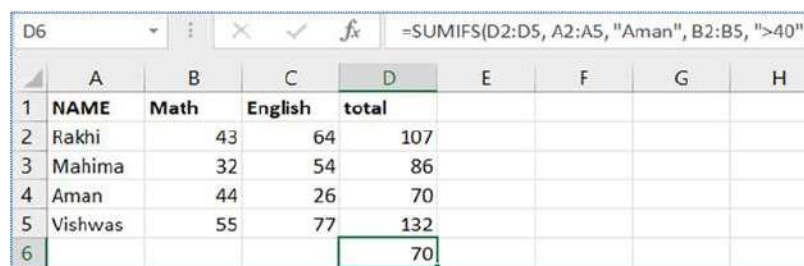
The condition/criteria in the formula are written in the double quote as the second parameter. Instead of passing the second parameter as less than or equal to, greater than, one can also put a condition based on other cell range.

For example, if one wants to add the D column value only when its corresponding A column value is "Aman". To do this, the formula will be written as `'=SUMIF(A2:A5, "Aman", D2:D5)'`. This means adding the value from D2 to D5 only when A2 to A5 value contains "Aman". In this case, the result will be 70. Here, A2:A5 is the range for criteria and D2:D5 is the range for Sum.

SUMIFS

While using the Sum function on a range of cells, if one needs to use multiple criteria, SUMIFS function can be used.

In the below example (fig 7.6), we want to sum the value of the D column only if the A column value is "Aman" and the B column value is greater than 40. This criteria will be written as `'=SUMIFS(D2:D5, A2:A5, "Aman", B2:B5, ">40")'`.



The screenshot shows the Excel formula bar with the formula `=SUMIFS(D2:D5, A2:A5, "Aman", B2:B5, ">40")`. Below it is a table with columns A to H. Column A contains names, B contains Math marks, C contains English marks, and D contains total marks. The table data is as follows:

	A	B	C	D	E	F	G	H
1	NAME	Math	English	total				
2	Rakhi	43	64	107				
3	Mahima	32	54	86				
4	Aman	44	26	70				
5	Vishwas	55	77	132				
6				70				

Fig 7.6

Here

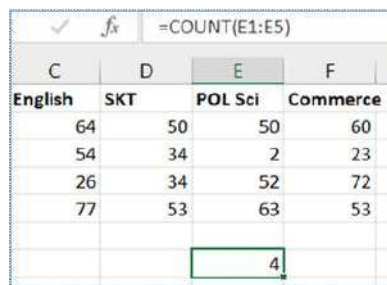
- D2 to D5 is the sum range
- A2:A5 is criteria 1 range
- “Aman” is the criteria value of A2:A5
- B2:B5 is the criteria 2 range
- “>40” is the criteria value of B2:B5

7.2.4 COUNT, COUNTIF, COUNTIFS Functions

COUNT Function

The count() function is used when one needs to count the number of cells. A range of cells is defined and the count function gives the total number of cells in that range. The result of count() does not include cells containing string data.

Let us take an example, Fig 7.7.



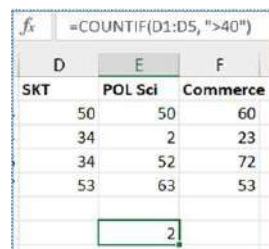
C	D	E	F
English	SKT	POL Sci	Commerce
64	50	50	60
54	34	2	23
26	34	52	72
77	53	63	53
		4	

Fig 7.7

Here cell E7 shows the result of Count() function for the range of cells E1 to E5. But here, the cell E5 contains the string value which gives the final result 4. As we have already discussed that this function does not count the cells with string values.

COUNTIF Function

When one needs to count the cells based on one criteria only, then the Countif function is used. The criteria can be less than, greater than, equal to, etc. The criteria can be used either for numerical or string values.



D	E	F
SKT	POL Sci	Commerce
50	50	60
34	2	23
34	52	72
53	63	53
	2	

Fig 7.8

Look at Fig 7.8. Here we are counting the number of cells containing a numerical value greater than 40. Thus, the final result we get is 2. The formula used in the example was =COUNTIF(D1:D5, ">40").

COUNTIFS Function

This function is used in the case where one needs to count the cells in a range based on more than one criteria. The multiple criteria are separated by a comma.

50\")'. The spreadsheet has columns A through G. Column A is 'NAME', B is 'Math', C is 'English', D is 'SKT', E is 'POL Sci', and F is 'Commerce'. The data rows are: Rakhi (Math: 43, English: 64, SKT: 50, POL Sci: 50, Commerce: 60), Mahima (Math: 32, English: 54, SKT: 34, POL Sci: 2, Commerce: 23), Aman (Math: 44, English: 26, SKT: 34, POL Sci: 52, Commerce: 72), and Vishwas (Math: 55, English: 77, SKT: 53, POL Sci: 63, Commerce: 53). The result of the formula, '1', is displayed in cell G1."/>

A	B	C	D	E	F	G
NAME	Math	English	SKT	POL Sci	Commerce	
Rakhi	43	64	50	50	60	
Mahima	32	54	34	2	23	
Aman	44	26	34	52	72	
Vishwas	55	77	53	63	53	
				1		

Fig 7.9

Let us say, in Fig 7.9 one needs to count the number of cells in column A for a value equal to AMAN and column E for values greater than 50. The formula will be, =COUNTIFS(A1:A5, "Aman", E1:E5, ">50").

- The count of cells in A1:A5 will give the value 1.
- The count of cells in column E will give a value of 2.

Only those cells are counted for which all criteria are met. In this case, only one value in column E satisfies the first criteria. So, the final result we get is 1.

7.2.5 AVERAGEIF Function

The AVERAGEIF function is a predefined function in Excel, which helps in calculating the average of a range of cells based on criteria. Criteria can be true or false conditions.

It is typed =AVERAGEIF and has three parts:=AVERAGEIF(**range**, **criteria**, **[average_range]**)

The condition in function syntax is referred to as criteria, which can check things like:

- If a number is greater than another number
- If a number is smaller than another number
- If a number or text is equal to something

The [average_range] is the range of cells where the function calculates the average. This value is optional. If not specified, the average will be calculated for the same range as the condition.

AVERAGEIF will automatically ignore the empty cells in the defined range, even when the criteria match. AVERAGEIF returns the value #DIV/0! if no cells in the range meet the criteria. While using the AVERAGEIF function one can only apply a single criteria. To apply multiple criteria, one can use the AVERAGEIFS function.

7.2.6 MAX and MIN Function

The MAX function is a predefined function in Excel, which helps in finding the highest number in a range. It does not take into account text or logical values and only works for the cells with numerical values.

It is typed as '=MAX'.

The MIN function is a predefined function in Excel, which helps in finding the lowest number in a range. It is the opposite of the MAX function. Same to the MAX function, it also ignores the cells with text values and only works on the cells with numerical values.

fx =MAX(D2:D5)		fx =MIN(D2:D5)	
D	E	D	E
SKT	POL Sci	SKT	POL Sci
50	50	50	50
34	2	34	2
34	52	34	52
53	63	53	63
53		34	

Fig 7.10

Fig 7.10 is showing examples of MAX and MIN functions.

7.2.7 CONCATENATE Function

The meaning of 'Concatenate' is to join or combine. The CONCATENATE function in Excel is used to combine the text values from two or more cells. There is one more way to combine the cells in Excel, i.e., merge cells. The main difference between these two methods is that merge cells combine the two cells 'physically' and shows the result in one larger cell spanning across multiple rows or columns. On the other hand, CONCATENATE function combines the text data value from two or more cell and show the result in another cell.

The Excel CONCATENATE function can be used to join up to 30 text items. It gives the result as a text.

The syntax of the function is as follows:

Syntax: **=CONCATENATE (text1, text2, [text3], ...)**

For example, the simple formula to CONCATENATE the values of two cells A3 and B3 are as follows:

=CONCATENATE(A3,B3)

fx =CONCATENATE(A3,B3)					
A	B	C	D	E	F
First name	Surname	Name			
Aman	Sharma	AmanSharma			

Fig. 7.11

See figure 7.11. The values will be combined without any delimiters. To separate the values with space, one can use " ". In that case, the formula will be like this,

=CONCATENATE(A3, " ", B3)

This will produce the result 'Aman Sharma'.

Concatenating a Text String and a Cell Value

There are more things that the Excel CONCATENATE function can do. One can use the function to join a cell value with a string as well. For example:

=CONCATENATE(A3, " ", B3, " Delhi")

The example is shown in fig 7.12.

	A	B	C	D	E	F	G
1	First name	Surname	Name				
2							
3	I	Love	I Love Delhi				

Fig 7.12

When using CONCATENATE function, there should be at least one "text" argument for it to work. In case of invalid arguments, the formula returns a #VALUE! Error. The result of the CONCATENATE function is always a text string.

7.3 Summary

In this chapter, we have studied:

- Conditional statements, IF statements, and their types
- Implementation of conditioning while adding cells.
- Implementation of conditioning while counting the cells.
- Some useful functions like AVERAGEIF and CONCATENATE.

7.4 Answer to Self-Check exercises

- Q.1 Define conditional statements and explain them with examples.
- Q.2 Discuss SUM, SUMIF and SUMIFS?
- Q.3 How to count the cell with specified conditions?
- Q.4 Explain the use of CONCATENATE function in Excel and how is it different from the 'merge cell'.

7.5 Suggested Readings

- Ron Mansfield, The Compact Guide to Microsoft office; BPB Publication
- Dienes, Sheila s., Microsoft office, Professional for windows 95

- Joan Lambert, Curtis Frye, Microsoft Office Step by Step (Office 2021 and Microsoft 375), Microsoft Press Publication
- Kenneth N. Berk, Data Analysis with Microsoft Excel
- Wayne L. Winston, Microsoft Excel 2013: Data Analysis and Business Modeling
- Danielle Stein, Using Excel for business analysis: a guide to financial modelling fundamentals
- Olafusi M., Microsoft Excel and Business Data Analysis for The Busy Professional
- Rajiv Mattus: Learning Excel 97 for windows step by step BPB publication.
- Lonnie. E. Moseley & David M. Boodey: Mastering office 97.
- Micro Soft Office 97: Unleashed: Techmedia.
- John Walken Bach: Excel 97.

Chapter-8

SORTING AND FILTERING OF DATA

STRUCTURE

- 8.0 Learning Objectives
- 8.1 Introduction
- 8.2 Sort
- 8.3 Filter
- 8.4 LOOKUP Function
- 8.5 Summary
- 8.6 Answer to Self-Check exercises
- 8.7 Suggested Readings

8.0 Learning Objectives

After going through this lesson, you will be able to:

- learn to sort the data in excel
- learn the sorting of data in different ways
- understand the concept of sorting and filtering
- learn the method to filter the text and numerical data
- understand the working of the Lookup function

8.1 Introduction

Excel provides various tools and functions for data analysis and management. Whenever we are dealing with a large volume of data, we need to apply some operations on the data to easily process the information. One of the most commonly applied such operations is Sort. Sort is basically used to arrange the data in a particular order like ascending or descending. Using sort we can arrange the text data values in alphabetical order. Another usually employed function is Filter. The filter is normally used to find the selected information from the heap of the data. To narrow down the search area and get the information more easily different filters are applied to the data. Excel has many predefined filters for the different data types. In addition to these two functions, in this chapter, we have also discussed the Lookup function.

8.2 Sort

There are many cases when you need to organize your data in a certain way which will help in better data analysis. This type of data organizing functionality is provided in the Excel with SORT function. This function allows the user to:

- Sort data A to Z or Z to A
- Sort data in Ascending or Descending order
- Sort data from lowest to highest or highest to lowest

The sort function can be applied either row-wise or column-wise. It allows the application of the sorting function to a particular column/row to expand the sorting to the whole data.

The sort function is given in the HOME tab under the Editing group with the label 'Sort & Filter'. You can see fig 8.1.

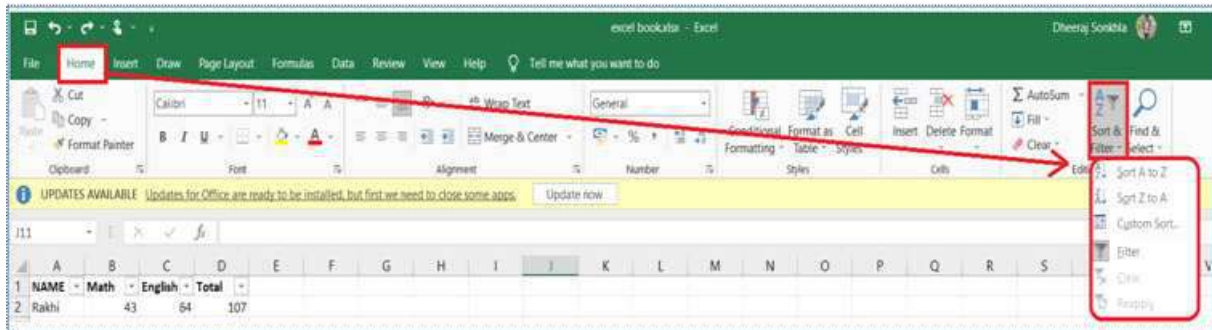


Fig 8.1

Ascending and Descending sorting

To sort the data in ascending or descending order, abide by the following steps.

- Select a column with a numerical value on which sorting is to be carried out.
- Then click on the Sort & Filter option in the editing group under the Home tab.
- Here, more options will appear. Click on 'Sort Smallest to Largest' for sorting the data in ascending order. In case of descending order sorting, click on 'Sort Largest to Smallest'.
- Next, a dialogue box will appear with two options;
 - o Expand the selection- will apply the sort to other data columns in the sheet concerning the selected column.
 - o Continue with the current selection- will apply the sort to the selected column only.
- After selecting the appropriate option, the user can get the sorted data in either ascending or descending order.

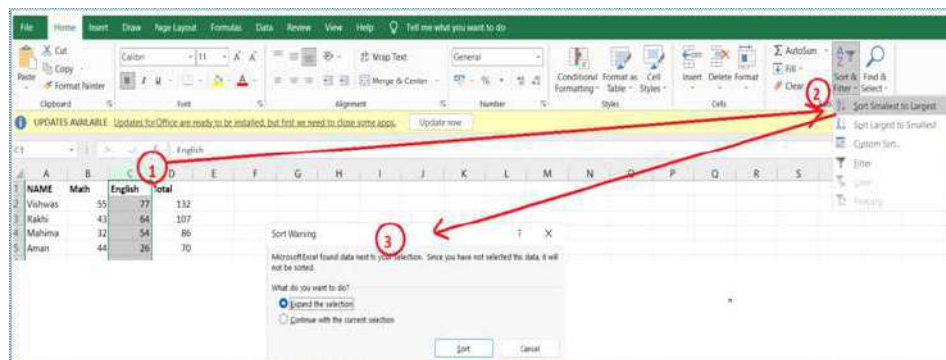


Fig 8.2

An example of steps included in sorting is shown in fig 8.2.

A-Z and Z-A Sorting

This type of sorting is used to sort text data alphabetically. For example, in figure 8.2 we have a column NAME with text values. We can sort the data in this column in alphabetic order, either A to Z or Z to A. To carry out this sorting follow the steps:

- Select a column with text value on which sorting is to be carried out. NAME in our example.
- Then click on the Sort & Filter option in the editing group under the Home tab.
- Here, more options will appear. Click on 'Sort A to Z' or 'Sort Z to A'.
- Next, a dialogue box will appear with two options;
 - o Expand the selection- will apply the sort to other data columns in the sheet concerning the selected column.
 - o Continue with the current selection- will apply the sort to the selected column only.
- After selecting the appropriate option, user can get the alphabetically sorted data.

	A	B	C	D	E
1	NAME	Math	English	Total	
2	Aman	44	26	70	
3	Mahima	32	54	86	
4	Rakhi	43	64	107	
5	Vishwas	55	77	132	

Fig 8.3

Fig 8.3 shows the result of the 'A to Z' sort on the NAME column. Refer the fig 8.2 to compare the data before and after sorting.

Users can apply the sorting function in both ways, column-wise or row-wise. Here, column-wise sorting is discussed. To apply the sort row-wise, follow the steps mentioned above after selecting a row you need to sort. The rest of the process is the same.

8.3 Filter

When working with a huge amount of data, one cannot find the specific data easily. To narrow down the search and find the information quickly, filters can be used. It helps to narrow down the data in your worksheet, which allows you to view only the information you need.

To filter data:

In our example, we are going to filter the data in the table shown below (Table 8.1) based on different cases. For correct working filtering, the worksheet should include a header row, which is used to distinguish each column. In our example, our worksheet is organized into four columns identified by the header cells in row 1. The data in the worksheet is shown in table 8.1.

Table 8.1 Example data

NAME	MATH	ENGLISH	TOTAL
Aman	44	26	70
Mahima	32	54	86
Rakhi	43	64	107
Vishwas	55	77	132

To get started with the filtering, follow the steps:

- o First, select any cell in the range of data.
- o Then go to the 'DATA' tab, and Click Filter in 'Sort & Filter' group.
- o In the header cell for each column, a drop-down arrow will appear.

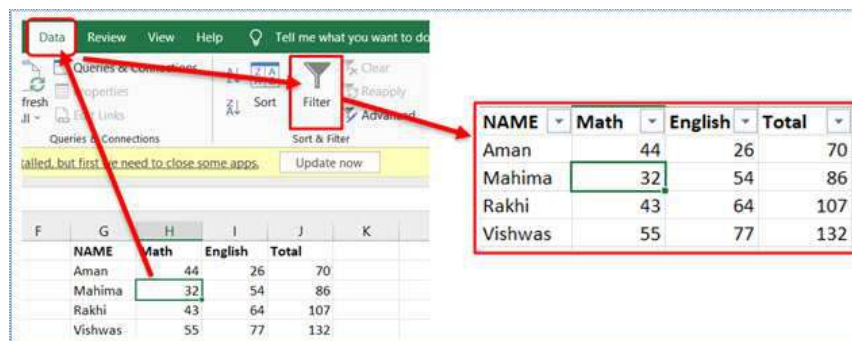


Fig. 8.4

- o Click the drop-down arrow for the column you want to filter. See fig 8.5.

Now, there are many options under the filter function. Users can also apply the sort operation on the data of the column. For example, in fig 8.5, one can see the different options available after clicking the down arrow in the 'Math' column.

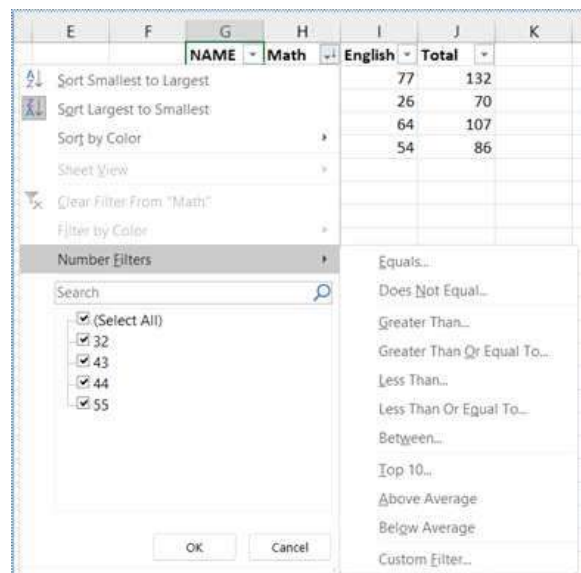


Fig 8.5

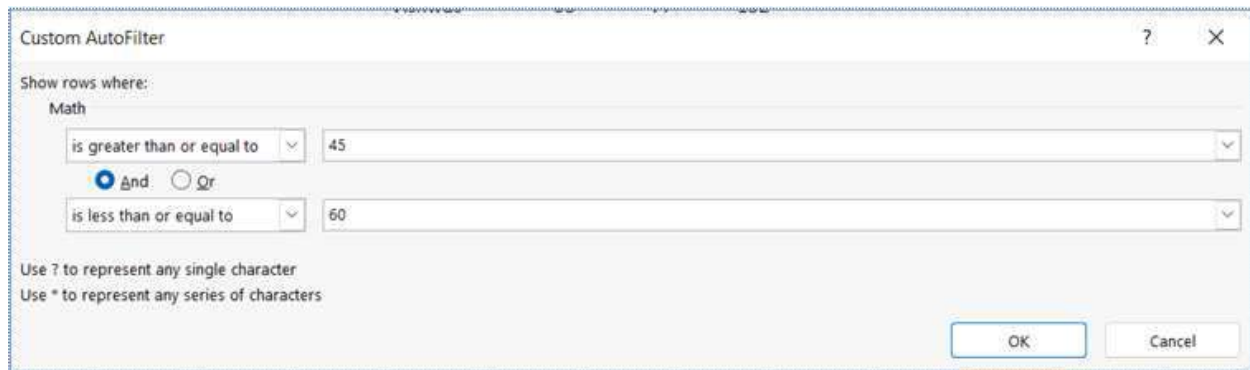
Number Filter

Based on the type of data in the column selected, the user will get the options as 'Number filters' or 'Text Filters'. In our example, we have selected the math column having numerical values. So, here the option available is 'Number Filters'. Number Filters allow the user to manipulate the numerical data in multiple ways. As you can see in fig 8.5, there are multiple conditions available under the category of 'Number filter'.

Let us say, we want to filter the marks in MATHS between 45 and 60.

So, the steps are:

- o First, select the 'Between' option under Number Filters.
- o A dialogue window will appear as shown below.



- o Now add the condition for filtering, that the value should be 'greater than or equal to' 45 and 'less than or equal to' 60. Then press OK.
- o The result you will get is shown below.

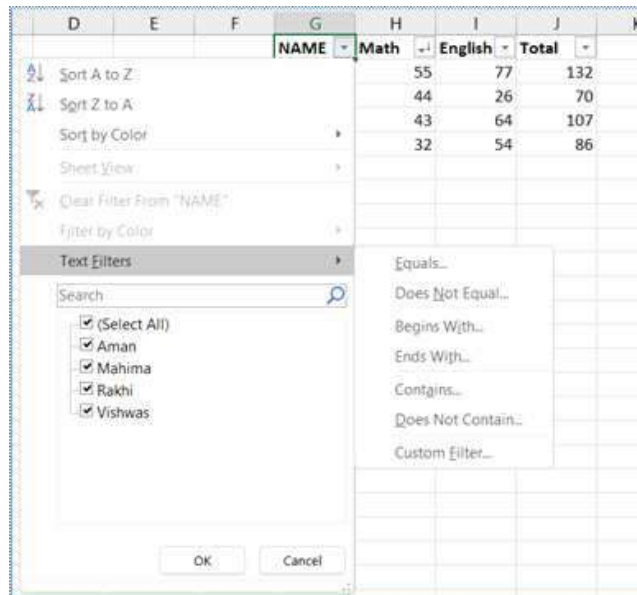
NAME	Math	English	Total
Vishwas	55	77	132

The filtering discussed above is advanced Number Filtering. You can try the different conditions available for 'Number Filter' using the same steps mentioned above.

Text Filter

Similarly, for the column with text data, we have Text Filters. Let us take an example to understand its works. If we want to find the entry of a person with a name starting with alphabet A, we will carry out the following steps.

- o First, click the down arrow on the Name column. We will get the options shown below.



- o After selecting the 'Text Filters', you can see the different conditions available.
- o We will choose the option, 'Begins With'. After selecting this, we will get a dialogue box as shown below.



- o Enter the Condition, i.e., A and press OK. It will give the following results.

NAME	Math	English	Total
Aman	44	26	70

- o This is the final result of Text Filtering.

There are also other types of filtering available, such as Date filtering. This option will appear for the column with the date as data.

Saving the Filtered Data

It may be the case, that the user does not want to modify the original data. In that case, the user can save the result of filtering to another worksheet. There are multiple ways to save the filtered data to a different worksheet. One of the most common ways is to:

- o Add a new worksheet by clicking the plus (+) sign near Sheet one above the status bar.
- o This will add a new worksheet. You can rename this sheet.
- o Now, copy the filtered data by selecting the data and pressing CTRL + C.
- o Go to the new sheet, select a cell where you want to paste, and press CTRL + V.
- o Save the worksheet.

This is the simplest way to save the filtered data into a different worksheet.

8.4 LOOKUP Function

Most of the time, the user wants to look up for a value in the data and need the results for the matching value to be returned. MS Excel provides the user with a handful of ways to do lookup. The lookup can be done in horizontal or vertical way. The function given in the Excel for this purpose is LOOKUP.

In simple terms, one can say that the LOOKUP function searches a value in one column or row and returns a corresponding value from the same position in another column or row. Lookup functions are generally very assistive with large sets of data to find specific information throughout many rows or columns.

There are two forms of LOOKUP in Excel. Namely, Vector and Array. We have discussed the Vector form in this book. In this context, a vector denotes a one-column or one-row range. The vector form of LOOKUP is used to search one row or one column of data for a specified value, and draw a value from the same position in another row or column. The syntax for vector Lookup is:

LOOKUP(lookup_value, lookup_vector, [result_vector])

Where,

- o Lookup_value (required value) – This is the value to search for. It can be a number, text, logical value of TRUE or FALSE, or a reference to a cell containing the lookup value.
- o Lookup_vector (required value) – This is the one-row or one-column range of data to be searched. It must be sorted in ascending order.
- o Result_vector (optional) – This is the optional one-row or one-column range of data from which you want to return the result - a value in the same position as the lookup value. Result_vector must be the same size as lookup_range. If excluded, the result is delivered from lookup_vector.

The following examples helps in understanding the working of Lookup function.

Vertical Lookup

We took a data of seller and their product, Fig 8.6. Here, we are doing a vertical lookup i.e., searching is done in column range. The list of sellers is in column D (D2:D5) and the products they sold in column E (E2:E5). We are making a dashboard where the user will enter the seller's name in B2 and the formula would pull a corresponding product in B3. To accomplish this task the formula needed will be :

=LOOKUP(B2,D2:D5,E2:E5)

	A	B	C	D	E	F
1	RESULT			Seller	Product	
2	Seller	tesla		Apple	Tech	
3	Product	Car		Classmate	Stationary	
4				Tesla	Car	
5				Tata	Steel	

Fig 8.6

Where,

- o B2 – is the lookup_value. The user will enter the name of seller here and we need to check the corresponding product.
- o D2:D5 – is the lookup_vector. This is the range of lookup_vector from which the user will enter the value.
- o E2:E5 – is the result_vector. This is the range of values from which the corresponding value will be searched for.

Horizontal Lookup

There may be the case that the source data has a horizontal layout, i.e., the entries reside in rows rather than columns, fig 8.7. In this case, issue one-row range in the lookup_vector and result_vector arguments. Thus, the formula will be like this:

=LOOKUP(B2,E1:H1,E2:H2)

	A	B	C	D	E	F	G	H
1	RESULT			Seller	Apple	Classmate	Tesla	Tata
2	Seller	apple		Product	tech	stationary	car	steel
3	Product	tech						

Fig 8.7

Uses of Lookup function

Some common uses of LOOKUP functions are listed below:

- o The users can find the exact or suitable match by using the lookup function.
- o The users can search for data both vertically (columns) and horizontally (rows) in the worksheet.
- o It is easy to use and does not require selecting the complete table.

There are some important points to remember while using the lookup function. They are listed below:

- o LOOKUP assumes that lookup_vector is sorted in ascending order which means from A to Z or from smallest to largest.
- o When lookup_value cannot be found, LOOKUP will match the next smallest value, i.e. the largest value in lookup_vector that is less than or equal to lookup_value.
- o When lookup_value is greater than all values in lookup_vector, LOOKUP matches the last value.
- o When lookup_value is smaller than the smallest value in lookup_vector, the Excel LOOKUP returns #N/A.
- o Result_vector must be the same size as lookup_vector.
- o LOOKUP is not case-sensitive, it does not differentiate uppercase and lowercase text.

8.5 Summary

In this chapter, we have studied:

- Concept of sort and filter
- Different methods in sorting and filtering.
- Lookup function.

8.6 Answer to Self-Check exercises

Q.1 Differentiate between sort and filter with examples.

Q.2 How to create a sub sample of dataset using filtering?

Q.3 What is text filter and its different conditions?

Q.4 Explain the concept of horizontal and vertical Lookup.

8.7 Suggested Readings

- Ron Mansfield, The Compact Guide to Microsoft office; BPB Publication
- Dienes, Sheila s., Microsoft office, Professional for windows 95
- Joan Lambert, Curtis Frye, Microsoft Office Step by Step (Office 2021 and Microsoft 385), Microsoft Press Publication
- Kenneth N. Berk, Data Analysis with Microsoft Excel

- Wayne L. Winston, Microsoft Excel 2013: Data Analysis and Business Modeling
- Danielle Stein, Using Excel for business analysis: a guide to financial modelling fundamentals
- Olafusi M., Microsoft Excel and Business Data Analysis for The Busy Professional
- Rajiv Mattus: Learning Excel 98 for windows step by step BPB publication.
- Lonnie. E. Moseley & David M. Boodey: Mastering office 98.
- Micro Soft Office 98: Unleashed: Techmedia.
- John Walken Bach: Excel 98.

Chapter-9

DESCRIPTIVE STATISTICS

STRUCTURE

- 9.0 Learning Objectives
- 9.1 Introduction
- 9.2 Descriptive Statistics
- 9.3 Univariate analysis
- 9.4 Multivariate analysis
- 9.5 Implementation with Excel
- 9.6 Summary
- 9.7 Answer to Self-Check exercises
- 9.8 Suggested Readings

9.0 Learning Objectives

After going through this lesson, you will be able to:

- Understand the basics of descriptive statistics
- Understand the univariate and multivariate data analysis
- Understand the concept of central tendency and measure of dispersion.
- Implementation of descriptive statistics in MS Excel.

9.1 Introduction

Statistics is the field concerned with collecting, organizing, analyzing, interpreting, and presenting data. Statistical formulas and methods are logical and very interesting to implement. The main reason to use statistical analysis is to make the data more informative and bring clarity to the research questions. The statistical procedures help in finding the hidden patterns in the data that are not directly observable. The discovery of these patterns helps in problem-solving and decision-making.

9.2 Descriptive Statistics

Descriptive statistics include numerical and graphical procedures to help the person to understand the data and discover the hidden patterns. In other words, it is the branch of statistics that focuses on measuring and describing the data in such a way that the person can find the patterns in the data which are usually not apparent. In this chapter, the numerical procedures for data analysis are discussed. Normally, a person gathers data, which is not examined and exists as a series of numerical values with no observable relationship. The use of descriptive analysis techniques helps the person to present the data in such a way that hidden patterns can be assessed numerically and visually.

Before analyzing the data, the motive of analysis is important. Next main point is to identify the type of analysis, i.e., univariate or multivariate analysis. Based on this, the methods are selected for final evaluation.

9.3 Univariate analysis

Univariate analysis is the simplest type of analysis where only a single variable is analyzed. There are many ways to analyze univariate data. The most common manner of analysis is to check the central tendency and measures of dispersion. These two are the popular types of summary statistics.

Central Tendency: the measure of central tendency is used to describe a set of data by identifying the central point of distribution or central position within that set of data. That's why it is often called a measure of central location. The mean, mode, and median are the most familiar measures of central tendency. Other measures are used. Let's discuss them briefly.

- **Mean** – It is the average value in a set of data.
- **Median**- It is the middlemost score in a set of data that has been arranged in the order of magnitude. The value of the median is not much affected by skewed data and outliers.
- **Mode**- The mode is the most commonly occurring value/score in the set of data.
- **Trim mean:** It is a method of statistics that is used to get a more realistic mean of the set of data. To use this method, a percentage of the highest and lowest values are cut and then the mean is calculated. This step of pre-calculation enhances the reliability of the mean.
- **Geometric mean** - A geometric mean is a mean or average which shows the central tendency of a set of numbers by using the product of their values. The geometric mean is mostly used to determine the performance results of an investment or portfolio. For a set of n observations, a geometric mean is the n th root of their product. The calculation of geometric mean does not accept negative or zero values, e.g. only positive values. When working with percentages derived from data values, the geometric mean is recommended to use.
- **Harmonic mean**- The use of harmonic mean is averaging of ratios. The ratios are speed and time, cost, etc. The harmonic mean is the most suitable mean if the data consists of rates. Thus, if the data values are rates, then use the harmonic mean. The very common use of harmonic mean is in averaging things like rates (e.g., the average travel speed given a duration of several travels).

Measures of dispersion – Sometimes, the measures of central tendency are not enough to describe data. Two sets of data may have the same mean but they can be completely different. Thus, there is a need to know about the extent of variability to describe data. And this is given by the measures of dispersion.

- **Range** - The range is the numerical difference between the largest and smallest values in the distribution. It is easy to calculate but susceptible to outliers and thus easily affected.
- **Quartile deviation** – The Quartile Deviation is used to judge the spread of a distribution about a measure of its central tendency (usually the mean). So, it generally provides an estimate of the range within which the central 50% of sample data lies.
- **SD**- The standard deviation is used to represent how spread out the values are. The SD represents a standard amount of distance between the mean and each score in the distribution. It is the square root of the variance (VAR). The variance is the average squared distance of the scores in a distribution from the mean.
- **CV** – CV refers to the coefficient of variation. CV is a standard measure of the dispersion of data points around the mean in a data series. It basically represents the ratio of standard deviation to the mean.
- **Gini coefficient**: it is used to measure the inequality among the values of frequency distribution. Normally, like the levels of the income distribution of a population. Value 0 of the Gini coefficient represents perfect equality and 1 (or 100%) represents maximum inequality among the values.

Moments and order statistics

- **Skewness** - A measurement of a data distribution that determines the extent to which it is “imbalanced” or “leaning” away from a standard bell shape (in the case of a normal distribution). Skewness is a term that describes whether, or to what extent, a set of values is not perfectly balanced but rather trails off to the left or right of the center.
- **Kurtosis** - Same as skewness the kurtosis is used to represent the distribution. Kurtosis describes the degree to which values cluster in the tails or the peak of a frequency distribution. The peak is the tallest part of the distribution, and the tails are the ends of the distribution.
- **Rank**-Rank is the data transformation in which the numerical values of the distribution are replaced with their ranks.
- **Percentile** – The use of Percentiles is to represent the point in a distribution of values below which a given percentage of values fall.

9.4 Multivariate data

Multivariate data – When data analysis involves three or more variables, it is categorized under multivariate.

- **Covariance**- the covariance value is the measure of the relationship between two random variables. Which to what level, do they change together.
- **Correlation matrix**- It is a tabular representation of correlation coefficients for different variables. The correlation matrix represents the correlation among all possible pairs of values in a table. It is a way to organize a large set of data which further helps to identify and visualize the patterns.

A correlation matrix constitutes the rows and columns that show the variables. Every single cell in the matrix contains the correlation coefficient. It is often implemented in conjunction with other statistical analyses. One of the major use of matrix is to analyze the multiple linear regression models.

- **Partial correlation** – Partial correlation is a method that is used to measure the strength of a relationship between two variables, while controlling for the effect of one or several other variables, on this relationship. For example, it can be used to measure the correlation between the amount of food eaten and blood pressure, while controlling for the variables like weight and frequency of exercise. It is noted that with the increase in the control variables the test becomes less reliable.
- **Rank correlation**- The Rank correlation is generally used in the case when data is not normally distributed or the presence of outliers affects the association between two random variables. Generally, with correlation, we only know about the relationship between two variables and how one affects the other. But we do not know about the quantity, that is how much they are related.

9.5 Implementation with Excel

Now, let us perform the descriptive analysis with MS Excel. First, we need to locate the Data analysis group in the Data tab as shown below (fig 9.1).



Fig 9.1

If you are not able to find this option then follow the steps mentioned below:

1. Click the File tab, click Options in the bottom left, and then click the Add-Ins category.
 2. In the Manage box, select Excel Add-ins and then click Go.
 3. In the Add-Ins dialogue box, check the 'Analysis ToolPack' check box, and then click OK.
- If Analysis ToolPak is not listed in the Add-Ins available box, you need to click 'Browse' to locate it.
 - If you are prompted that the Analysis ToolPak is not currently installed on your computer, click Yes to install it.

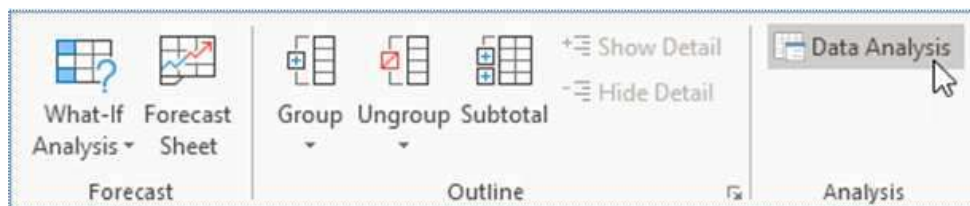
After this step, you will be able to locate the Data Analysis in the Data tab. This Analysis Toolpack is used to generate descriptive statistics. Here is an example.

We took the scores of 10 participants in a test as data, as shown below (fig 9.2). To generate the descriptive analysis for these data values follow the steps given below.

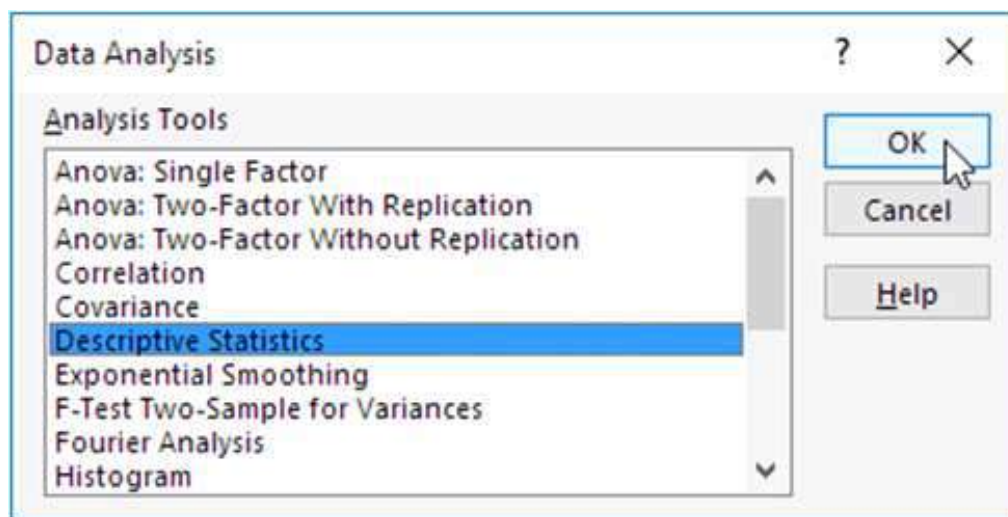
Scores
56
63
77
34
66
98
65
85
71
77

Fig 9.2

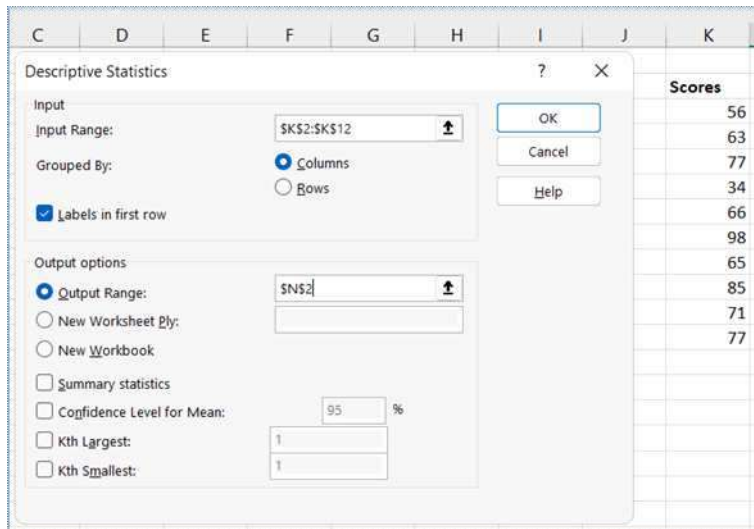
1. Go to the Data tab, in the Analysis group, and click **Data Analysis**.



2. Select Descriptive Statistics from the list and click OK.



3. Select the range K2:K12 (range of input data) as the Input Range.
4. Select cell N2(you can choose any blank cell) as the Output Range.
5. Make sure Summary statistics are checked.



6. Click OK.

Result:

K	L	M	N	O
Scores			Scores	
56				
63			Mean	69.2
77			Standard Error	5.460566
34			Median	68.5
66			Mode	77
98			Standard Deviation	17.26782
65			Sample Variance	298.1778
85			Kurtosis	1.368501
71			Skewness	-0.47131
77			Range	64
			Minimum	34
			Maximum	98
			Sum	692
			Count	10

In the above image, you can check the various statistical information about the data values we have taken (fig 9.2).

In addition to these measures, we can also evaluate other measures in excel.

- **Harmonic mean:** HARMEAN function is available in Excel for harmonic mean. The HARMEAN function takes multiple arguments. The arguments are in the form number 1, number 2, number 3, etc. up to 255. Arguments can be numerical values, cell references, or a range.

Example: =HARMEAN(1,2,6)

It will return 1.8.

- **Geometric mean:** The GEOMEAN function is available in Excel for geometric mean. The GEOMEAN function takes multiple arguments. The arguments are in the form number1, number2, number3, etc. up to 255. Arguments can be numerical values, cell references, or a range.

Example: =GEOMEAN(4,9)

It will return 6.

- **Trim Mean:** The Excel TRIMMEAN function is used to calculate the mean while excluding outliers. The number of data points that will be excluded by the function is provided in the form of a percentage. This percentage can be either in decimal format or percent format.

For the data in fig 9.2, the TRIMMEAN will be: =TRIMMEAN (K3:K12,0.2) and it will return the value 70.

- **Quartile Deviation:** The QUARTILE function is used to calculate the quartile deviation.
- First, you need to enter the data set in a single column of the excel sheet. For example, fig 9.2.
- Click on any of the empty cells of the excel sheet for the result display.
- If you want to find the first quartile then, you have to type “=QUARTILE(K3:K12,1)” and hit enter.
- Just like this, you need to type “=QUARTILE(K3:K12,3)” to find the third quartile.
- **Coefficient of Variance-** To calculate the coefficient of variance in Excel, you need to apply the following formula:

Formula: ‘=STDEV(K3:K12)/AVERAGE(K3:K12)’

Where STDEV is the function to calculate Standard Deviation and Average is used to calculate Average or Arithmetic Mean. You can also use the value of Standard Deviation and Mean from the statistical information returned from Data Analysis.

- **Gini Coefficient:** To calculate the Gini coefficient in Excel follow the steps given below:
 - Enter the data. Two columns namely, the cumulative population % and cumulative income % of individuals. Shown in the figure below.

	A	B
1	Cummulative population %	Cummulative Income %
2	0	0
3	20	15
4	50	36
5	60	54
6	100	100

The meaning of these values is,

- The bottom 20 % individuals of in the country account for 15 % of the total income
 - The bottom 50 % individuals of in the country account for 36 % of the total income
 - The bottom 60 % individuals of in the country account for 54 % of the total income
 - 100% of individuals in the country account for 100% of the total income.
- The next step is to calculate the Area under Lorenz Curve. This curve is used to visualize the distribution of income in a country. We will use the formula $=(A3-A2)*(B3+B2)*0.5$ to calculate the area under Lorenz Curve in column C.

SUM					$=(A3-A2)*(B3+B2)*0.5$
	A	B	C		
1	Cummulative population %	Cummulative Income %	Area under Lorenz curve		
2	0%	0%			
3	20%	15%	$=(A3-A2)*(B3+B2)*0.5$		
4	50%	36%			
5	60%	54%			
6	100%	100%			

- The last step is to calculate the Gini coefficient. For that, we will type the formula $'=1-2*SUM(C3:C6)'$ in D2.

D2					$=1-2*SUM(C3:C6)$
	A	B	C	D	
1	Cummulative population %	Cummulative Income %	Area under Lorenz curve	Gini Coefficient	
2	0%	0%		0.111	
3	20%	15%	0.015		
4	50%	36%	0.0765		
5	60%	54%	0.045		
6	100%	100%	0.308		

So, the Gini Coefficient for this population turned out to be **0.111**. One can use the same method to calculate the Gini coefficient for a large dataset also.

- **Rank:** The RANK function is available in Excel which returns the statistical rank of a given value within a provided array of values.

Formula: $=RANK(\text{number}, \text{ref}, [\text{order}])$

C2						$=RANK(B2, \$B\$2: \$B\$5)$
	A	B	C	D	E	F
1	Name	English	Rank			
2	Vishwas	77	1			
3	Aman	26				
4	Rakhi	64				
5	Mahima	54				

As seen above, the formula is applied in the C2 cell. And same is dragged to the last value in the adjacent column. So the final result will be like this:

	A	B	C
1	Name	English	Rank
2	Vishwas	77	1
3	Aman	26	4
4	Rakhi	64	2
5	Mahima	54	3

The final result after using the RANK function can be seen above.

- **Percentile:** The PERCENTILE function returns the Kth percentile of the selected values in a range. After MS EXCEL 2010 more functions have been introduced. PERCENTILE.INC() works the same as the PERCENTILE. You can still use the PERCENTILE().

Formula: PERCENTILE(array,k)

Where,

1. Array: It is a required argument. The array or range of data that defines relative standing.
2. k: It is required. The percentile value is in the range of 0...1, inclusive. You can either use percentage or decimal values.
3. For example,

=PERCENTILE(E2:E5,0.5)		
	E	F
	Scores	
	10	
	20	
	40	
	30	
	25	

4. As you can see in the above image, we have used the value of k as 0.5 i.e., 50%
5. The value of k cannot be non-numeric otherwise, it will return #VALUE! error value.
6. Fork is < 0 or if k > 1, PERCENTILE returns the #NUM! error value.
- **Covariance:** There are two ways to calculate Covariance in Excel.
1. **COVARIANCE.P():** The first method is to use the Covariance.P() function. It has two arguments, Array1 and array 2. These are range of cells for two variables for which Covariance is to be calculated.

- For example,

Data1	Data2
3	9
2	7
4	12
5	15
6	17

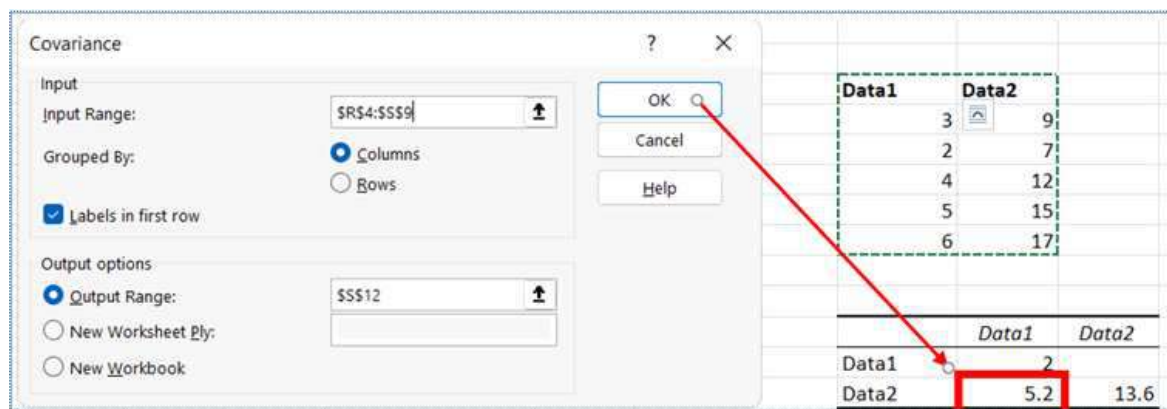
Covariance	=COVARIANCE.P(R5:R9,S5:S9)
------------	----------------------------

COVARIANCE.P(array1, array2)

Here, you can see two variables Data 1 and Data 2 are selected as the two arguments. The value it will return is **5.2**.

2. Using Data Analysis tool.

- Just go to the Analysis group in the Data tab and click Data Analysis.
- Select Covariance from the list.
- Select the input data range.
- Select output range.



- Check the box for Labels in the first row (if any).
- Click OK. The result will be shown in a table form.
- An example is shown above. The result is highlighted in the rectangular box, i.e., 5.2.
- **Correlation Matrix:** There are two ways to calculate Correlation in Excel.
 - o **CORREL():** This function is used to calculate the correlation. It has two arguments Array 1 and Array 2. These are range of cells for two variables for which Correlation is to be calculated.
 - o **Using Data Analysis tool.**
 - Just go to the Analysis group in the Data tab and click Data Analysis.
 - Select Correlation from the list.
 - Select the input data range.

- $$\text{Partial Corr (X,Y).Z} = (\text{Corr (X,Y)} - \text{Corr (X,Z)} * \text{Corr (Y,Z)}) / (\sqrt{1 - (\text{Corr (X,Z)})^2} * \sqrt{1 - (\text{Corr (Y,Z)})^2})$$

5	65	6	87	Partial Correlation (X,Y).Z	-0.6385
---	----	---	----	-----------------------------	---------

- apply the traditional formula step-by-step in excel.

9.6 Summary

In this chapter, we have studied:

- About the descriptive statistics.
- Univariate and multivariate data analysis.
- Implementation of various statistical methods in excel to perform descriptive data analysis like the measure of central tendency and measures of dispersion.

9.7 Answer to Self-Check exercises

- Q.1 Define descriptive statistics.
- Q.2 Discuss the Data Analytical Toolpack in Excel.
- Q.3 How to perform the data analysis in Excel?
- Q.4 Explain the various measures of descriptive statistics available under data analysis.

9.8 Suggested Readings

- Ron Mansfield, The Compact Guide to Microsoft office; BPB Publication
- Dienes, Sheila s., Microsoft office, Professional for windows 95
- Joan Lambert, Curtis Frye, Microsoft Office Step by Step (Office 2021 and Microsoft 395),Microsoft Press Publication
- Kenneth N. Berk, Data Analysis with Microsoft Excel
- Wayne L. Winston,Microsoft Excel 2013: Data Analysis and Business Modeling
- Danielle Stein, Using Excel for business analysis: a guide to financial modelling fundamentals
- Olafusi M., Microsoft Excel and Business Data Analysis for The Busy Professional
- Rajiv Mattus: Learning Excel 99 for windows step by step BPB publication.
- Lonnie. E. Moseley & David M. Boodey: Mastering office 99.
- Micro Soft Office 99: Unleashed: Techmedia.
- John Walken Bach: Excel 99.

Chapter-10

GRAPHICAL PRESENTATION OF DATA

STRUCTURE

- 10.0 Learning Objectives
- 10.1 Introduction
- 10.2 Charts
 - 10.2.1 Types of Charts
- 10.3 Create a chart with Insert Charts
- 10.4 Scatter Chart
- 10.5 Curve Fitting
- 10.6 Summary
- 10.7 Answer to Self-Check exercises
- 10.8 Suggested Readings

10.0 Learning Objectives

After going through this lesson, you will be able to:

- Understand the importance of the graphical presentation of data
- Understands the different types of charts
- Understand the importance and applications of charts

10.1 Introduction

It is sometimes not easy to interpret Excel data due to the complexity and size of the data. So, charts are a way to represent the data graphically and interpret the data easily. Charts are the visual representation of data.

Excel provides charts to take advantage of graphical representation. The data represented through charts is more understandable than the data stored in an Excel table. This makes the process of analyzing data fast. Excel users can fast analyze the data.

Graphical representation of data using charts makes complex data analysis easier to understand. Excel has a variety of charts, each with its different functionality and representation style.

10.2 Charts

A chart is a graphical representation of your data in a worksheet. Using charts to represent worksheet data often creates a better understanding of your data rather than simply presenting the numbers in a spreadsheet. They are also a great way to add branding to your presentations.

There is a list of basic and advanced levels of charts used for different scenarios to analyze the data.

Excel offers various charts to represent the data in different manners, such as - Pie charts, Bar charts, Line charts, Stock charts, Surface charts, Radar charts, and many more. You can utilize them as per your data analysis needs.

10.2.1 Types of Charts

I. Clustered column and 3-D clustered column



A clustered column chart shows values in 2-D columns. A 3-D clustered column chart shows columns in 3-D format, but it doesn't use a third value axis (depth axis). Use this chart when you have categories that represent:

- Ranges of values (for example, item counts).
- Specific scale arrangements (for example, a Likert scale with entries like Strongly agree, Agree, Neutral, Disagree, strongly disagree).
- Names that are not in any specific order (for example, item names, geographic names, or the names of people).
- **Stacked column and 3-D stacked column:** A stacked column chart show values in 2-D stacked columns. A 3-D stacked column chart shows the stacked columns in 3-D format, but it doesn't use a depth axis. Use this chart when you have multiple data series and you want to emphasize the total.



- **100% stacked column and 3-D 100% stacked column:** A 100% stacked column chart shows values in 2-D columns that are stacked to represent 100%. A 3-D 100% stacked column chart shows the columns in 3-D format, but it doesn't use a depth axis. Use this chart when you have two or more data series and you want to emphasize the contributions to the whole, especially if the total is the same for each category.

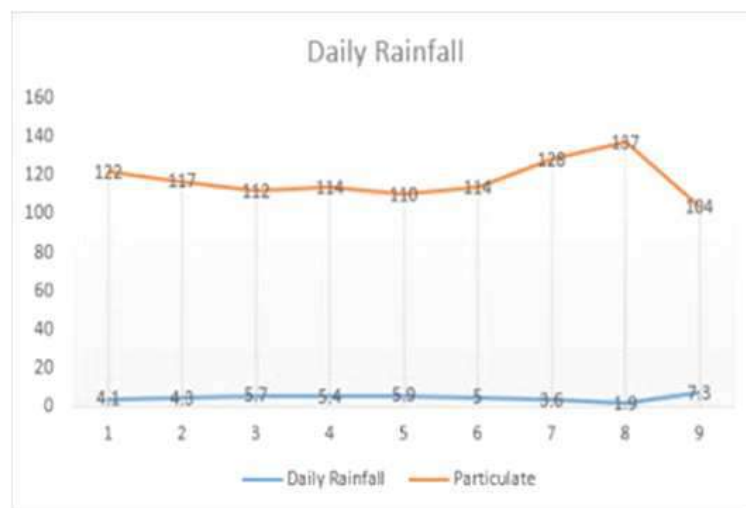


- **3-D column:** 3-D column charts use three axes that you can change (a horizontal axis, a vertical axis, and a depth axis), and they compare data points along the horizontal and the depth axes. Use this chart when you want to compare data across both categories and data series.



II. Line Charts

Data that's arranged in columns or rows on a worksheet can be plotted in a line chart. In a line chart, category data is distributed evenly along the horizontal axis, and all value data is distributed evenly along the vertical axis. Line charts can show continuous data over time on an evenly scaled axis, so they're ideal for showing trends in data at equal intervals, like months, quarters, or fiscal years.



III. Pie Chart

Data that's arranged in one column or row on a worksheet can be plotted in a pie chart. Pie charts show the size of items in one data series, proportional to the sum of the items. The data points in a pie chart are shown as a percentage of the whole pie.



Consider using a pie chart when:

- You have only one data series.
- None of the values in your data are negative.
- Almost none of the values in your data are zero values.
- You have no more than seven categories, all of which represent parts of the whole pie.

Types of pie charts

- **Pie and 3-D pie** Pie charts show the contribution of each value to a total in a 2-D or 3-D format. You can pull out slices of a pie chart manually to emphasize the slices.



- **Pie of pie and bar of pie** Pie of pie or bar of pie charts show pie charts with smaller values pulled out into a secondary pie or stacked bar chart, which makes them easier to distinguish.



IV. Doughnut Chart

Data that's arranged in columns or rows only on a worksheet can be plotted in a doughnut chart. Like a pie chart, a doughnut chart shows the relationship of parts to a whole, but it can contain more than one data series.



- V. **Bar Chart:** Data that's arranged in columns or rows on a worksheet can be plotted in a bar chart. Bar charts illustrate comparisons among individual items. In a bar chart, the categories are typically organized along the vertical axis and the values along the horizontal axis.



Consider using a bar chart when:

- The axis labels are long.
- The values that are shown are durations.

Types of bar charts

- **Clustered bar and 3-D clustered bar** A clustered bar chart shows bars in 2-D format. A 3-D clustered bar chart shows bars in 3-D format; it doesn't use a depth axis.



- **Stacked bar and 3-D stacked bar** Stacked bar charts show the relationship of individual items to the whole in 2-D bars. A 3-D stacked bar chart shows bars in 3-D format; it doesn't use a depth axis.



- **100% stacked bar and 3-D 100% stacked bar** A 100% stacked bar shows 2-D bars that compare the percentage that each value contributes to a total across categories. A 3-D 100% stacked bar chart shows bars in 3-D format; it doesn't use a depth axis.



VI. Area Chart

Data that's arranged in columns or rows on a worksheet can be plotted in an area chart. Area charts can be used to plot change over time and draw attention to the total value across a trend. By showing the sum of the plotted values, an area chart also shows the relationship of parts to a whole.

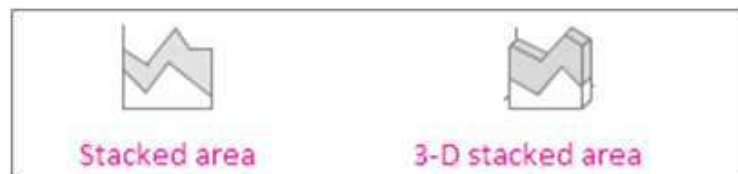


Types of area charts

- **Area and 3-D area** Shown in the 2-D or 3-D format, area charts demonstrate the trend of values over time or other category data. A 3-D area chart uses three axes (horizontal, vertical, and depth) that one can change.



- **Stacked area and 3-D stacked area** Stacked area charts show the trend of the contribution of each value over time or other category data in 2-D format. A 3-D stacked area chart does the same, but it shows areas in the 3-D format without using a depth axis.



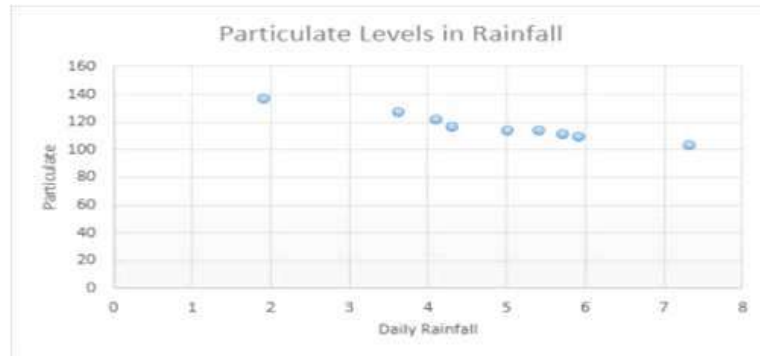
- **100% stacked area and 3-D 100% stacked area** 100% stacked area charts demonstrate the trend of the percentage that each value contributes over time or other category data. A 3-D 100% stacked area chart does the same, but it presents areas in the 3-D format without using a depth axis.



VII. Scatter Chart

Data that's arranged in columns and rows on a worksheet can be plotted in an XY (scatter) chart. Place the x values in one row or column, and then enter the corresponding y values in the adjacent rows or columns.

A scatter chart has two value axes: a horizontal (x) and a vertical (y) value axis. It combines x and y values into single data points and shows them in irregular intervals, or clusters. Scatter charts are typically used for showing and comparing numeric values, like scientific, statistical, and engineering data.

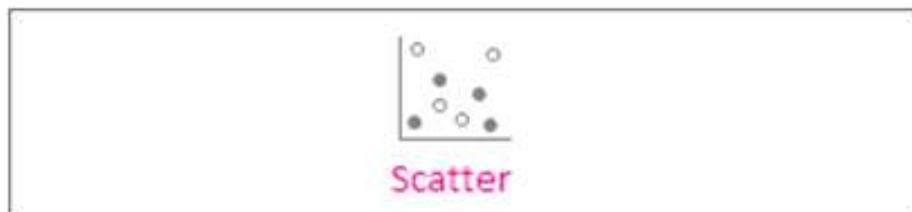


Consider using a scatter chart when:

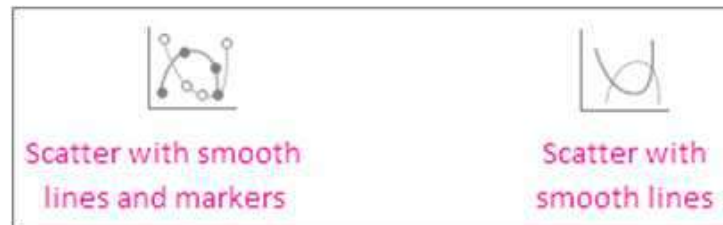
- You need to change the scale of the horizontal axis.
- You need to make that axis a logarithmic scale.
- Values for the horizontal axis are not evenly spaced.
- There are many data points on the horizontal axis.
- You want to adjust the independent axis scales of a scatter chart to reveal more information about data that includes pairs or grouped sets of values.
- You want to show similarities between large sets of data instead of differences between data points.
- You want to compare many data points without regard to time—the more data that you include in a scatter chart, the better the comparisons you can make.

Types of scatter charts

- **Scatter** This chart shows data points without connecting lines to compare pairs of values.



- **Scatter with smooth lines and markers and scatter with smooth lines** This chart shows a smooth curve that connects the data points. Smooth lines can be shown with or without markers. Use a smooth line without markers if there are many data points.

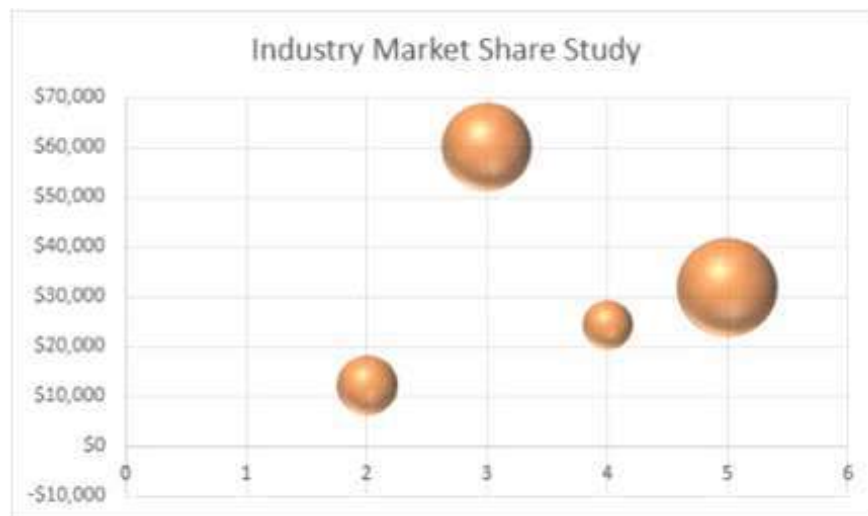


- **Scatter with straight lines and markers and scatter with straight lines** This chart shows straight connecting lines between data points. Straight lines can be shown with or without markers.



VIII. Bubble Chart

Much like a scatter chart, a bubble chart adds a third column to specify the size of the bubbles it shows to represent the data points in the data series.



Type of bubble charts

- **Bubble or bubble with 3-D effect** Both of these bubble charts compare sets of three values instead of two, showing bubbles in 2-D or 3-D format (without using a depth axis). The third value specifies the size of the bubble marker.



IX. Radar

A radar chart compares the aggregate values of many data series.

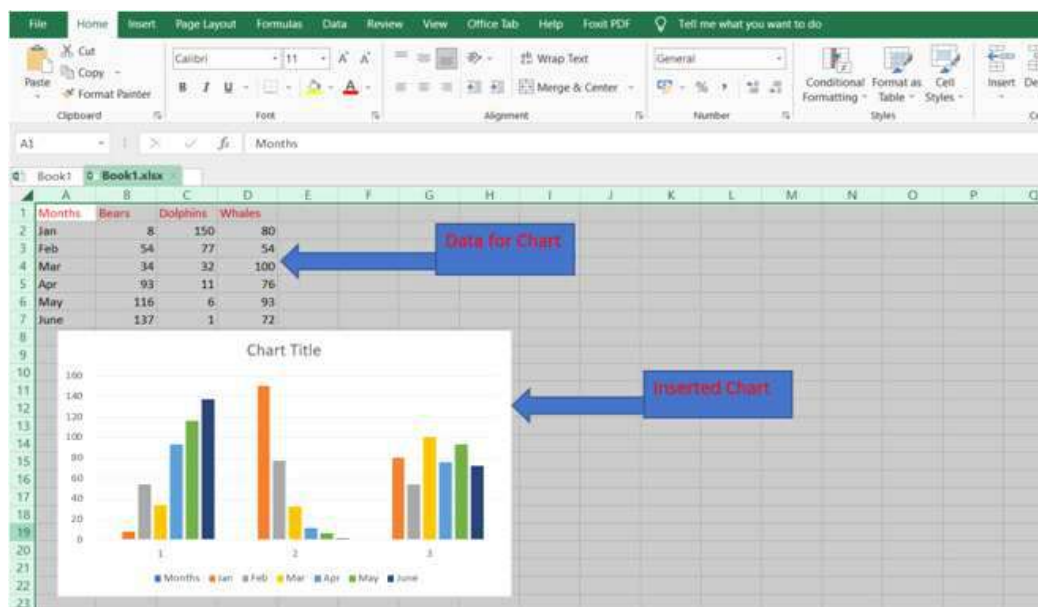
10.3 Create a chart with Insert Charts

Follow the steps:

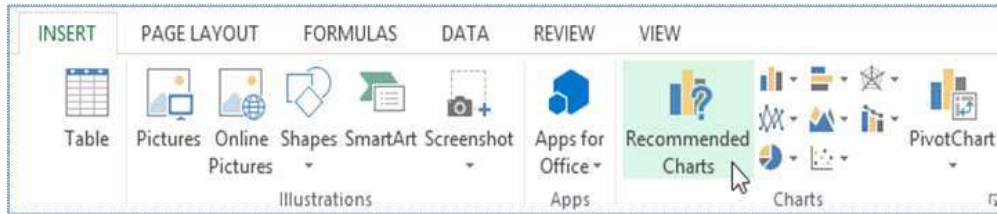
- Step 1 " Select the data for the chart.
- Step 2 " Click the Insert tab on the Ribbon.
- Step 3 " Click the Desired Chart on the Ribbon.
- Step 4. Select Ok.



To create a chart with a short method, just select the data you want in the chart and press ALT+F1. It will immediately create a chart. The disadvantage of this method is that the resultant chart may not be the best fit for your data or as per your requirement.



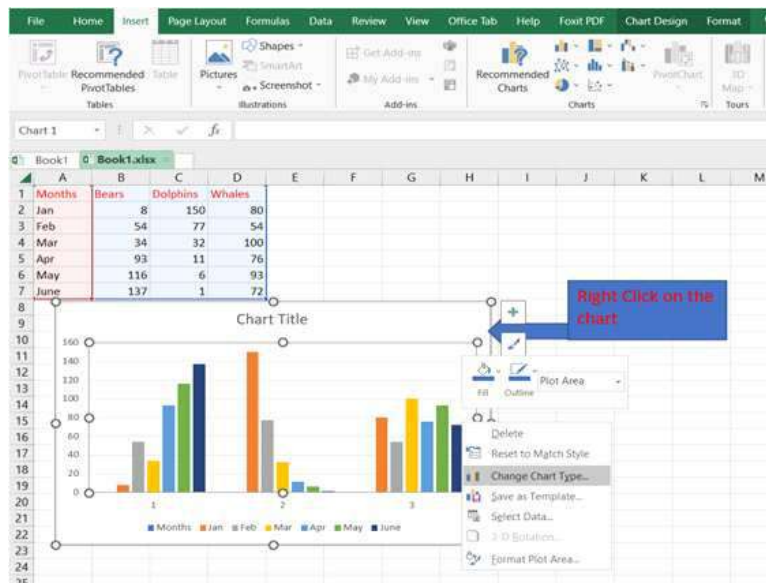
If you're not sure which type of chart to use, the **Recommended Charts** command will suggest several different charts based on the source data.



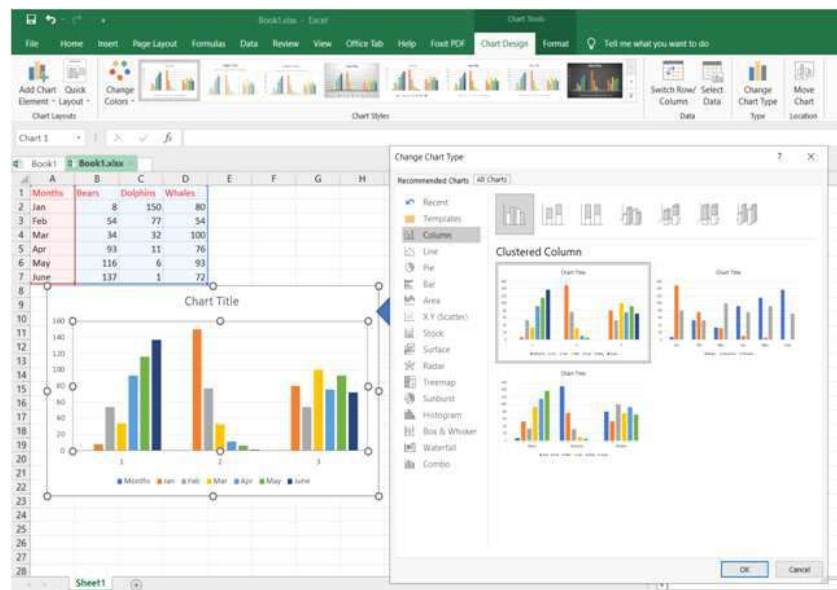
Change Chart Type

You can easily change to a different type of chart at any time.

1. Select the chart.
2. Right Click on the chart, and click Change Chart Type.

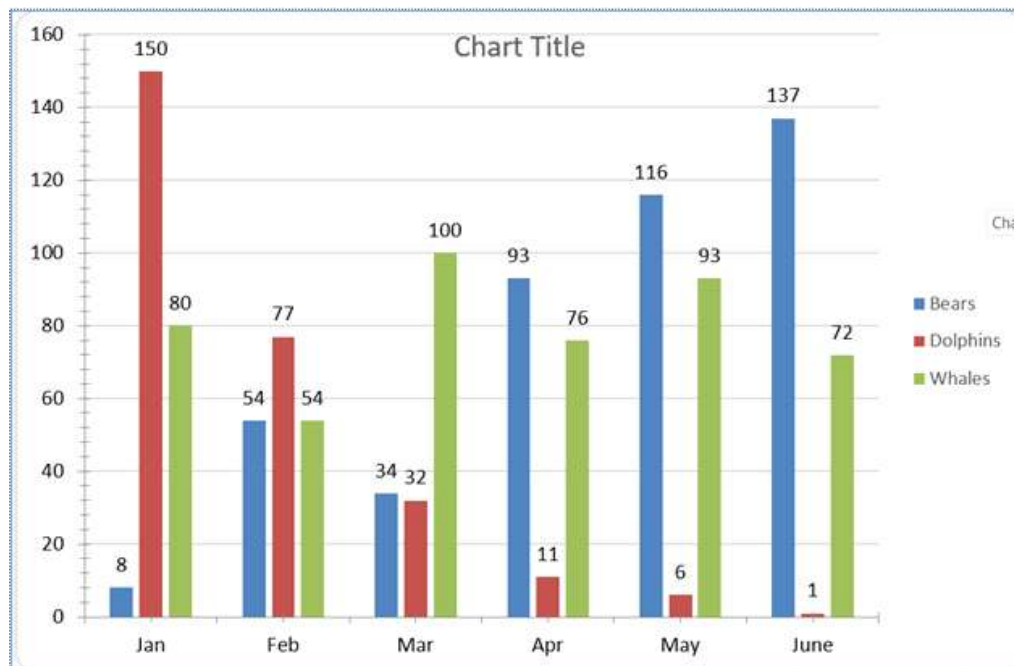


Note: On the Chart Design tab, in the Type group, click Change Chart Type.



3. Select the Chart then click Ok

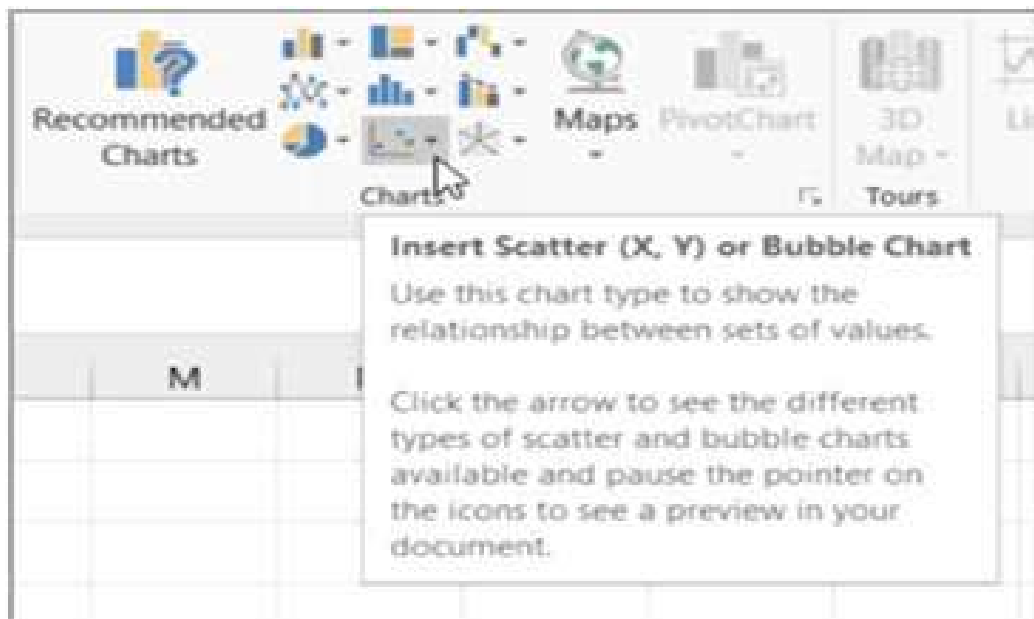
The final result will be like this,

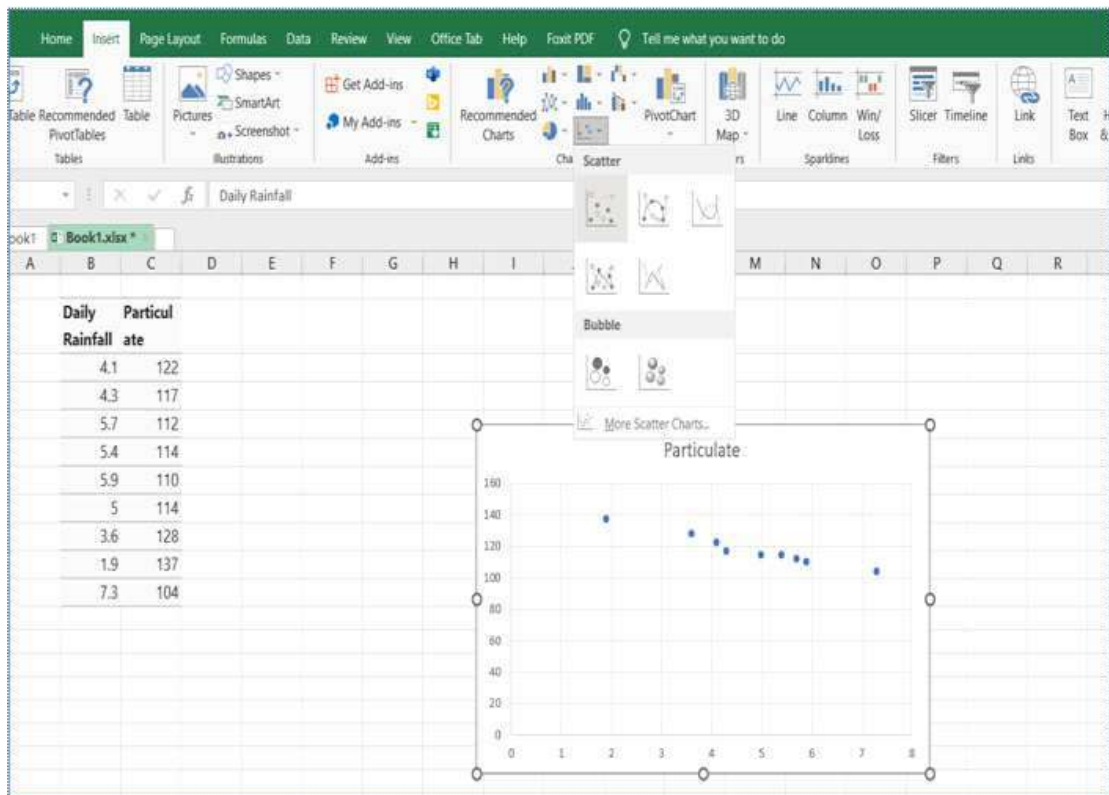


10.4 Scatter Chart

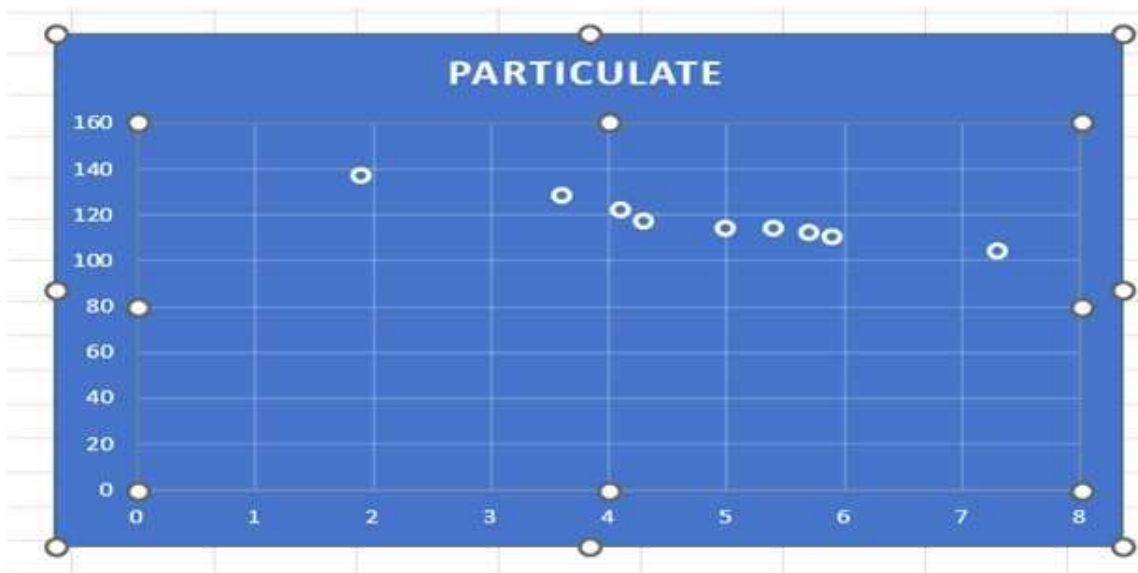
Now, let us see the steps to create a scatter chart. The following procedure will help you create a scatter chart in easy steps. For this chart, we used the example worksheet data.

1. Select the data you want to plot in the scatter chart.
2. Click the **Insert** tab, and then click **Insert Scatter (X, Y) or Bubble Chart**.





The final result will be like this:



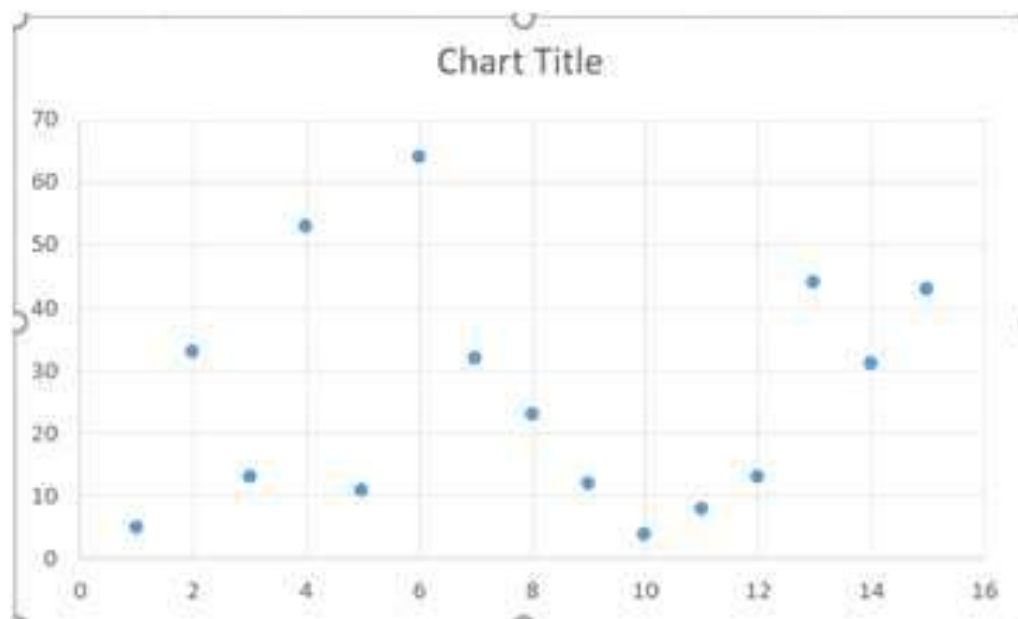
10.5 Curve Fitting

Excel charts provide a very convenient way to fit a curve into the experimental data.

- First, create a data set of two variables

	A	B
1	x	y
2	1	5
3	2	33
4	3	13
5	4	53
6	5	11
7	6	64
8	7	32
9	8	23
10	9	12
11	10	4
12	11	8
13	12	13
14	13	44
15	14	31
16	15	43
17		

- Create a **scatter chart**.

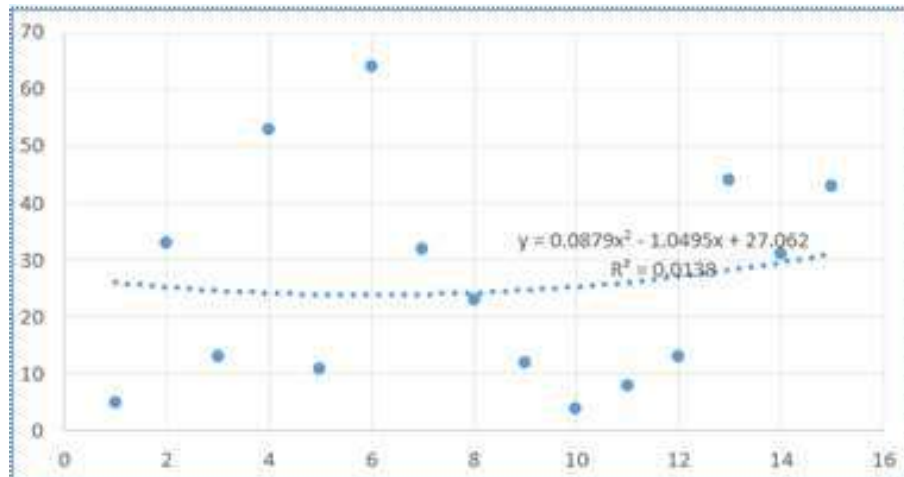


- Then right-click on the data series and select **"Add Trendline."** and then click **More Options**.
- In the window that appears to the right, click the button next to **Polynomial**. Then check the boxes next to **Display Equation on chart** and **Display R-squared value on chart**.

The equation of the curve is as follows:

$$y = 0.0879x^2 - 1.0495x + 27.062$$

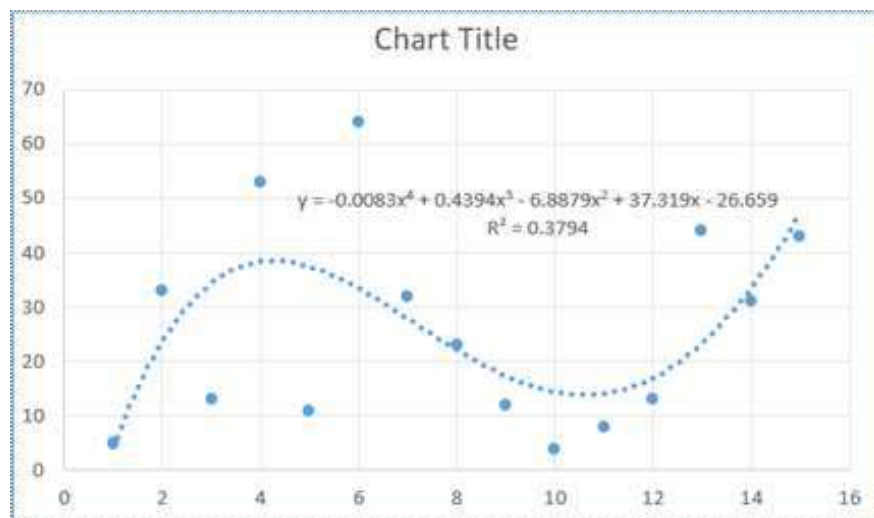
The R-squared value state the percentage of variation in the response variable. The value of R-squared is 0.0138 for this curve.



The curve we get for this value is not flexible. To make it more flexible we can try to increase the polynomial order to 4 from 2. This will give the following equation.

$$y = -0.0083x^4 + 0.4394x^3 - 6.8879x^2 + 37.319x - 26.659$$

and the resultant curve will be like this,



The R-squared value for this particular curve is 0.3794. As you can see that this R-squared value is comparatively higher than that of the previous curve, which shows that it fits the dataset much more closely.

10.6 Summary

In this chapter, we have studied:

- About the concept of graphs presentation in excel
- Different types of charts available in Excel with their creation
- Modification of charts as per user requirement
- Best Curve fitting for the given set of data

10.7 Answer to Self-Check exercises

- Q.1 What is graph presentation? Give examples.
- Q.2 Discuss 5 types of charts with their applications?
- Q.3 How to change a chart type?
- Q.4 Explain the role of polynomial order in curve fitting.

10.8 Suggested Readings

- Ron Mansfield, The Compact Guide to Microsoft office; BPB Publication
- Dienes, Sheila s., Microsoft office, Professional for windows 105
- Joan Lambert, Curtis Frye, Microsoft Office Step by Step (Office 2021 and Microsoft 3105),Microsoft Press Publication
- Kenneth N. Berk, Data Analysis with Microsoft Excel
- Wayne L. Winston,Microsoft Excel 2013: Data Analysis and Business Modeling
- Danielle Stein, Using Excel for business analysis: a guide to financial modelling fundamentals
- Olafusi M., Microsoft Excel and Business Data Analysis for The Busy Professional
- Rajiv Mattus: Learning Excel 1010 for windows step-by-step BPB publication
- Lonnie.E. Moseley & David M. Boodey: Mastering office 1010
- Micro Soft Office 1010: Unleashed: Techmedia.
- John Walken Bach: Excel 1010.
